ARMY, MARINE CORPS, NAVY, AIR FORCE

SCAR

MULTI-SERVICE TACTICS, TECHNIQUES, AND PROCEDURES FOR STRIKE COORDINATION AND RECONNAISSANCE

ATP 3-60.2
MCRP 3-20D.1
NTTP 3-03.4.3
AFTTP 3-2.72

JANUARY 2018

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*Supersedes ATP 3-60.2/MCRP 3-20D.1/NTTP 3-03.4.3/AFTTP 3-2.72, dated 10 January 2014.

MULTI-SERVICE TACTICS, TECHNIQUES, AND PROCEDURES

FOREWORD

This multi-Service tactics, techniques, and procedures (MTTP) publication is a project of the Air Land Sea Application (ALSA) Center in accordance with the memorandum of agreement between the Headquarters of the Army, Marine Corps, Navy, and Air Force doctrine commanders directing ALSA to develop MTTP publications to meet the immediate needs of the warfighter.

This MTTP publication has been prepared by ALSA under our direction for implementation by our commands and for use by other commands as appropriate.

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PREFACE

1. Purpose

This publication provides strike coordination and reconnaissance (SCAR) multi-Service tactics, techniques, and procedures (MTTP) to the military Services for coordinating interdiction.

Note: For the Army, the term “command and control” was replaced with “mission command.” Mission command now encompasses the Army’s philosophy of command (still known as mission command) as well as the exercise of authority and direction to accomplish missions (formerly known as command and control). The Army still uses the term “command and control” as it pertains to airspace command and control.

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2. Scope

This publication describes tactics, techniques, and procedures (TTP) for tactical level planning and execution of the SCAR mission, and addresses operational considerations for effective application. SCAR interdiction assets include fixed-wing (FW), rotary-wing (RW), and surface-to-surface fires. This MTTP publication is not applicable for targets in close proximity to friendly forces, where detailed integration with fire and maneuver is required. These targets require close air support (CAS) TTP.

3. Applicability

This publication applies to operational and tactical-level planners and operators. This MTTP publication is intended as a reference document for mission planning and execution of SCAR missions with allied, coalition, and multi-Service forces.

4. Implementation Plan

Participating Service command offices of primary responsibility will review this publication; validate the information; and, where appropriate, reference and incorporate it in Service manuals, regulations, and curricula as follows:

Army. Upon approval and authentication, this publication incorporates the TTP contained herein into the US Army Doctrine and Training Literature Program as directed by the Commander, US Army Training and Doctrine Command (TRADOC). Distribution is in accordance with applicable directives listed on the authentication page.

Marine Corps.[1] The Marine Corps will incorporate the procedures in this publication in US Marine Corps doctrine publications as directed by the Deputy Commandant, Combat Development and Integration (DC, CD&I). Distribution is in accordance with the Marine Corps Publication Distribution System.

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[1] Marine Corps PCN: 144 000170 00

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5. User Information

a. US Army Combined Arms Center; HQMC, DC, CD&I; NWDC; Curtis E. LeMay Center for Doctrine Development and Education (LeMay Center); and Air Land Sea Application (ALSA) Center developed this publication with the joint participation of the approving Service commands. ALSA will review and update this publication as necessary.

b. This publication reflects current joint and Service doctrine, command and control organizations, facilities, personnel, responsibilities, and procedures. Changes in Service protocol, appropriately reflected in joint and Service publications, will be incorporated in revisions to this document.

c. We encourage recommended changes for improving this publication. Key your comments to the specific page and paragraph and provide a rationale for each recommendation. Send comments and recommendations directly to:

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SUMMARY OF CHANGES

ATP 3-60.2/MCRP 3-20D.1/NTTP 3-03.4.3/AFTTP 3-2.72, *Multi-Service Tactics, Techniques, and Procedures for Strike Coordination and Reconnaissance (SCAR)*.

This revision:

Updates include:

- Modifying this publication to include “interdiction”, which is broader than the term “air interdiction” used in the last version of this publication.
- Modifying the scope of this publication to include interdiction from surface-to-surface (S/S) fires, and air interdiction from fixed-wing and rotary-wing assets.
- Restructuring Chapter I, “Fundamentals”, and Chapter II, “Command and Control”, for clarity.
- Modifying Chapter III, “Planning”, and Chapter IV, “Execution”, to include S/S fires integration tactics, techniques, and procedures (TTP) and Army aviation integration requirements.
- Better identifying and defining SCAR aircrew authorities, responsibilities, and functions than in the last version of this publication. SCAR responsibilities serve as the minimum requirements for an aircrew, of any Service, to fulfill the role of the SCAR aircrew.
- Clarifying the difference between the SCAR aircrew and the SCAR mission by associating the word “aircrew” with “the SCAR” when referring to the individual aircrew (i.e., “the SCAR aircrew”).
- Incorporating the term “strikers” when referring to aircraft capable of conducting air interdiction (AI), airborne alert air interdiction (XAI), and armed reconnaissance (AR).
- Updating the appendices to reflect SCAR multi-Service tactics, techniques, and procedures modifications and coordinating measure definition changes.
- Adding description, location, elevation, remarks, and restrictions (DLER) as the preferred attack briefing format.

Adds:

- Army aviation integration requirements to support and conduct SCAR missions.
- Army, Marine Corps, and Navy S/S fires integration TTP. The SCAR aircrew has the ability to coordinate for S/S fires, but will not be expected to conduct a call for fire unless specifically trained to do so by the owning Service.
- Relevant Joint Air Attack Team (JAAT) TTP thereby retaining JAAT capabilities while ensuring the Army maintains the ability to “mass fires”. Incorporated JAAT TTP includes S/S fires integration, Army aviation integration requirements, and “coordinated attacks”.
- Command and control, and deconfliction methods figures for clarity.

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NTTP 3-03.4.3	**Navy Warfare Development Command** **Norfolk, Virginia**
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31 Jan 2018

SCAR

MULTI-SERVICE TACTICS, TECHNIQUES, AND PROCEDURES FOR STRIKE COORDINATION AND RECONNAISSANCE

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*Supersedes ATP 3-60.2/MCRP 3-20D.1/NTTP 3-03.4.3/AFTTP 3-2.72, dated 10 January 2014.

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EXECUTIVE SUMMARY

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ATP 3-60.2/MCRP 3-23C/NTTP 3-03.4.3/AFTTP 3-2.72, *Multi-Service Tactics, Techniques, and Procedures for Strike Coordination and Reconnaissance (SCAR).*

Chapter I Fundamentals

Chapter I provides an overview of the SCAR mission, defines SCAR terminology, and lists a SCAR aircrew's authorities, responsibilities, and functions.

Chapter II Command and Control

Chapter II describes operational and tactical level command and control (C2) agencies and airspace control constructs that enable SCAR missions.

Chapter III Planning

Chapter III is a planning guide to aid planners during SCAR mission development. This chapter includes planning considerations at the operational and tactical levels.

Chapter IV Execution

Chapter IV provides methods to deconflict assets in support of a SCAR mission and describes how to use the find, fix, target, track, engage, and assess (F2T2EA) process locate and engage targets efficiently while meeting the supported commander's intent.

Appendix A

Strike Coordination and Reconnaissance (SCAR) Mission Planning Guide

Appendix A is a condensed planning guide to aid planners during SCAR mission development.

Appendix B

Strike Coordination and Reconnaissance (SCAR) Mission Briefing Guide

Appendix B provides essential information to guide SCAR mission briefings.

Appendix C

Strike Coordination and Reconnaissance (SCAR) Product Examples

Appendix C provides examples for planning and executing SCAR missions.

Appendix D

Coordination Measures

Appendix D provides airspace coordinating measures (ACMs) and fire support coordination measures (FSCMs) commonly associated with SCAR missions.

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Chapter I
FUNDAMENTALS

1. Introduction

a. This publication provides strike coordination and reconnaissance (SCAR) multi-Service tactics, techniques, and procedures (MTTP) for coordinating interdiction. The end state is a streamlined interdiction process that meets the commander's intent by providing executable tasks to exploit and destroy targets. This publication serves as a reference for planners, and a checklist for operators to execute SCAR missions.

b. SCAR missions are flown in a specific geographic area and are elements of command and control (C2) used to match weapons effects with targets per the supported commander's prioritized target list through proper real-time allocation of interdiction assets. A SCAR mission is designed to effectively and efficiently destroy targets and conduct associated battle damage assessments (BDAs).

c. Specific focus areas of this publication include C2, planning, and execution for engaging interdiction targets. This MTTP publication is not applicable for targets in close proximity to friendly forces, where detailed integration with fire and movement is required. These targets require close air support (CAS) tactics, techniques, and procedures (TTP). The TTP in this document are consistent with Army techniques publication (ATP) 3-09.34; Marine Corps reference publication (MCRP) 3-31.4; Navy tactics, techniques, and procedures (NTTP) 3-09.2.1; Air Force tactics, techniques, and procedures (AFTTP) 3-2.59, *Multi-Service Tactics, Techniques, and Procedures for Kill Box Planning and Employment* (hereafter referred to as *Kill Box MTTP*). Also, it is consistent with dynamic targeting in accordance with (IAW) ATP 3-60.1, MCRP 3-31.5, NTTP 3-60.1, AFTTP 3-2.3, *Multi-Service Tactics, Techniques, and Procedures for Dynamic Targeting*; and are IAW joint publication (JP) 3-09, *Joint Fire Support*; JP 3-03, *Joint Interdiction*; and JP 3-60, *Joint Targeting*.

2. Definitions

a. SCAR. A mission flown for the purpose of detecting targets and coordinating or performing interdiction or reconnaissance on those targets. (DOD Dictionary of Military and Associated Terms, Source: JP 3-03)

b. The SCAR Aircrew. The SCAR aircrew refers to an aircrew tasked to perform SCAR mission responsibilities. See paragraph 6 for the SCAR aircrew's responsibilities.

c. Interdiction. An action to divert, disrupt, delay, or destroy the enemy's military surface capability before it can be used effectively against friendly forces, or to achieve enemy objectives. (DOD Dictionary of Military and Associated Terms, Source: JP 3-03)

d. Air Interdiction (AI). Air operations conducted to divert, disrupt, delay, or destroy the enemy's military surface capabilities before it can be brought to bear effectively against friendly forces, or to otherwise achieve objectives that are conducted at such distances from friendly forces that detailed integration of each air mission with the fire and movement of friendly forces is not required. (DOD Dictionary of Military and Associated Terms, Source: JP 3-03)

e. Airborne Alert Air Interdiction (XAI). Airborne alert air interdiction identifies an AI airborne alert mission tasked for on-call targets that may be retasked during execution for targets of opportunity (also referred to as armed reconnaissance). (Source: Air Force Doctrine, Annex 3-03)

f. Armed Reconnaissance (AR). A mission with the primary purpose of locating and attacking targets of opportunity (i.e., enemy materiel, personnel, and facilities) in assigned general areas or along assigned communications routes, and not for the purpose of attacking specific briefed targets. (Source: Marine Corps tactical publication (MCTP) 3-20D, *Offensive Air Support*)

g. Striker. An aircraft capable of conducting AI, XAI, or AR.

h. Kill Box. A three-dimensional, permissive fire support coordination measure (FSCM) with an associated airspace coordinating measure (ACM) used to facilitate the integration of joint fires and the coordination of the associated airspace. A kill box is a measure, not a mission. (Source: *Kill Box MTTP*)

i. Kill Box Coordinator (KBC). The asset or agency assigned responsibility to deconflict aircraft and airspace within a kill box. (Source: *Kill Box MTTP*)

3. Brevity Codes

The following is a list of brevity codes that the SCAR aircrew should use to task interdiction assets.

Note: The following are definitions for nomenclature used within brevity code definitions: (Source: ATP 1-02.1, MCRP 3-30B.1, NTTP 6-02.1, AFTTP 3-2.5, *Multi-Service Brevity Codes*, hereafter referred to as *Brevity MTTP*)

[A/S]: Brevity code meaning applies to air-to-surface (A/S) operations/communications.

[S/S]: Brevity code meaning applies to surface-to-surface (S/S) operations/communications.

[AIR-MAR]: Brevity code meaning applies to maritime air (AIR-MAR) operations/communications.

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a. INVESTIGATE.

(1) [A/S] Verify specified element(s) of rules of engagement (ROE), positive identification (PID), collateral damage estimation (CDE), and/or coordination of forces on the referenced target or track. (Source: *Brevity MTTP*)

(2) [AIR-MAR] Verify specified element(s) of ROE, PID, clear field of fire, or coordination of forces on the referenced target or track. (Source: *Brevity MTTP*)

b. TARGET.

(1) [A/S] [S/S] ROE, PID, coordination of forces, and commander's guidance requirements on the referenced target/track have been satisfied. Target/track

correlation and CDE must be accomplished prior to employing ordnance/fires. (Source: *Brevity MTTP*)

(2) [AIR-MAR] ROE, PID, coordination of forces, and commander's guidance requirements on the referenced target/track have been satisfied. Target/track correlation and clear field of fire must be accomplished prior to employing ordnance/fires. (Source: *Brevity MTTP*)

c. SMACK.

(1) [A/S] [S/S] Clearance to employ ordnance/fires on surface targets' coordinates. ROE, PID, CDE, coordination of forces, and commander's guidance requirements on the referenced target/track have been satisfied. Not to be utilized in a close air support engagement. (Source: *Brevity MTTP*)

(2) [AIR-MAR] Clearance to employ ordnance/fires on surface target coordinates/Link-16 track. ROE, PID, clear field of fire, coordination of forces, and commander's guidance requirements on the referenced target/track have been satisfied. (Source: *Brevity MTTP*)

Note: Usage of the term SMACK within Service S/S fires organizations may be limited. Refer to chapters III and IV for the request process to engage targets with S/S fires.

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4. Assignment of the SCAR Aircrew Role

a. The SCAR aircrew's authority to facilitate interdiction (S/S and A/S fires), comes from the joint force commander (JFC) via the air tasking order (ATO) or a C2 element. The airspace control order (ACO) or a C2 element defines the airspace within which the SCAR aircrew will operate. Striker capabilities (e.g., speed, on station time, or sensors) should be matched to the mission and the size of the area in which the SCAR aircrew is assigned responsibility.

b. In the absence of a designated SCAR aircrew, the first striker on station that is capable of performing the SCAR aircrew responsibilities, as outlined in paragraph 6, should be prepared to assume the role of the SCAR aircrew, if tasked by a C2 element. The SCAR aircrew role may be transferred at an appropriate time to any SCAR capable asset by applicable C2 elements. Considerations for transferring the SCAR aircrew role include:

(1) Situational awareness.

(2) Sensor capability.

(3) Communication capability.

(4) Time on station (TOS).

5. The SCAR Aircrew Authorities

a. The SCAR aircrew has the authority to direct strikers in support of a SCAR mission to proceed against appropriate targets using INVESTIGATE, TARGET and SMACK taskings. A SMACK tasking may be restricted by the joint force air component commander (JFACC) via the ATO, special instructions (SPINS), or C2 elements. Strikers in direct support of a SCAR aircrew retain the ability to deny a

SCAR tasking due to extenuating circumstances, such as systems limitations, operator limitations, or safety of flight.

b. Unless specifically authorized, the SCAR aircrew does not have the authority to direct mission changes for strikers tasked to engage scheduled targets. Strikers tasked to engage scheduled targets are assigned to an AI mission listed in the ATO. If a SCAR mission includes assets that have planned targets, the SCAR aircrew retains the ability to facilitate, but not limit, mission accomplishment of those assets. This may include updating known surface-to-air threats, assisting with deconfliction, updating target, preventing redundant target engagement, or assisting with BDA.

c. The SCAR aircrew may establish and adjust coordination measures (e.g., ACM or FSCM) within the SCAR aircrew's assigned airspace to maximize flexibility during mission execution. Airspace changes that require higher headquarters approval must be requested by the SCAR aircrew.

d. When a KBC is operating concurrently with a SCAR aircrew, the SCAR aircrew will maintain authority over the KBC. The KBC will support the SCAR aircrew by assuming the SCAR aircrew's aircraft deconfliction responsibility within the assigned volume of airspace.

e. When a SCAR aircrew is assigned a SCAR mission, whether via the ATO or C2 element, no further authorization is required to task supporting assets to engage targets.

Note: It is important to understand that joint interdiction can be conducted inside an area of operations (AO) in direct response to a JFC tasking and may not be in support of the AO commander. The JFC may, for example, have designated certain high-payoff targets that are located inside a subordinate commander's AO. Any commander executing such a mission within a land or maritime AO must coordinate the operation with the affected component commander to avoid adverse effects and fratricide. See JP 3-03 for more information.

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6. The SCAR Aircrew's Responsibilities

a. The following subparagraphs provide a common understanding of what a SCAR aircrew is expected to perform, regardless of aircraft type or aircrew experience level. The Services are responsible for conducting training to qualify SCAR aircrews to meet these responsibilities and maintain proficiency. These responsibilities are:

b. Deconflict all assets within the assigned airspace volume.

(1) Aircraft deconfliction includes deconfliction of supporting aircraft, regardless of whether the platform is a fixed-wing (FW), rotary-wing (RW), tiltrotor, or unmanned aircraft (UA) aviation asset.

(2) Weapons deconfliction includes deconfliction of fires delivered from A/S and S/S platforms.

c. Coordinate the location, identification, engagement, and assessment of targets to achieve mission objectives and the commander's intent.

(1) Ensure ROE, SPINS, PID, CDE, clear field of fire, and commander's guidance requirements on the referenced targets have been satisfied through available means.

(2) Find, fix, track, target, engage, and assess (F2T2EA) targets using supporting assets and own-ship capabilities.

(3) Prioritize targets IAW the commander's guidance.

(4) Assess threats to assets in support of a SCAR mission through available means and mitigate them.

(5) Conduct reconnaissance and provide situation updates to supporting assets.

(6) Pass target information to supporting assets using either the preferred description, location, elevation, and remarks and restrictions (DLER) attack brief format, or a 9-line CAS brief format.

Note: During SCAR MTTP publication revision, a North Atlantic Treaty Organization (NATO) standardization agreement (STANAG) was in place, which prescribes the 9-line CAS brief format as the agreed upon attack brief format for passing target information during a SCAR mission supported by NATO assets. While use of the DLER attack brief format is preferred during SCAR mission operations, the 9-line CAS brief format may be required when operating with NATO supporting assets.

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(7) Coordinate, sequence, and direct aircraft.

(8) Match desired weapons effects with targets.

(9) Integrate supporting asset capabilities, as required.

(10) Coordinate BDA of targets and reattack, as required.

(11) Mitigate redundant strikes against previously attacked targets.

(12) Coordinate for additional assets, as required.

Note: The SCAR aircrew is responsible for coordinating all fires, including S/S, but is not expected to be capable of conducting a call for fire (CFF). S/S fires requests will be coordinated through the appropriate element (e.g., joint air-ground integration center (JAGIC), fire support coordination center (FSCC), etc.). Some SCAR aircrew may be capable of conducting a CFF based on Service-specific training. This must be discussed during the planning phase or upon airspace "check-in" to clarify the SCAR aircrew's capabilities and limitations. See ATP 3-09.32, MCRP 3-31.6, NTTP 3-09.2, AFTTP 3-2.6, *Multi-Service Tactics, Techniques, and Procedures for the Joint Application of Firepower*, for a detailed description of a CFF.

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d. Provide a battlefield handover briefing to the follow-on SCAR aircrew, and provide reports to higher headquarters and supported agencies.

7. The SCAR Aircrew's Functions

The SCAR aircrew's functions serve to identify what the SCAR aircrew should be able to provide based on aircrew and platform capabilities. The SCAR aircrew:

a. Provides target marks (e.g., infrared (IR) pointer, laser, or rockets).

b. Serves as an intelligence, surveillance, and reconnaissance (ISR) collection platform.

c. Performs AI, as required.

d. Monitors any target area of interest (TAI) and named area of interest (NAI) in the vicinity to provide timely warning of enemy location and activity.

e. Provides terminal guidance for laser-guided weapons.

f. Assists with BDA.

8. Target Priority Establishment

The ATO or C2 element establishes target priority for the SCAR aircrew. Since there may be multiple supported commanders within a joint operations area (JOA), using the joint integrated prioritized target list (JIPTL) ensures the SCAR aircrew has an ATO or C2 element-assigned SCAR target priority. The SCAR aircrew will follow the target priorities established by the ATO or C2 element. The following list describes establishing target priorities at the operational level.

a. JFC. The JFC establishes target priority via the JIPTL using the joint targeting coordination board (JTCB). The targets selected for the JIPTL include the component commanders' target nominations, which are combined and prioritized for approval by the JTCB.

b. JFACC. The JFACC is normally the supported commander for the JFC's theater-wide, air interdiction effort. The JFACC implements the JFC's target priorities by executing the air operations directive (AOD), ATO, SPINS, ROE, and exercising C2 (typically tactical control (TACON)) over air assets made available for tasking. Targets may include targets of opportunity that emerge during execution and are not on the ATO.

c. Joint Force Land Component Commander (JFLCC), Joint Force Maritime Component Commander (JFMCC), or Joint Force Special Operations Component Commander (JFSOCC). Component commanders nominate interdiction targets and affect the SCAR mission's target priority process via the JTCB and JIPTL. Once a JFLCC, JFMCC, or JFSOCC is established and an AO is assigned, that commander becomes the supported commander for interdiction within that AO and has the authority to designate target priority, effects, and timing within their AO. If the component commander further delegates AO assignment to subordinate commanders, there may be multiple AOs with varying target priorities. The SCAR aircrew will change the assigned target priorities only if specific changes are approved by C2 elements to ensure unity of command and unity of effort.

Chapter II
COMMAND AND CONTROL

1. Theater Air-Ground System (TAGS)

a. The TAGS provides the JFC the means to control, integrate, coordinate, and deconflict all air operations within the JOA, including multinational forces. In the joint or combined environment, TAGS consists of the Air Force theater air control system (TACS), Army air-ground system (AAGS), Marine air command and control system (MACCS) and the Navy tactical air control system (NTACS). Refer to JP 3-30 *Command and Control in Joint Air Operations*; JP 3-52, *Joint Airspace Control;* and ATP 3-52.2, MCRP 3-20.1, NTTP 3-56.2, or AFTTP 3-2.17, *MTTP for the Theater Air-Ground System* for more details on the TAGS.

b. The SCAR aircrew must have more than a basic understanding of the airspace C2 architecture to fully comprehend the SCAR aircrew role. This understanding will allow the SCAR aircrew to manage the assigned airspace and effectively coordinate A/S and S/S fires when required. The airspace C2 construct will vary based on where the SCAR mission is conducted.

2. Operational C2

a. Operational C2 Elements within the TAGS.

(1) The following is an overview of the joint air operations center (JAOC).

(a) The JAOC is the JFACC's senior C2 node of the TACS and is responsible for planning and executing SCAR, armed reconnaissance (AR), AI, airborne alert air interdiction (XAI) mission, and supporting sorties made available for tasking on an ATO. Service and functional component SCAR requirements are submitted to the JAOC as air support requests, or may be determined during ATO and targeting processes based on target nominations. The JAOC is a jointly manned organization and should include liaison representation from each Service component. These liaisons work to ensure the intents and priorities of their commanders are represented in the JAOC.

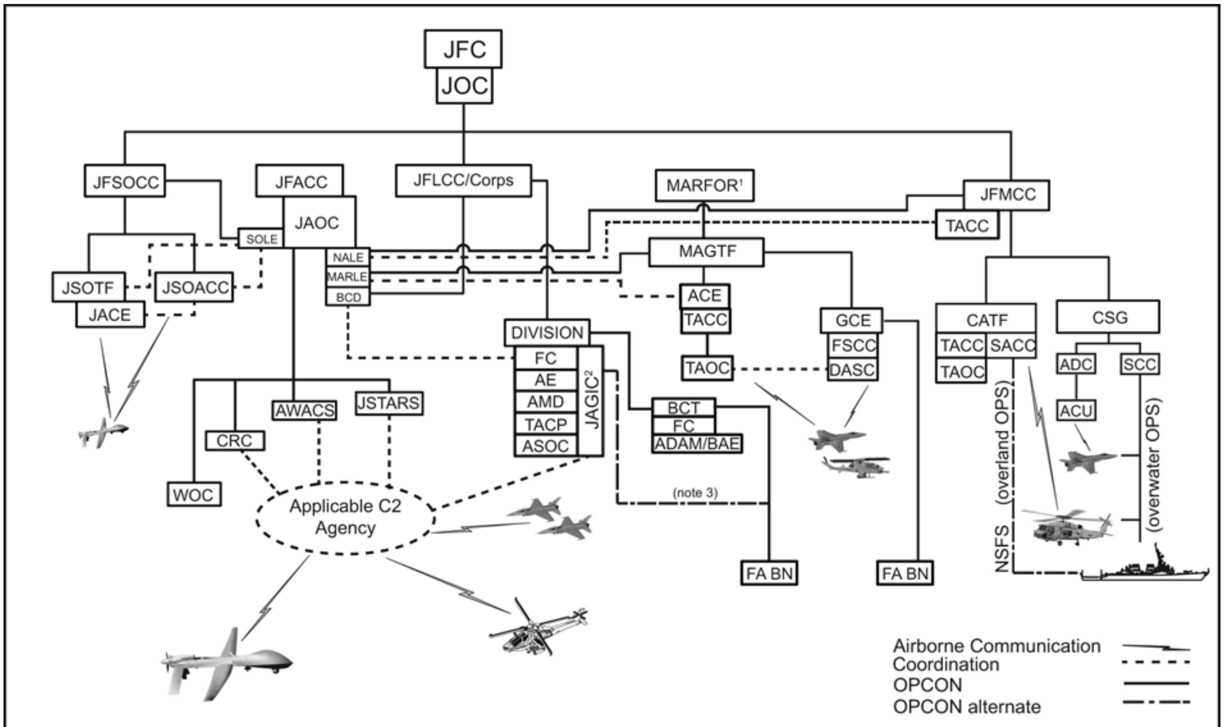
(b) The JFC may delegate development of the JIPTL to the JFACC. Based on JFC priorities, the JAOC combines and prioritizes component target nominations into the JIPTL. Prioritized targets approved by the JTCB are then allocated to a SCAR/AR/AI/XAI asset by JAOC planners based on the priorities established in the JFC's air apportionment decision. The JAOC produces the ATO that tasks these allocated sorties for SCAR/AR/AI/XAI missions. For requirements that emerge too late for inclusion in the ATO, the JAOC exercises centralized control to redirect available air assets to meet JFC priorities. This centralized control also may be accomplished by subordinate TACS airspace control elements, such as the air support operations center (ASOC), Airborne Warning and Control System (AWACS), and control and reporting center (CRC).

(c) The JFC designates the JFACC as the airspace control authority responsible to develop the airspace control plan and ACO. JAOC airspace control teams coordinate airspace requests from requesting components to ensure kill boxes with associated ACMs are established and published on the ACO to meet SCAR mission requirements. The JAOC also may coordinate airspace requests that emerge too late for inclusion in the ACO to ensure airspace is allocated, integrated and deconflicted across the JOA.

(d) Within the JAOC, the senior offensive duty officer is responsible for SCAR mission oversight during execution. The offensive operations team will monitor the location and status of each SCAR mission. Regardless of mission location, critical situation updates and mission results (i.e., inflight reports) must be reported via the tactical C2 element to the appropriate operational C2 to enable mission assessment and provide resources. For more information on the JAOC, TACS, and joint airspace control, see: JP 3-30, *Command and Control of Joint Air Operations*; JP 3-52, *Joint Airspace Control*; ATP 3-52.2, MCRP 3-20.1, NTTP 3-56.2, or AFTTP 3-2.17, *MTTP for the Theater Air-Ground System;* and ATP 3-52.1, MCRP 3-20F.4, NTTP 3-56.4, AFTTP 3-2.78, *Multi-Service Tactics Techniques, and Procedures for Airspace Control.*

(2) The tactical air command center (USMC) (TACC (USMC)), Army command posts (CPs), tactical air control center (Navy) (TACC (USN)), and joint special operations task force (JSOTF) joint fires element (JFE) are the operational C2 elements within each Service that enable SCAR mission accomplishment. For additional information, see ATP 3-52.2, MCRP 3-20.1, NTTP 3-56.2, or AFTTP 3-2.17.

(3) Figure 1 depicts a possible TAGS C2 organizational structure's relationship specific to a SCAR mission. Additional Service-specific AO C2 diagrams for SCAR missions follow in notional figures in this chapter.



Notes:

1. Depending on the phase of the operation, designated Marine Corps forces may be TACON to either the JFMCC, JFLCC, or JFACC.
2. The JAGIC is the physical integration of Air Force TACP and ASOC personnel, and selected Army division current operations staff members. Refer to ATP 3-91.1, AFTTP 3-2.86 *The Joint Air Ground Integration Center* for a breakdown of staffing within a JAGIC.
3. The division fires cell can directly contact the FA battalion to send a fire mission.

Legend:

ACE—air combat element (USMC)
ACU—aircraft control unit
ADC—air defense commander
ADAM—air defense airspace management
AE—airspace element
AFTTP—Air force tactics, techniques, and procedures
AMD—air and missile defense
ASOC—air support operations center
ATP—Army tactical publication
AWACS—Airborne Warning and Control System
BAE—brigade aviation element
BCD—battlefield coordination detachment
BCT—brigade combat team
BN—battalion
C2—command and control
CATF—commander, amphibious task force
CRC—combined reporting center
CSG—carrier strike group
DASC—direct air support center
FA—field artillery
FC—fires cell (Army)
FSCC—fire support coordination center
GCE—ground combat element
JACE—joint air coordination element
JAGIC—joint air-ground integration center
JAOC—joint air operations center
JFACC—joint force air component commander
JFC—joint force commander
JFLCC—joint force land component commander
JFMCC—joint force maritime component commander
JFSOCC—joint force special operations component commander
JOC—joint operations center
JSOACC—joint special operations air component commander
JSOTF—joint special operations task force
JSTARS—Joint Surveillance and Target Attack Radar System
MAGTF—Marine air-ground task force
MARFOR—Marine Corps forces
MARLE—Marine liaison element
NALE—naval and amphibious liaison element
NSFS—naval surface fire support
OPCON—operational control
OPS—operations
SACC—supporting arms coordination center
SCC—sea combat commander
SOLE—special operations liaison element
TACC—tactical air command center (USMC)
TACC—tactical air command center (USN)
TACON—tactical control
TACP—tactical air control party
TAOC—tactical air operations center
TADC—tactical air direction center
USMC—United States Marine Corps
USN—United States Navy
WOC—wing operations center

Figure 1. Joint Force SCAR Command, Control, and Coordination

b. ATO.

(1) During ATO development, the master air attack plan (MAAP) cell within the JAOC will analyze component target nominations on the JIPTL and allocate SCAR missions to interdict targets, as required.

(2) If an ATO-assigned SCAR mission requires modification, proper approval and coordination must be accomplished before making adjustments to the mission working area or assigned targets. Mission changes will require coordination between the component liaisons (i.e., battlefield coordination detachment (BCD), naval and amphibious liaison element (NALE), Marine liaison element (MARLE), and special operations liaison element (SOLE)), and the JAOC's combat plans division (CPD) or combat operations division (COD) personnel. Once the changes are approved, the JAOC must pass the updates to subordinate C2 elements and the SCAR aircrew.

3. Tactical C2

a. Tactical C2 Systems and Agencies.

(1) Army. The JAGIC is a collection of specific Army and Air Force personnel designed to support and enable Army division level, current operations through the rapid execution of fires, and clearance of fires and airspace. The JAGIC organizes complementary and reinforcing functions resident in the ASOC, tactical air control party (TACP), and division current operations integration cell (COIC) to best achieve air-ground integration. The Air Force contribution to the JAGIC is a blend of Air Force C2 capabilities and personnel from the ASOC and division TACP. The Air Force also supplies an interdiction coordinator, who coordinates executing air interdiction operations in division-assigned airspace and joint fires in kill boxes. The Army contributes current operations personnel representing Army fires, air and missile defense (AMD), aviation, and airspace elements. Located in the Army division COIC, the JAGIC provides commanders a technique to coordinate, integrate, and control operations in division-assigned airspace. The JAGIC colocates decision-making authorities from the land and air components to support the supported maneuver commander's objectives and intent. For additional information, see ATP 3-91.1, AFTTP 3-2.86, *The Joint Air Ground Integration Center*.

(2) Marine Corps.

(a) Tactical Air Operation Center (TAOC). The TAOC is subordinate to the TACC (USMC) and performs air surveillance, direction, and control and serves as the Marine air-ground task force (MAGTF) primary air and missile defense agency. The TAOC also performs a variety of delegated air direction functions, including coordination and deconfliction of airspace, anti-air warfare, tanker management, asset management, and coordination and control of fires in the deep area of the operational environment. See MCRP 3-20F.6, *Tactical Air Operations Center Handbook*, for additional information.

(b) Direct Air Support Center (DASC). The DASC is the principal air control agency of the MACCS responsible for the direction and control of air operations directly supporting the ground commander and is subordinate to the TACC (USMC). The DASC processes immediate air support requests, integrates aircraft employment with other supporting arms, manages terminal control assets supporting ground combat element and logistics combat element forces, and procedurally controls all aircraft transiting through DASC controlled airspace. See MCRP 3-20F.5, *Direct Air Support Center Handbook*, for a detailed discussion of the DASC.

(3) Navy.

(a) Sea Combat Commander (SCC). The SCC (call sign “ZULU”) is the embarked destroyer squadron commander in the composite warfare commander (CWC) organization that consolidates the following command and coordination functions under one commander: surface warfare commander (SUWC), antisubmarine warfare commander (ASWC), and maritime interdiction operations commander. The SCC may designate subordinate commanders to act as deputy SCCs to execute these warfare tasks. See Navy tactical reference publication (NTRP) 1-02, *Navy Supplement to the DoD Dictionary of Military and Associated Terms*, for a description of the SCC.

(b) Supporting Arms Coordination Center (SACC). The SACC is a central location on board an amphibious command ship for all communication facilities incident to the coordination of fire support of the artillery, air, and naval gunfire. This is the naval counterpart to the fire support coordination center used by the landing force.

(c) Aircraft Control Unit (ACU). An ACU is a platform equipped with sensors, communications equipment, and personnel trained to provide safe and effective positive or procedural air control in a specified area of operations. There can be multiple ACUs depending on the operational environment. ACUs perform battle management functions and can be assigned air intercept control (AIC) or maritime air control (MAC) duties. See ATP 3-04.18, MCRP 3-20.2, NTTP 3-20.8, AFTTP 3-2.74, *Multi-Service Tactics, Techniques, and Procedures for Air Operations in Maritime Surface Warfare*, for a detailed description of ACU, AIC, and MAC.

(d) MAC. An ACU conducting MAC is an extension of the SCC/SUWC charged with providing C2 to airborne assets operating within a designated maritime area of operations. MAC duties are assigned by the SCC/SUWC and similar to the responsibilities of Air Force tactical C2, Army tactical operations center, or MACCS agencies like the DASC or TAOC. Responsibilities for an ACU conducting MAC will be determined by mission requirements, threat environment, and commander's intent, and may include:

- Coordinating with warfare commanders.
- Locating and tracking potential or known threats.
- Fulfilling PID and ROE requirements.

- Managing airborne assets to sanitize the battle space.
- Providing air control of assigned air assets.
- Coordinating target prosecution.
- Assisting in collecting and relaying inflight reports from participating assets.
- Establishing and maintaining the common operational picture (COP) via integration with ISR platforms and tactical aircraft.
- Providing over-the-horizon target information for surface and airborne assets.

(4) Air Force.

(a) AWACS. Air surveillance and control may be provided by an AWACS aircraft equipped with search and height-finding radar. The E-3 Sentry is an AWACS aircraft that provides all-weather surveillance, C2, airborne battle management, and communications to commanders of United States (US), NATO, and other allied air defense forces.

(b) Joint Surveillance Target Attack Radar System (JSTARS). JSTARS aircraft provide ground and air commanders with situation development, attack planning, targeting, and limited post-attack assessment information. JSTARS can provide ground surveillance, situational awareness, and target information directly to the SCAR aircrew, or can relay information through an applicable C2 element. JSTARS can provide limited ASOC and joint air coordination element (JACE) back-up functions. For additional information, see ATP 3-55.6, MCRP 2-10A.4, NTTP 3-55.13, AFTTP 3-2.2, *Air-to-Surface Radar System Employment.*

(c) CRC. Air Force CRCs are ground-based airspace control and air defense, air battle management centers that provide the commander, Air Force forces (COMAFFOR) with a decentralized execution capability. Critical core competencies of the CRC include air battle execution, surveillance, combat identification, data link management, cyberspace security, and theater air defense. The CRC provides a robust systems and communications hub capability that connects lateral and subordinate C2 nodes to the JAOC.

(5) Special Operations. The JACE, provided by the COMAFFOR, is the tactical air C2 element of the special operations air ground system responsible for directing and controlling air operations directly supporting special operations forces. The JACE will align with the JSOTF JFE to facilitate deconfliction and coordination of FSCMs with special operations forces (SOF) and conventional air support.

b. When conducting a SCAR mission, aircrew will check in with the tactical C2 element as prescribed within the theater airspace control plan (ACP), ACO, and SPINS. Those C2 elements should provide a situation update to the SCAR aircrew upon check-in. When available, the C2 element should provide sensor data or tracks of interest applicable to the SCAR aircrew’s assigned area. This will allow the SCAR aircrew to combine real-time intelligence from outside agencies with onboard sensors.

c. Tactical C2 also provides a means for the SCAR aircrew to request additional assets, either air or S/S fires assets, and conduct additional airspace coordination, when required (e.g., when employing stand-off weapons or long range artillery). Depending on capabilities and delegated authorities and responsibilities, tactical C2 will coordinate with operational C2 and fire support agencies to facilitate these requirements. It is critical that once coordination is complete, the tactical C2 element ensures the SCAR aircrew is informed of all changes and updates.

4. Coordination Measures

a. ACMs and FSCMs facilitate efficient use of airspace to accomplish missions, expedite target attacks (e.g., permissive FSCMs), and provide safeguards for friendly forces (e.g., restrictive FSCMs). Additional ACMs and FSCMs information can be found in ATP 3-52.1, MCRP 3-20F.4, NTTP 3-56.4, AFTTP 3-2.78, *Multi-Service Tactics Techniques, and Procedures for Airspace Control* (hereafter referred to as the *Airspace Control MTTP*).

b. ACMs facilitate SCAR mission effectiveness by ensuring the safe, efficient, and flexible use of airspace. ACMs can assist planners in designating SCAR mission working areas. These areas may be formal (e.g., established in the ACO) or informal (e.g., established by tactical C2 during execution). The SCAR aircrew may establish additional ACMs, within the assigned area, to facilitate target engagement. Airspace changes or additions shall be coordinated with tactical C2. See chapter IV for a discussion of SCAR airspace management and control considerations.

c. FSCMs also facilitate SCAR effectiveness, particularly when operating inside a land component AO. They can reduce and expedite coordination requirements and provide approval of fires for SCAR and striker aircraft. When established within a land or maritime AO, kill boxes are the preferred FSCMs for SCAR missions because they are three-dimensional measures that account for airspace and surface engagement areas. Detailed discussion of kill boxes can be found in the *Kill Box MTTP.*

d. A SCAR aircrew must understand which ACMs and FSCMs are being used to prevent conflicts and potential fratricide between friendly aircraft and S/S fires. Appendix D outlines ACMs and FSCMs that are applicable to SCAR missions.

5. AO-specific Airspace Control Elements

a. Army AO. See figure 2 for a depiction of the airspace control elements within an Army AO. Airspace within the Army AO is controlled as follows:

(1) Short of the fire support coordination line (FSCL), Beneath the Coordinating Altitude (CA).

(a) Contact the JAGIC for routing to the kill box or designated area.

(b) Contact the SCAR aircrew for deconfliction and tasking within the kill box.

(2) Short of the FSCL, above the CA.

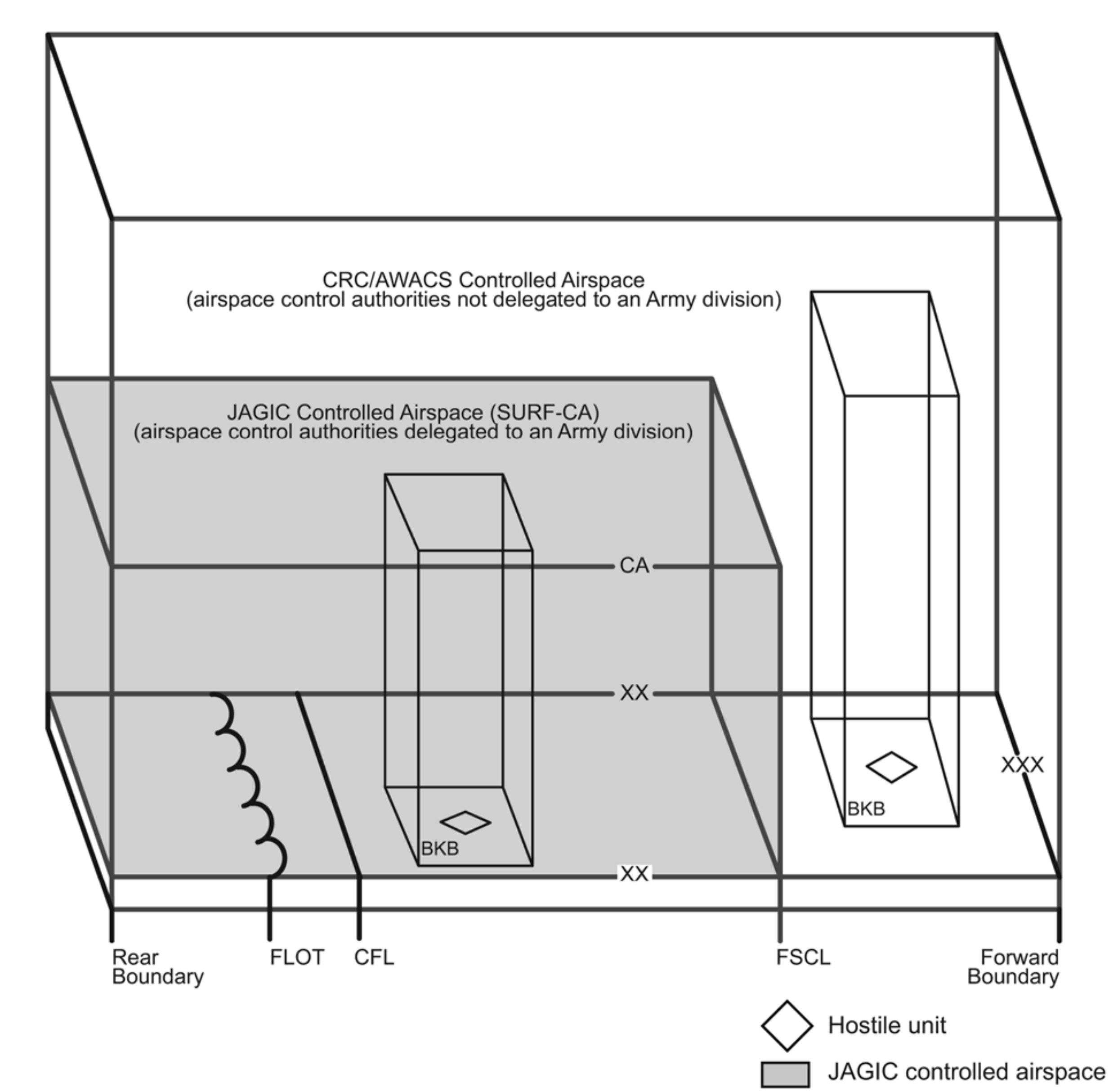
(a) Contact the CRC or airborne airspace control element (e.g., AWACS) for routing to the kill box or designated area.

(b) Contact the SCAR aircrew for deconfliction and tasking within the kill box.

(3) Beyond the FSCL.

(a) Contact CRC or airborne C2 for routing to the kill box or designated area.

(b) Contact the SCAR aircrew for deconfliction and tasking within the kill box.



Legend:
AWACS—Airborne Warning and Control System
BKB—blue kill box
CA—coordinating altitude
CFL—coordinated fire line
CRC —control and reporting center
FLOT—forward line of own troops
FSCL—fire support coordination line
JAGIC—joint air ground integration center
SURF—surface
XX—division area of operations boundary
XXX—corps area of operations boundary

Figure 2. Notional Army Division AO

b. MAGTF AO. See figure 3 for a depiction of airspace control elements within a MAGTF AO. Airspace within a MAGTF AO is controlled as follows:

(1) Short of the Battlefield Coordination Line (BCL), beneath the CA.

(a) Contact the DASC for routing to the kill box or designated area.

(b) Contact the SCAR aircrew for deconfliction and tasking within the kill box.

(2) Short of the BCL, Above the CA.

(a) Contact the TAOC for routing to the kill box or designated area.

(b) Contact the SCAR aircrew for deconfliction and tasking within the kill box.

(3) Beyond the BCL.

(a) Contact the TAOC for routing to the kill box or designated area.

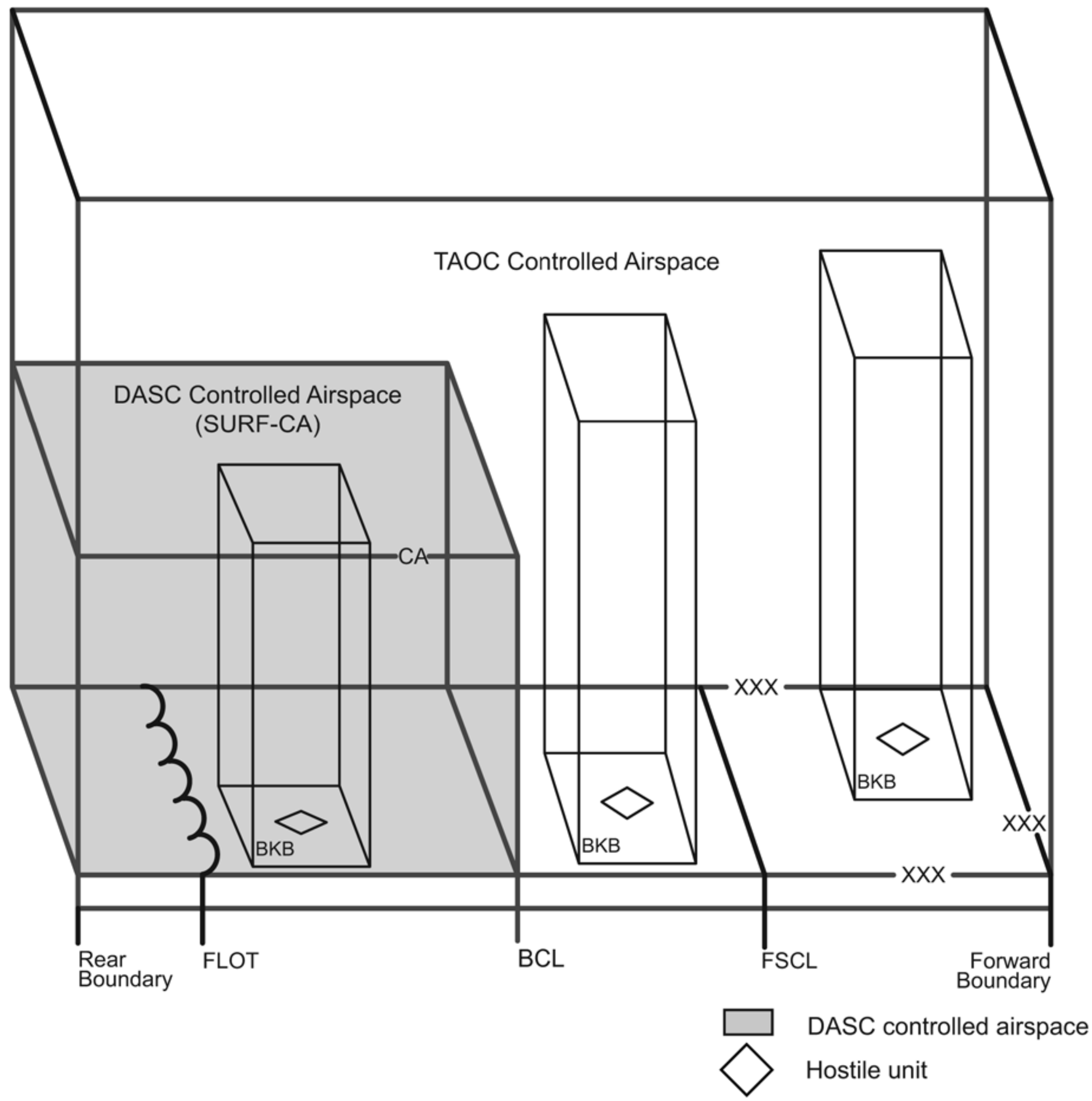
(b) Contact the SCAR aircrew for deconfliction and tasking within the kill box.

c. Maritime AO. See figure 4 for a depiction of the airspace control elements within a maritime AO. An ACU conducts maritime airspace C2, which may be an air or surface platform. Contact the ACU for routing to the designated area. The SCAR aircrew may receive MAC from the ACU during a SCAR mission. Assets in support of the SCAR aircrew should contact the SCAR aircrew for deconfliction and tasking within the designated area. Operations in support of a carrier strike group (CSG) or surface action group (SAG) may require direct communications with the SCC (i.e., call sign “ZULU”). Operations in support of an amphibious ready group (ARG) may require direct communications with the SACC or TACC (USN). See ATP 3-04.18, MCRP 3-20.2, NTTP 3-20.8, AFTTP 3-2.74, *MTTP for Air Operations in Maritime Surface Warfare,* for additional information.

d. Joint Special Operations Area (JSOA). See figure 5 for a depiction of the airspace control elements within a JSOA.

(1) Contact the JACE for routing to the kill box or designated area.

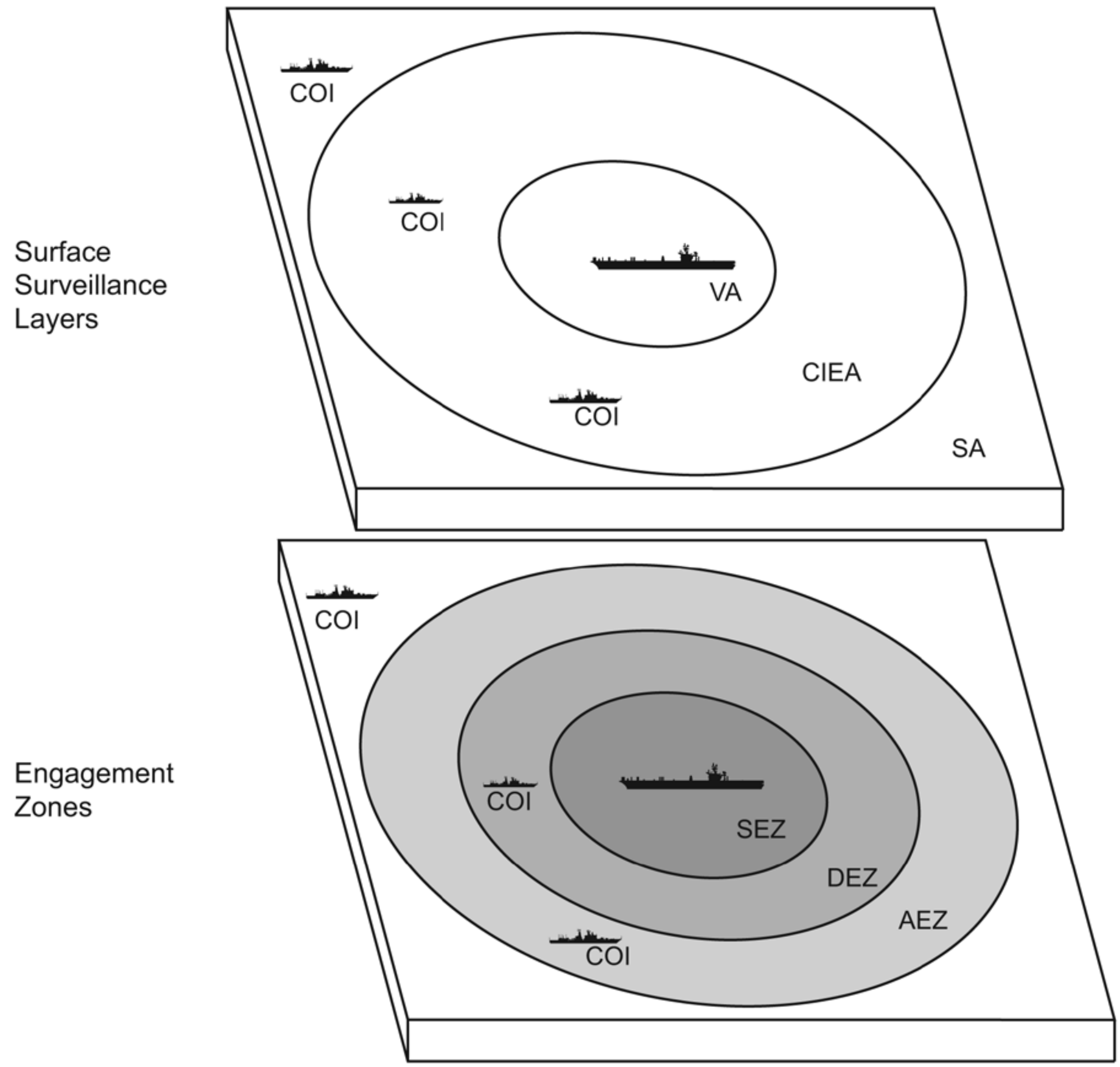
(2) Contact the SCAR aircrew for deconfliction and tasking within the kill box.



Note: Airspace control authorities have been delegated to the MAGTF.

Legend:
BCL—battlefield coordination line
BKB—blue kill box
CA—coordinating altitude
DASC—direct air support center
FLOT—forward line of own troops
MAGTF—Marine air-ground task force
SURF—surface
TAOC—tactical air operations center
XXX—corps area of operations boundary

Figure 3. Notional MAGTF AO



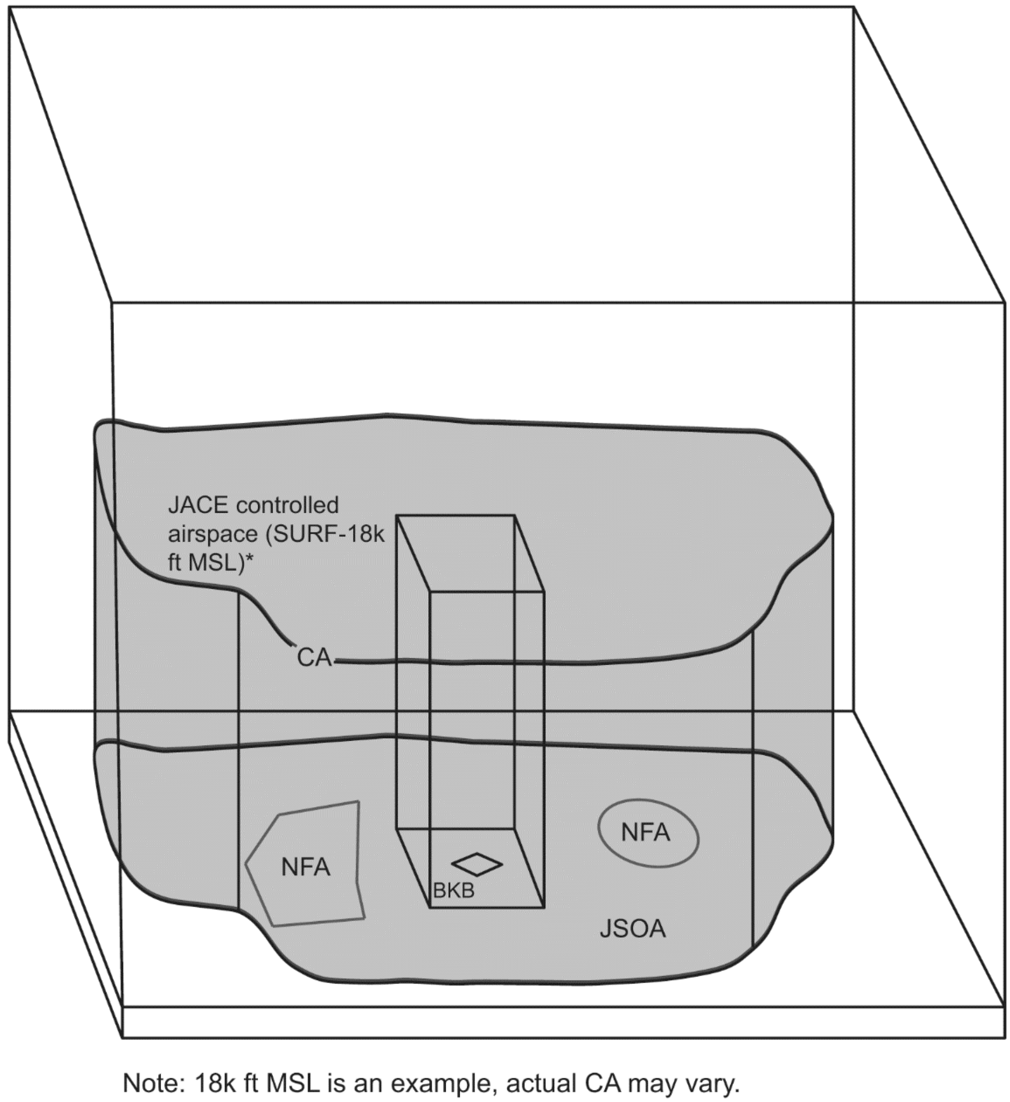
Notes:

1. Airspace is controlled by an ACU (e.g., normally callsign "RED CROWN").
2. When naval forces transition from open ocean through near-land to littoral joint or coalition operations, these areas merge with, and are superseded by, jointly defended airspace defined by the area air defense commander (AADC) and airspace control authority (ACA).

Legend:

AADC—area air defense commander
ACA—airspace control authority
ACU—aircraft control unit
AEZ—air engagement zone
CIEA—classification, identification, and engagement area
COI—contact of interest
DEZ—dual engagement zone
SA—surveillance area
SEZ—ship engagement zone
VA—vital area

Figure 4. Notional Open Ocean Maritime AO



Legend:
CA—coordinating altitude
BKB—blue kill box
ft—feet
JACE—joint air coordination element
JSOA—joint operations area
k—thousand
MSL—mean sea level
NFA—no-fire area
SURF—surface
Hostile unit

Figure 5. Notional Joint Special Operations Area

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Chapter III
PLANNING

1. Introduction

SCAR operations can be complex due to a potentially large number of targets and assets. The amount of time allotted prior to the mission will determine the breadth and depth of the planning process.

2. SCAR Mission Elements

A SCAR mission consists of five elements: the SCAR aircrew; interdiction assets; C2; reconnaissance, surveillance, and target acquisition (RSTA); and support. When planning a SCAR mission, consider each element in context of the mission and resources available to best meet the commander's intent. Depending upon the situation, every element may not be necessary for mission accomplishment.

a. The SCAR Aircrew. Identify a SCAR-qualified aircrew to conduct the SCAR mission.

b. Interdiction Assets. Identify A/S and S/S assets capable of conducting the required interdiction. Consider varying weapons and delivery platforms to increase flexibility.

c. C2. As discussed in chapter 2, determine the airspace control elements the SCAR aircrew will interact with to ensure SCAR mission success.

d. RSTA. Identify all available RSTA assets, in addition to the SCAR aircrew, to ensure the SCAR mission is capable of determining target location, and assessing associated PID, collateral damage, and ROE.

e. Support. It is critical that SCAR mission support assets are requested and made available, such as aerial refueling or airborne C2. Flexibility to maintain a high operations tempo should be designed into the plan by having on-call assets available, as required.

3. SCAR Planning Considerations

a. General Mission-Planning Considerations.

(1) The objective of a SCAR mission is to increase interdiction mission efficiency and effectiveness.

(2) SCAR mission planning occurs within the ATO cycle. ATO planning processes are discussed in JP 3-30, *Command and Control for Joint Air Operations*. SCAR missions are identified during development of the MAAP, which is constructed by the CPD in the JAOC. When identified, SCAR missions and supporting assets will be assigned on the ATO.

(3) The SCAR aircrew should be familiar with ROE and SPINS to ensure proper application of SCAR tasking types (i.e., INVESTIGATE, TARGET, and SMACK).

(4) The SCAR aircrew should be familiar with available interdiction asset capabilities, limitations, and caveats to provide the best possible asset and weapon-to-target pairing. The SCAR aircrew will receive target priorities via the ATO or applicable C2 element, which will be used to task the best available assets within the SCAR aircrew's designated area.

(5) SCAR and striker assets must be free from prohibitive enemy air threats within the working area. The counterair mission integrates offensive and defensive operations to attain and maintain the JFC's desired protection and degree of control of the air by neutralizing or destroying enemy aircraft and missiles. Counterair operations are conducted to ensure access and freedom of action. The required degree of air control may vary over time from no control, to parity, to local air superiority, or to air supremacy depending upon the situation and the JFC's concept of operations (CONOPS). If required, counterair should be provided by additional aircraft tasked with combat air patrol. Based on allowable risk, SCAR and striker assets with air-to-air capability may be used to respond to unplanned air threats.

b. Aviation Asset Mission Types in Support of a SCAR Mission.

(1) Aviation assets in support of a SCAR mission will be assigned a geographic area versus specific targets. Mission types that will support a SCAR mission include:

(a) AR (Marine Corps, Navy). AR is a mission with the primary purpose of locating and attacking targets of opportunity (e.g., enemy materiel, personnel, and facilities) in assigned general areas or along assigned communications routes, and not for the purpose of attacking specific, briefed targets. (MCTP 3-20D, *Offensive Air Support*)

Note: Army units may be tasked to conduct AR. However, Army units will determine which mission best supports the AR tasking (e.g., movement to contact or zone reconnaissance) based on mission considerations. Army aviation does not operate under the same "mission" construct as many Air Force, Navy, and Marine Corps assets. Instead, Army aviation utilizes "mission type orders" in the form of a task and purpose, which includes the "five W's" (i.e., who, what, when, where, and why). For example, an Army aviation unit's mission type orders may be "On order, 1-151 Attack Reconnaissance Battalion attacks to destroy Company A, 1st Battalion, 173rd Motorized Rifle Regiment in the vicinity of blue kill box (BKB) 096CH to ensure freedom of movement of 1st Armored Brigade Combat Team (ABCT)."

(b) XAI (Air Force). XAI identifies an AI airborne alert mission tasked for on-call targets that may be retasked during execution for targets of opportunity (also referred to as armed reconnaissance). (Air Force Doctrine, Annex 3-03)

Note: While AR and XAI are the primary missions that support SCAR, these nomenclatures are not compatible with theater battle management core system (TBMCS) at the time this publication was developed and will not display as such on the ATO. Currently, AR in TBMCS stands for aerial refueling. Alert interdiction (XINT) is the mission type that is compatible with TMBCS and is most reflective of AR and XAI. Planners and operational units are responsible for understanding this TBMCS deficiency and must coordinate appropriately.

(2) Aviation assets tasked to engage planned targets, such as during AI missions, can be incorporated into the SCAR aircrew's plan, but are not available for the SCAR aircrew to task unless specifically authorized. In this case, the SCAR aircrew will need to facilitate, but not limit, mission accomplishment for these assets. This may include updates to known surface-to-air threats, assistance with deconfliction, updates to target location and identification, prevention of redundant target engagement, or assistance with BDAs. AI is the mission type associated with assets tasked to engage planned targets. If assets tasked to engage planned targets affect a SCAR mission's airspace or engagement area, the SCAR aircrew must establish or ensure an appropriate deconfliction plan.

c. S/S Fires Assets in Support of SCAR.

(1) The baseline expectation for the SCAR aircrew is to be responsible for coordinating fires, including S/S fires, but is not expected to be capable of conducting a CFF. S/S fires requests will be coordinated through the appropriate element (e.g., JAGIC, FSCC, etc.). Although some SCAR aircrew may be capable of conducting a CFF, based on Service-specific training, this is not a SCAR aircrew responsibility. SCAR aircrew capable of conducting a CFF must be determined during the planning phase or upon airspace "check-in" to clarify the SCAR aircrew's capabilities and limitations.

(2) The SCAR aircrew needs to be familiar with S/S weapon systems capable of reaching the SCAR mission working area. With increased capabilities in long-range fire and ability to engage point and area targets, long-range artillery systems and naval surface fire support (NSFS) are viable attack options. The SCAR aircrew's information regarding the capabilities of these assets should come from the BCD, ground liaison detachment (GLD), or other Service liaison officers prior to the mission.

Note: Artillery units are tasked to suppress, neutralize, or destroy; but ammunition capability, quantity, accuracy, and target type may limit the effects artillery can create.

d. ATO Coordination.

(1) The ATO is the primary source for asset information relevant to a SCAR mission. The SCAR aircrew should refer to the ATO for supporting asset and geographic location information relevant to the SCAR mission. ATO planners should ensure RSTA assets are available, as required, to facilitate SCAR operations. Once the SCAR aircrew identifies asset composition, the SCAR

aircrew should attempt to contact the identified assets as part of mission planning.

(2) While Army aviation sorties should be listed on the ATO, the aviation unit's parent headquarters operation order (OPORD) may contain more current and accurate information on the Army aviation asset availability and mission specifics. The SCAR aircrew can request this information from the GLD.

e. SCAR Aircraft Considerations. SCAR-capable units should request the most suitable load out for SCAR missions. SCAR mission tasks include finding and identifying targets, handing targets off to striker assets, interdicting a variety of tactical and potentially mobile targets, and performing BDAs. Aircraft should be configured with the following capabilities:

(1) Aircraft Sensors. Examples include synthetic aperture radar (SAR) mapping capability, full-motion video, and a moving target indicator.

(2) Data Link Capability. Examples include Link 16 or situation awareness data link (SADL), improved data modem, digital communication suite, remote viewing terminal, and secure communication suites.

(3) Target designation and marking capability. Examples include rockets, bombs, guns, a laser designator spot tracker, and IR pointers.

(4) Weapons Loads. Examples include precision-guided munitions and area munitions.

(5) Multiple Secure Voice Radios.

f. Intelligence. The SCAR aircrew is expected to integrate all available intelligence sources relevant to the assigned AO to mitigate threats and facilitate rapid targeting. Accurate and timely intelligence is critical to the SCAR mission to maximize the use of on-station time and minimize time-to-kill. When executing SCAR missions, threat location information, whether general or specific, can minimize an aircrew's threat exposure. Accurate and timely intelligence of adversary air-to-air and surface-to-air capabilities are essential in planning support assets to protect airborne assets and employ appropriate tactics. Sharing threat detection information is critical. Disseminate threat detection to affected airborne assets as expeditiously as possible.

g. Locating and Identifying Targets. Several planning considerations assist with locating and identifying a target. The SCAR aircrew should gather information on the types and locations of targets through intelligence briefs, develop different search strategies, and understand the target prioritization for the AO.

(1) Target Development. The type and location of targets can best be determined through intelligence briefs and information provided by the ground liaison officer or other liaison officers. In addition, these liaison officers may be able to detail force disposition, enemy tactics, and likely avenues of approach. The modified combined obstacle overlay should be the reference to identify doctrinal enemy templates over avenues of approach in the expected AO. Using this technique will aid in developing specific search plans. Details of friendly

activity in the area should be considered at this time. The SCAR aircrew should analyze the AO to develop specific point of interest (POI) lists, as well as mission materials that will facilitate effective searching once airborne. Mission materials may include:

(a) Overlay of the AO common grid reference system. The Global Area Reference System (GARS) is the preferred system.

(b) Detailed aircraft mission load.

(c) Elevations per keypad and minimum safe altitude.

(d) ACMs.

(e) Threats and threat maximum effective range scale.

(2) Imagery. The imagery of specific waypoints or POI can be printed and carried in flight to help verify search areas. In addition, an “information package” with photos of the enemy order of battle and a visual identification packet may be helpful for the SCAR aircrew and strikers to achieve PID.

(3) Time. The time required to search terrain for targets varies and is influenced by the following considerations:

(a) Sensor capabilities and limitations.

(b) Speed and altitude.

(c) Time of day and environmental conditions.

(d) Terrain (e.g., shadow, line-of-sight restrictions, movement indicators, or sea state).

(e) Urban terrain and population density.

(f) Logical search flow.

(g) Number of targets.

(h) Level of detail required per target.

(4) Search Techniques. Common techniques to use as rules of thumb for target searching include the following.

(a) Use big-to-small focused patterns while using all available sensors.

(b) Analyze in wide field of view, confirm in narrow.

(c) Assess the benefits of revisit plans versus overhead loiter.

(d) Use multiple sensor sources and angles to build a better operational picture.

(5) Search Considerations.

(a) Altitude. The search altitude may vary based on threat, sensor selection, and time of day. A search altitude is a trade-off between threat mitigation and sensor performance. A visual search should not be overlooked when locating the enemy due to the large field of view that can be covered, the ability to perceive movement when searching along lines of communications (LOCs), and the ability to capture glint and contrast.

(b) Sensor-Aided Search. Sensor search techniques will vary based on the environment within the assigned geographic area. Premission planning may be aided by software programs such as target acquisition weapons software (TAWS) to build an effective sensor search plan. Additional consideration should be given to maximize crew search capability (e.g., multi-crewed aircraft should split the task loads of sensor and visual searches). The SCAR aircrew should plan to use all available assets to aid in searching the SCAR mission working area.

(c) POIs. An additional technique is to split up several POIs between flights and supporting assets or within the SCAR element/section. For example, with 10 POIs, the SCAR aircrew would search POIs 1–5, and the additional flight would search POIs 6–10. The SCAR aircrew could also assign POI geographically. For example, the SCAR aircrew could search a valley while another flight searches a plateau. Either way, a systematic method will reduce search time and increase the area covered by the SCAR aircrew and supporting assets.

(d) Communication. While the SCAR aircrew is on station, other assets (such as JSTARS or SOF) may find potential targets and pass information to the SCAR aircrew. This information may be passed via voice or data transmission in the form of an attack brief, as described in chapter 4. At a minimum, information should include a target location and a brief description. Also, the SCAR aircrew must know the proper communication chain to request additional assets, if required.

(e) Understanding Threat Force Doctrine. Threat recognition and an understanding of enemy ground force doctrine are crucial to finding a target. Knowledge of how threat ground forces operate will assist the SCAR aircrew and supporting assets in recognizing patterns and searching areas to locate targets. The best way to achieve this knowledge is through an intelligence brief received before a mission. The intelligence brief can provide potential targets in the AO and estimates on how these targets will operate. Additionally, target imagery should enable the SCAR aircrew and supporting assets to anticipate the potential target arrangement.

Note: The SCAR aircrew should be familiar with the time-sensitive target (TST) list and the commander’s critical information requirements in the event they are detected or observed in the SCAR mission working area.

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h. Coordinating Strikes to Maximize Effects on the Targets.

(1) From a planning perspective, the first step is to understand the prioritized target list. Table 1 is an example of a prioritized target list.

Note: The term “prioritized target list” is non-specific and not necessarily assumed to be a target priority list, high-payoff target list, high-value target list, or JIPTL; but could be aligned with, or identical to, any of these lists, if required.

Table 1. Prioritized Target List Example

Precedence	Target
1	Rocket artillery
2	Armor
3	Cannon artillery
4	Command and control nodes
5	Logistics/supplies
6	Field fortifications
7	Mechanized/armored personnel carriers
8	Infantry

Note: More than one target can have the same precedence.

(2) After targets are identified, the SCAR aircrew will match strike asset capabilities to priority targets.

(3) The SCAR aircrew should be familiar with the general capabilities and limitations for interdiction assets. This includes standard weapons load out, data link capabilities, sensor capabilities, and loiter potential. Time permitting, the SCAR aircrew can directly contact assigned assets for an expanded capabilities brief specific to the ATO-tasked mission.

(4) The SCAR aircrew should integrate and leverage all available fire support assets during mission execution. This will help increase battlefield tempo and on-station time efficiency, and mass fires.

(5) The SCAR aircrew should maximize all available capabilities to perform accurate and timely BDAs. This includes using all available sensors from UA, reconnaissance aircraft, SOF, and electronic intelligence. This will assist in confirming strike effectiveness.

(6) Timely BDAs are required after each engagement and enable the SCAR aircrew to optimize asset allocation. This is essential and enables responsive reporting to C2. Through coordination with the JAOC, the SCAR aircrew’s BDAs will assist the supported commander with conducting their assessment of that day's ATO missions.

i. Suppression of Enemy Air Defense (SEAD) Considerations.

(1) The requirement for SEAD operations in support of a SCAR mission are based on threats present in the SCAR mission AO. Any valid surface-to-air threat necessitates that SEAD must be an integral part of SCAR mission planning and

operations. SEAD requirements may vary according to mission objectives, system capabilities, and threat complexity.

(2) Threat assessment and levels of allowable risk determine allocation of SEAD assets to support a SCAR mission. Service components with dedicated and specialized SEAD aircraft can provide special detectors, electronic warfare equipment, antiradiation missiles (ARMs), and precision-guided munitions optimized for SEAD. If SEAD assets are not tasked to support a SCAR mission, other fighter-attack and multirole fighter aircraft may be utilized to support the SEAD mission. Also, SEAD assets may be re-tasked from the JOA SEAD support plan by the appropriate echelon's C2 element. Tactical SEAD planning requires a detailed understanding of the capabilities and limitations of SEAD assets supporting a SCAR mission.

(3) Air defense threats can encompass many national or multinational systems, which are combined into an integrated air defense system (IADS). An adversary's IADS needs to be analyzed in depth, focusing on potential strengths and weaknesses. The goal is to identify command structure, air defense doctrine, early warning and tracking capabilities, C2 reliability and redundancy, and defensive weapons systems. SEAD operations target the following components of an IADS:

(a) C2 nodes.

(b) Surface-to-air missile (SAM) sites.

(c) Air defense artillery.

(d) Early warning and fire control radars and ground control intercept sites.

(e) SAM carriers and storage bunkers.

(f) Air defense operations and maintenance personnel.

(4) SEAD support to SCAR missions may include opportune suppression, whereby suppression may be unplanned or include aircrew self-defense against surface-to-air targets of opportunity. SEAD targets of opportunity are enemy air defense systems detected by surface or airborne sensors, or observers within range of available weapons and not yet targeted.

(5) The JFC will establish ROE for opportune suppression: because SEAD operations require correct identification of enemy systems to prevent fratricide; to avoid law of armed combat (LOAC) violations; and to ensure compliance with the commander's intent. This ROE is especially important when launching ARMs against sources of unknown, spurious electronic signals.

(6) The following are threat mitigation concerns.

(a) SCAR missions inhibited by enemy threat activity may need to suspend interdiction until sufficient SEAD or strike assets can be requisitioned from a tactical C2 element.

(b) If the threat is known, it may be necessary to deconflict assets to remain clear of the enemy weapons engagement zone.

(c) In the event of a pop-up threat, the SCAR aircrew may need to direct assets to “SLIDE” or “SCRAM”. For additional information, see the *Brevity MTTP*.

j. C2.

(1) Operational C2.

(a) Operational planners can use SCAR missions to improve the efficiency and effectiveness of interdiction missions.

(b) Planners can request or assign S/S fires to support SCAR missions to increase the available options for target engagement. If S/S fires are to be integrated into the SCAR mission working area, all players need to understand the C2 structure involved in coordinating S/S fires.

(c) Planners should give the SCAR aircrew sufficient time to find and fix mobile interdiction targets prior to the arrival of striker assets.

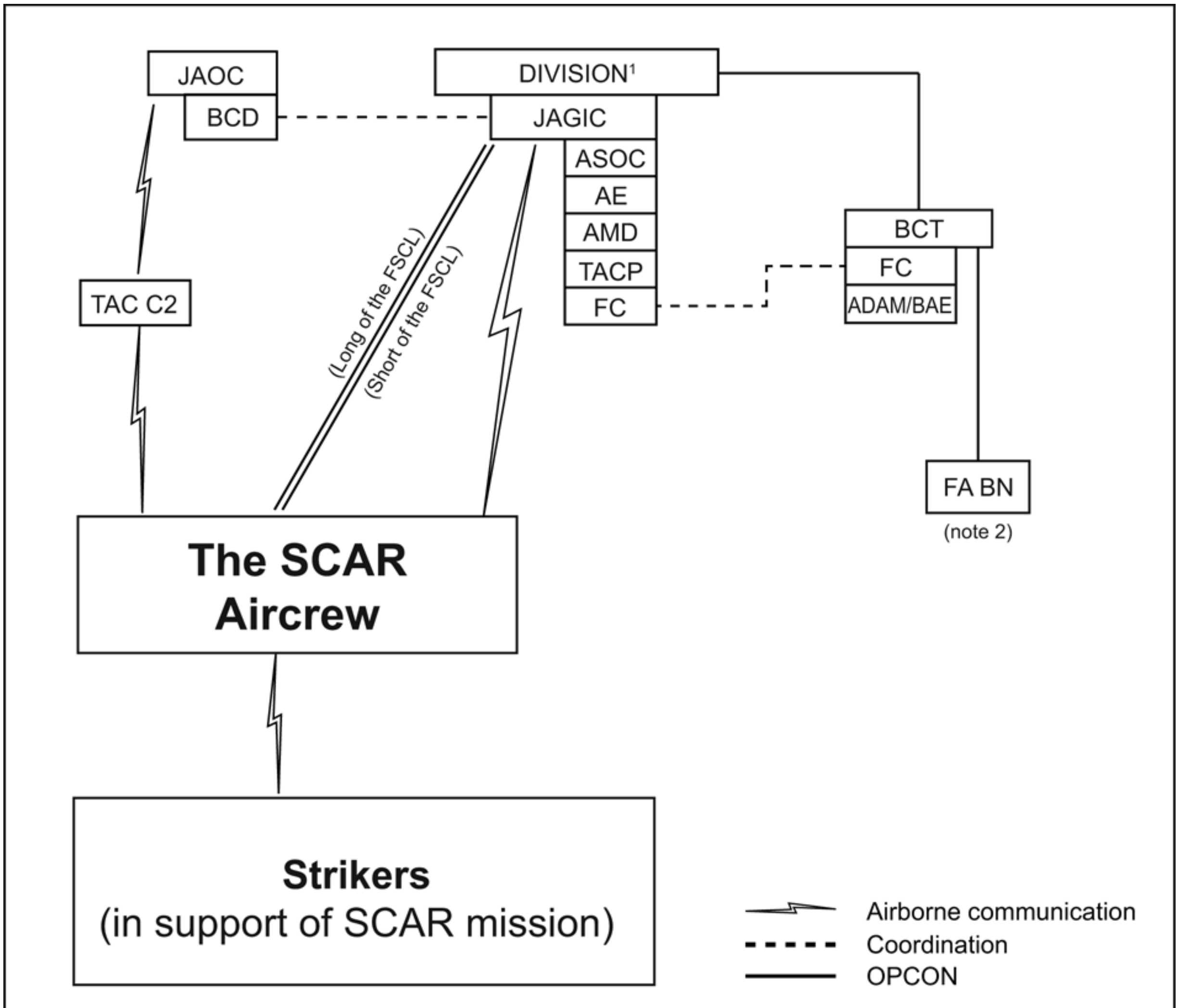
(2) Tactical C2. Tactical C2 plays an integral role in facilitating airspace, communications, and assets for SCAR missions.

(a) Dedicated Communications. SCAR missions require dedicated communication networks between C2 elements and SCAR mission assets. SPINS will assign SCAR mission radio networks. Ideally, there will be a frequency associated with each SCAR mission that enables SCAR mission assets to have a common frequency for coordination. The SCAR aircrew must consider potential communication limitations, such as the effects of terrain on line-of-sight communications and cyberspace threats, and plan to maintain communication between SCAR mission assets and C2 elements. C2 elements are still responsible for contacting SCAR mission assets with urgent threat warnings. Unmanned aircraft system (UAS) data link frequencies must be sanitized to reduce electromagnetic interference. The SCAR aircrew should consider potential data link capabilities of participating assets. See ATP 3-09.32, MCRP 3-31.6, NTTP 3-09.2, AFTTP 3-2.6, *MTTP for Joint Application of Firepower* for more information.

(b) S/S Fires Communication. C2 elements should be prepared to coordinate S/S fire support. Refer to figures 6, 7, 8, and 9 for communication architectures and coordination requirements.

Note: High altitude, long range Army S/S fires are controlled at echelons above brigade. Thus, it is unlikely the SCAR aircrew will be able to receive responsive S/S fires beyond the FSCL if not planned.

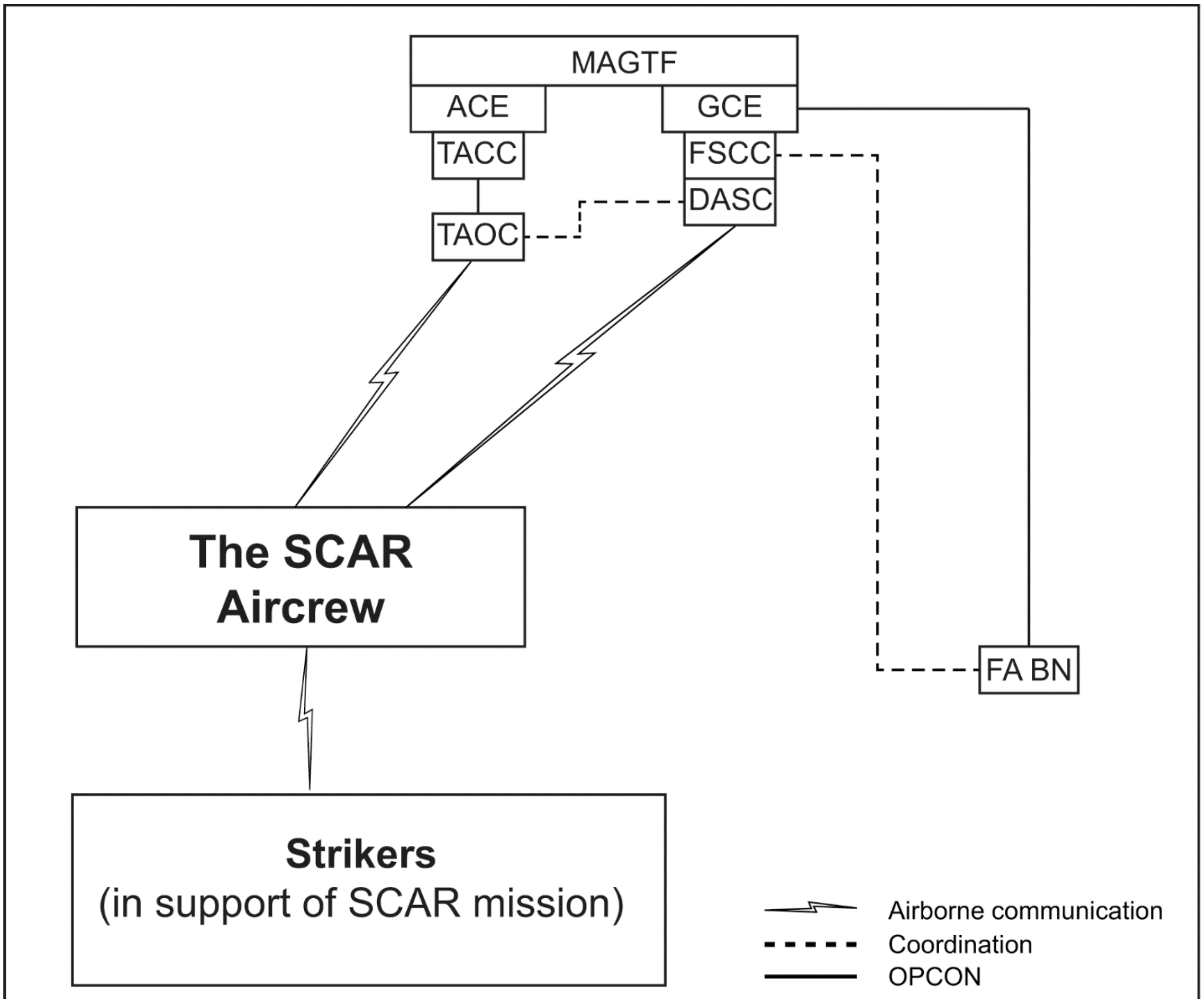
(c) General. Tactical C2 elements should maintain communication channels to operational C2 in the event additional interdiction assets are needed to support a SCAR mission, or changes to the friendly scheme of maneuver need to be conveyed.



Notes:
1. A SCAR mission may be supported by S/S fires provided by the division.
2. The field artillery battalion may be tasked to support division missions, in which case the division will maintain OPCON of the FA BN.

Legend:
ADAM—air defense airspace management
AE—airspace element
AMD—air and missile defense
ASOC—air support operations center
BAE—brigade aviation element
BCD—battlefield coordination detachment
BCT—brigade combat team
BN—battalion
FA—field artillery
FC—fires cell (Army)
FSCL—fire support coordination line
JAGIC—joint air-ground integration center
JAOC—joint air operations center
OPCON—operational control
S/S—surface-to-surface
SCAR—strike coordination and reconnaisance
TAC C2—tactical command and control
TACP—tactical air control party

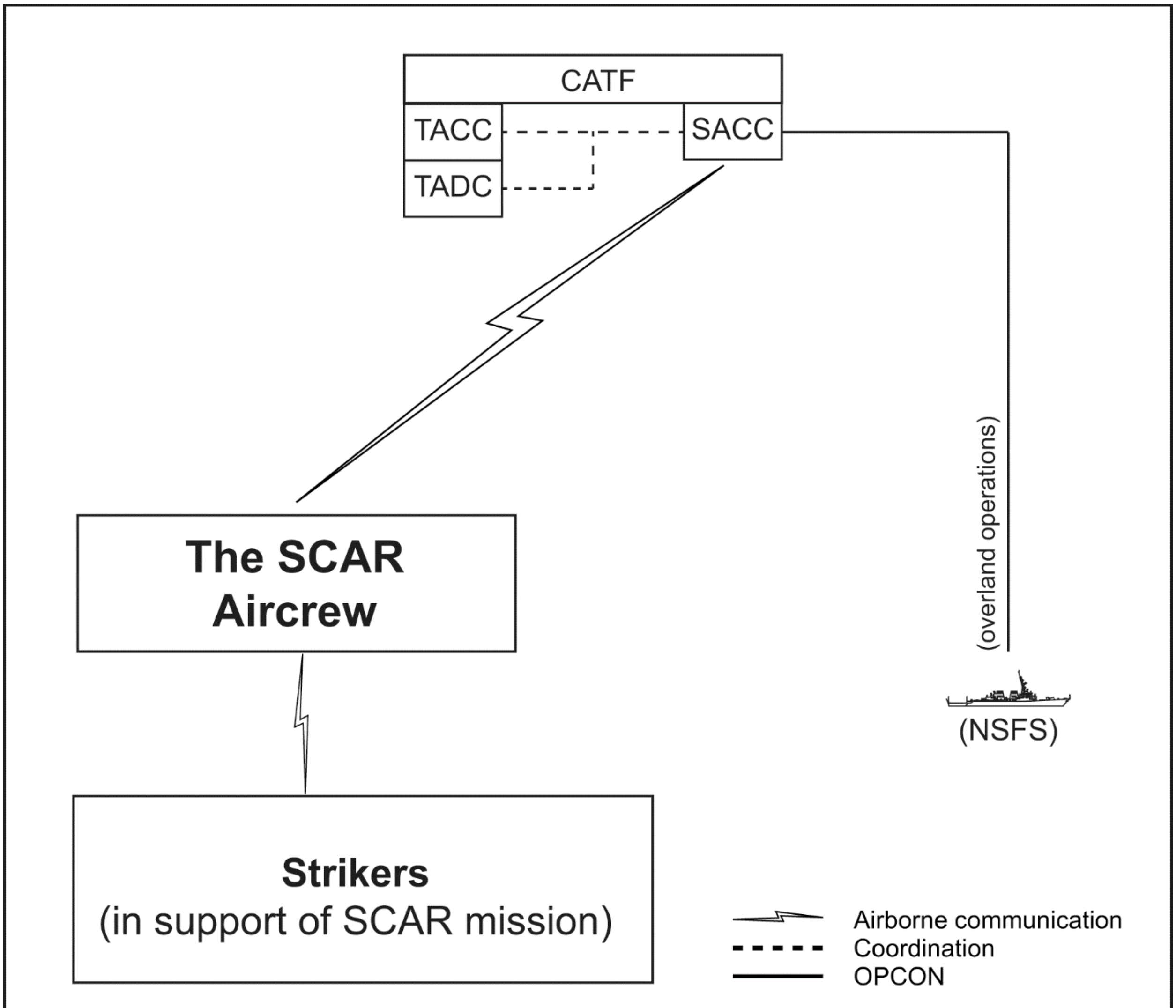
Figure 6. Army AO S/S Fires Communications



Legend:

ACE—aviation combat element (USMC)
BN—battalion
DASC—direct air support center
FA—field artillery
FSCC—fire support coordination center
GCE—ground combat element
MAGTF—Marine air-ground task force
OPCON—operational control
SCAR—strike coordination and reconnaissance
TACC—tactical air command center (USMC)
TAOC—tactical air operations center
USMC—United States Marine Corps

Figure 7. Marine Corps AO S/S Fires Communications



Legend:
CATF—commander, amphibious task force
NSFS—naval surface fire support
OPCON—operational control
SACC—supporting arms coordination center
SCAR—strike coordination and reconnaissance
TACC—tactical air control center (USN)
TADC—tactical air direction center
USN—United States Navy

Figure 8. Navy AO S/S Fires Communications Overland

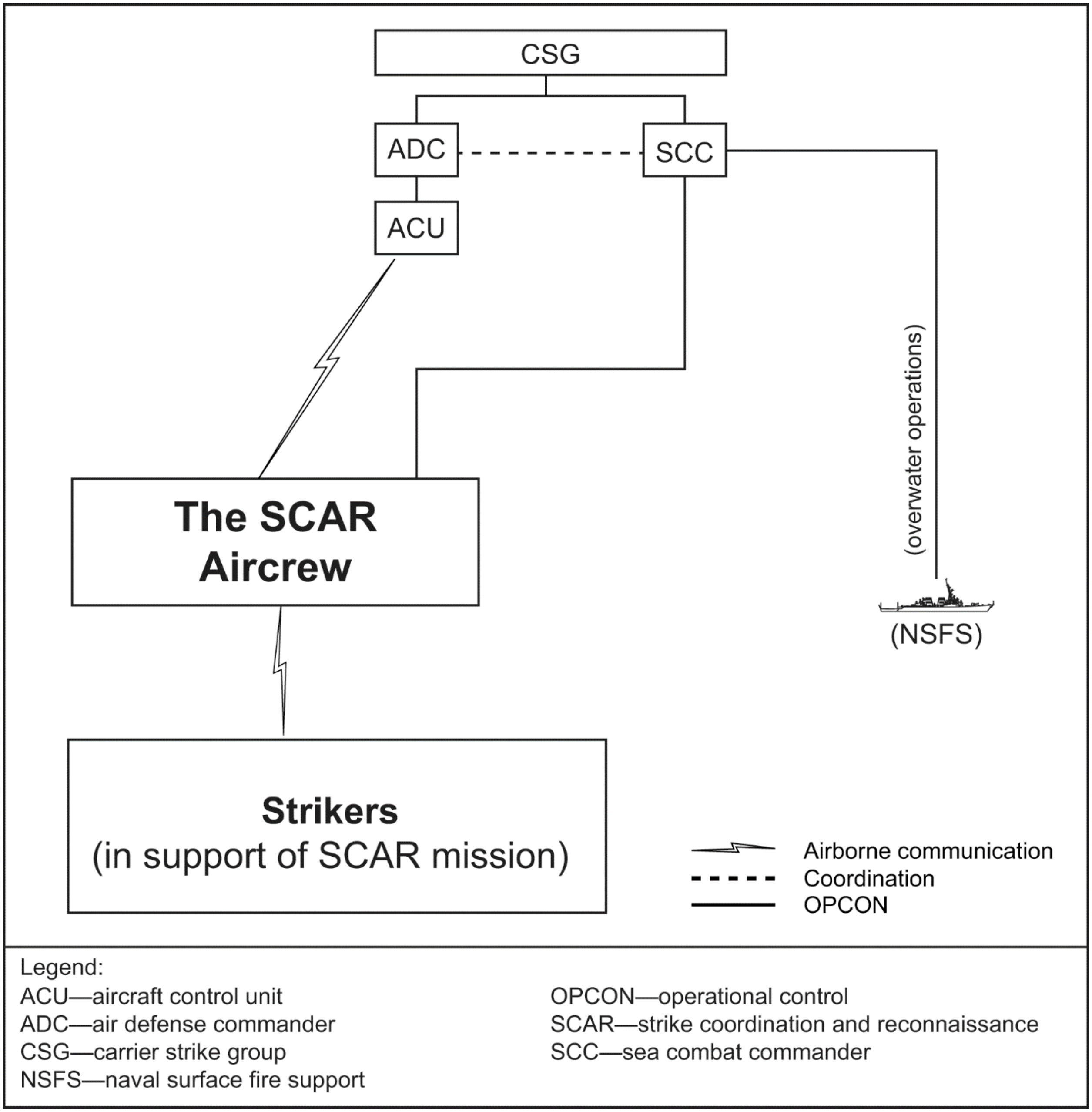


Figure 9. Navy AO S/S Fires Communications Overwater

k. SCAR Mission Airspace.

(1) Airspace Assignment. The SCAR aircrew must be delegated enough airspace to allow for SCAR assets to hold, maneuver, conduct RSTA, and deliver ordnance. The kill box FSCM is the preferred method to achieve this, but is not required. Regardless of the method to delegate airspace, the control measure should be activated during the entire SCAR mission, methods of entry and exit should be defined in the SPINS.

(2) ACM and FSCM Use. The airspace for the SCAR mission is defined through appropriate ACMs and FSCMs.

(a) Land AO. The preferred FSCM for over-land SCAR missions is a kill box. Kill boxes normally use GARS as the common reference system.

(b) Maritime AO. While kill boxes can be used within a maritime AO, maritime operations in support of a SAG, CSG, or ARG will typically use fleet-centric FSCMs depending upon the AO, ROE, or other considerations. For additional information, see ATP 3-04.18, MCRP 3-20.2, NTTP 3-20.8, AFTTP 3-2.74, *MTTP for Air Operations in Maritime Surface Warfare,* and *Airspace Control MTTP.*

(c) Airspace for S/S Fires Assets. Airspace for S/S fires assets supporting the SCAR mission must be coordinated. When the SCAR aircrew requires control of S/S fires assets using kill boxes, a BKB is preferred. A BKB will prohibit entry of S/S fires unless they are requested, coordinated, and approved by the SCAR aircrew. For additional kill box information, see the *Kill Box MTTP*.

Note: A BKB is preferred when the SCAR aircrew requires control of S/S fires engagement since a purple kill box (PKB) does not require coordination with the SCAR aircrew during mission execution.

(d) Mission Support Airspace. The dimensions of the airspace required to support the SCAR mission must be able to accommodate aviation assets and include all target areas. Airspace requirements that drive a ceiling above the CA must be coordinated with the controlling airspace authority prior to conducting the SCAR mission. S/S fires munition flight paths that extend into another component commander's controlled airspace must be coordinated. There is no CA beyond the FSCL.

(3) SCAR Asset Deconfliction Measures. Asset deconfliction measures available to the SCAR aircrew include the following. (Refer to chapter IV for additional deconfliction information.)

(a) Lateral.

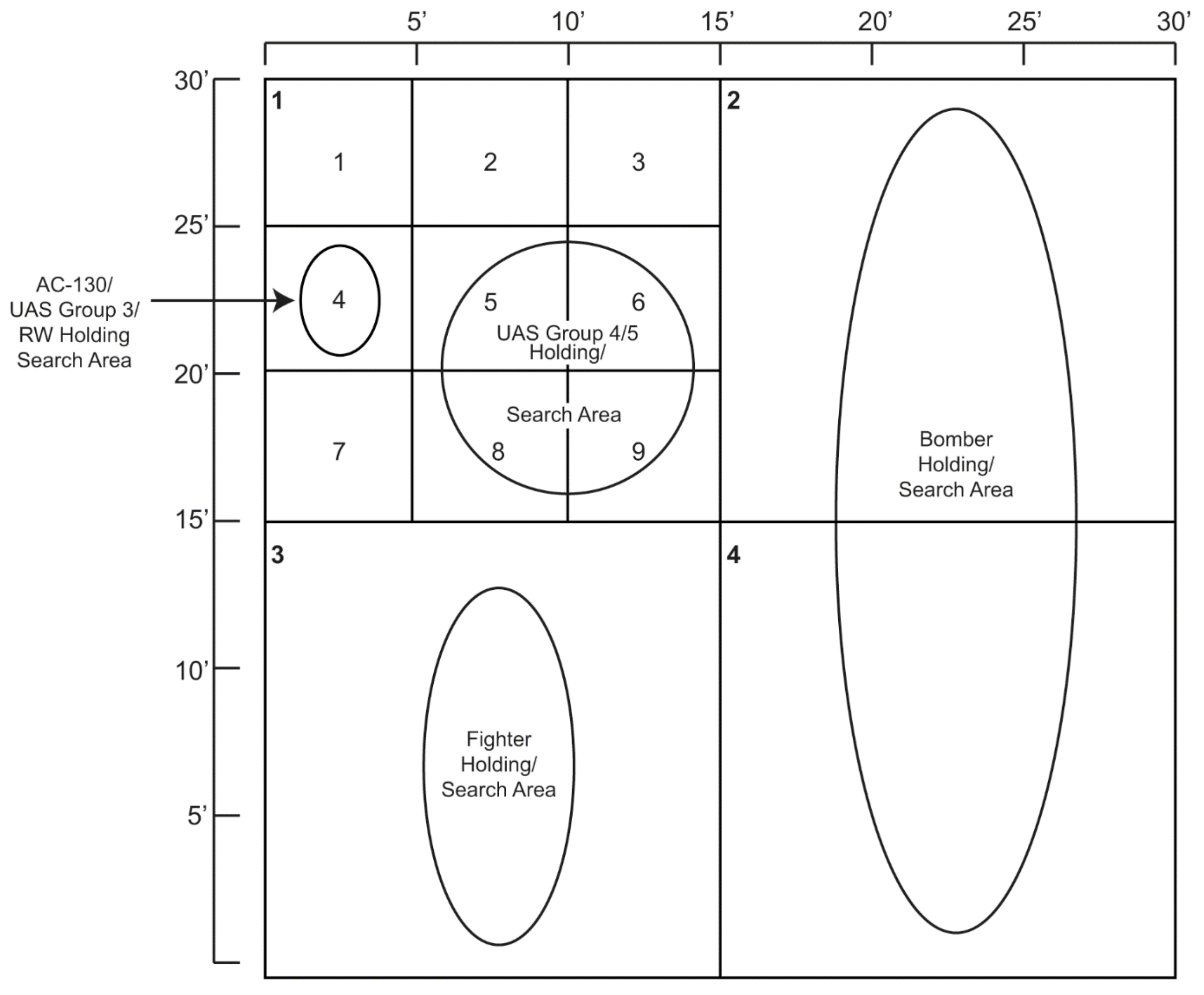
(b) Altitude.

(c) Time.

(d) A combination of lateral and altitude.

(4) Aircraft Holding and Deconfliction Planning Considerations. During mission planning, the SCAR aircrew should examine available airspace and ATO-tasked striker assets to formulate a holding and deconfliction plan. This plan should be as permissive as possible and as restrictive as necessary. Figure 10 depicts an example of sectored deconfliction with GARS where each 30-by-30 minute area defines one kill box. The kill box is further divided into four, 15-by-15 minute quadrants (labeled 1–4), with nine, 5-by-5 minute keypads within each quadrant. If attacking a target in quadrant 1, the other quadrants may be used to laterally deconflict additional aviation assets. Altitude deconfliction, as illustrated in figure 11, is an additional deconfliction method that may be used concurrently with lateral deconfliction. Refer to chapter IV for additional deconfliction specifics.

Note: A 30-by-30 minute kill box is 30 nautical miles (nm) by 30 nm only at the equator. Keep in mind, the nm dimensions of the kill box, or subset of the kill box, will be increasingly less as latitude increases/decreases away from the equator.



Legend:
RW—rotary wing
UAS—unmanned aircraft system

Figure 10. GARS Enabled Kill Box Lateral Deconfliction.

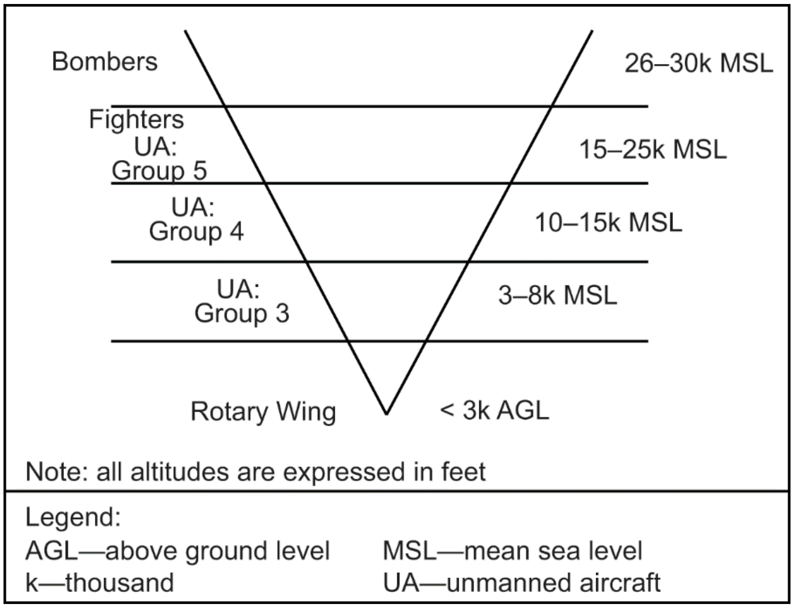


Figure 11. Altitude Deconfliction V-diagram

(a) Fighters. Plan a minimum of a 2,000 foot (ft) holding and deconfliction block for a section of fighters (i.e., two ship) and 4,000 ft for a division (i.e., four ship). The desired block altitude varies based on threats, average terrain, and munitions; but, typically, is in the 15,000–25,000 ft altitude block. This altitude block is a good tradeoff between on-station time and sensor capability. A 15-by-15 nm holding area (i.e., one GARS quadrant) is sufficient airspace for most fighters, but may not provide enough range to maximize sensor search in the holding area.

(b) Bombers. Bombers have a large turn radius. Plan a minimum holding area of 30-by-15 nm (i.e., two GARS quadrants), depending on altitude. On-station time is longer for bomber aircraft.

(c) Helicopters. Helicopters typically operate below 10,000 ft mean sea level (MSL) and at a groundspeed of 70–150 knots (kts). A high threat altitude profile may be 50–200 feet above ground level (AGL) while a medium/low threat altitude profile may be 1,500–2,500 feet AGL. Helicopter holding areas are established at distances from the target area that provide adequate threat stand-off. The SCAR aircrew should consider the fragmentation envelope and bomb fall line of concurrent strike missions in relation to the helicopter holding position to prevent fratricide.

Note: Helicopter deconfliction should not be treated any differently than deconfliction for other aircraft. Deconfliction will be based on helicopter performance and capabilities.

(d) AC-130. AC-130s will hold between 8,000–12,000 ft AGL at 200 kts, and need an area of 5 by 5 nm (i.e., one GARS keypad).

(e) UA. UA holding airspace will vary based on the group category. UA cannot see and avoid manned aircraft, but may contain other onboard systems (e.g., radio; tactical data links; or identification, friend or foe) which allow the SCAR aircrew to use standard deconfliction measures (e.g., altitude, lateral, timing, or a combination thereof). The following are UA altitude and separation requirements.

- Group 3 UA typically require 3-by-3 nm holding airspace (i.e., one GARS keypad) and altitudes between 3,000–8000 ft MSL.
- Group 4 UA can operate as high as 25,000 ft MSL, but are most effective between 10,000 and 15,000 ft above the target area.
- Group 5 UA can operate as high as 50,000 ft MSL, but are most effective between 15,000 and 25,000 ft MSL.
- Groups 4 and 5 UA typically require 10-by-10 nm holding airspace(i.e., four GARS keypads).
- UA require 500 ft altitude separation between other UA, and 1,000 ft altitude separation between UA and manned aircraft. Detailed mission planning is required to determine proper holding airspace and optimal UA operating altitudes.
- A lost link plan is a preprogrammed route an UA will fly if it loses contact with its control station. The UAS aircrew is responsible for coordinating with, and deconflicting, the lost link routing and altitude with the SCAR aircrew.

(5) S/S Fires Integration.

(a) Long-range Missile Systems and Artillery. The joint force has numerous high altitude, long-range indirect fire systems which use large volumes of airspace, such as the High Mobility Artillery Rocket System (HIMARS). These systems are coordinated through the Advanced Field Artillery Tactical Data System (AFATDS). Properly integrating long-range fires requires careful planning at the beginning of an operation. Because long-range fires use significant volumes of airspace, failure to integrate fires in the airspace control plan and daily airspace control order will result in significant deconfliction issues and, most likely, delay fires execution. For more information, see the *Airspace Control MTTP.*

(b) Army Mortars. Mortars are decentralized from traditional fires coordination filling immediate fire support needs for their commanders. Unlike other airspace users, mortars are deconflicted by segmenting the airspace for mortars to use, rather than integrating them into the airspace plan. Consequently, not all mortars will be coordinated using AFATDS. For more information, see the *Airspace Control MTTP.*

(c) Marine Corps Cannon Artillery and Mortars. Marine Corps fires are integrated by the MACCS with the FSCC, at the lowest level possible; and, when required, referred to the MAGTF force and effects coordination center (FECC) for resolution. For more information, see the *Airspace Control MTTP.*

• Fires information comes to the FSCC via AFATDS, voice, or tactical chat. The DASC, colocated with the senior FSCC, receives the information, and plots the fire mission on the tactical data system. For more information, see the *Airspace Control MTTP.*

• The DASC checks the fire mission for conflicts with aviation and informs other affected users (e.g., TACP, and MACCS agencies like TAOC or Marine air traffic control detachment (MATCD)), as applicable. If there are conflicts, the DASC and TAOC, with the TACC (USMC) and affected FSCC, may direct aircraft to move based on the priority of each mission. The MACCS informs all affected aircraft to avoid the trajectory of the fire mission. For more information, see the *Airspace Control MTTP.*

(d) Naval Guns. Five-inch projectiles fired from US Navy vessels have an effective range of, approximately, 9 miles and require airspace deconfliction. Request airspace deconfliction and naval gunfire during amphibious operations through the SACC or FSCC. During operations when the SACC and FSCC are not active, make requests through the fleet commander. For more information, see the *Airspace Control MTTP.*

(e) S/S Fires Deconfliction.

• When planning for S/S fires in support of a SCAR mission, the JAGIC/DASC/SACC must develop a deconfliction plan for the flight path of each round using lateral, altitude, or timing methods, or a combination thereof. Lateral deconfliction measures include establishing a geographic area (e.g., kill box quadrant, terrain features, or military grid reference system (MGRS) location) or a gun-target line (GTL) that the SCAR aircrew and supporting assets will avoid. Altitude deconfliction measures include setting an airspace floor 1,000 ft above the maximum ordinate of the artillery round within the affected airspace volume. When prosecuting targets and deconflicting airspace, planners must know the firing unit's location and the range fan in relation to the SCAR mission working area. Refer to chapter IV for additional deconfliction considerations.

• When conducting a SCAR mission within a BKB, the SCAR aircrew can deconflict S/S fires by changing the airspace status of particular quadrants or keypads within the assigned BKB to CLOSED, while keeping the fires status HOT. For example, if S/S fires employment is desired in quadrant 2, the SCAR aircrew would change quadrant 2's status to CLOSED, HOT until weapons impact. This would prohibit aircraft from entering quadrant 2, yet permit fires to be employed into quadrant 2. While a PKB may be used to facilitate S/S fires employment, there is no coordination requirement between the SCAR aircrew and S/S fires assets prior to S/S fires employing beneath the PKB floor. If operating within a PKB, the SCAR aircrew and strikers in support of the SCAR aircrew will not be able to descend below the PKB floor to engage targets or perform reconnaissance without the additional coordination.

l. Weather Considerations. Weather conditions will impact the level of SCAR mission effectiveness. Poor weather conditions may decrease the target search and identification ability for SCAR aircraft. To effectively search, PID, and assess collateral damage concerns, SCAR aircraft should be able to operate beneath the weather. Planners and operators also should consider the following based on weather conditions:

(1) Terrain elevation.

(2) Maximum altitude for the majority of threat systems.

(3) Altitude requirement for effective threat reactions and mission performance.

(4) Airspace requirements based on the number of striker assets and deconfliction plan.

(5) Weapon and sensor capability. Specific aircraft capabilities may allow assets to operate in and above weather.

m. Multinational Operations Considerations. During multinational operations, SCAR assets must have knowledge of multinational partner system capabilities to ensure close coordination. Units need to solicit platform subject matter experts in-theater to discuss cross platform capabilities with emphasis on sensors, communications, weapons, and airframe capabilities. Additionally, the SCAR aircrew must be familiar with national caveats to the SPINS when integrating partner forces.

n. Contingencies.

(1) Adverse Weather. Weather can severely affect aspects of a SCAR mission including deconfliction, target location, employment, and coordination. The SCAR aircrew should consider how to overcome weather conditions that are worse than projected. Some aircraft may provide an all-weather search and target capability (e.g., SAR map capable aircraft). Mitigation measures available to the SCAR aircrew may include high/low split war tactics where strike assets drop inertial aided munitions (IAMs) or laser-guided weapons (LGWs) through the clouds with the SCAR aircrew “buddy lasing” below the weather. Other options include increasing the working area through the C2 element to allow lateral separation, reducing the number of aircraft in the working area, or requesting an increased acceptable level of risk (ALR) for the mission to operate within the weapon engagement zones of surface threats.

(2) Prohibitive Interference. Based on ALR, the SCAR aircrew must plan for potential prohibitive interference on the battlefield. Based on the prohibitive threat, the SCAR aircrew can suppress the threat, reestablish sanctuary by altitude or lateral separation, or terminate the mission. An increased ALR can only be approved by the issuing authority.

(3) Contested Degraded Operations. The SCAR aircrew must have contingencies for communications degradation, Global Positioning System (GPS) jamming/unavailability, and data link denial. Mitigation measures may include using active and secure networks, having a chattermark plan, and adjusting weapons loadout and employment.

(4) Asset Fallout. Coordination and dissemination of an overall SCAR mission plan prior to execution will reduce the likelihood of reduced mission effectiveness due to:

(a) SCAR Fallout. Identify the best qualified asset to assume the SCAR aircrew role in case of SCAR aircrew fallout.

(b) Capability Fallout. Identify specific capabilities required to accomplish the mission and identify back-up assets to create the desired effect.

(c) Tanker Fallout. If the tanker plan changes or tanker fallout occurs, the SCAR aircrew may need to prioritize SCAR assets based on capabilities, weapons load, and fuel requirements.

(5) Non-PID or CDE Capable Aircraft. SCAR aircraft may need to satisfy PID and assess collateral damage for striker aircraft that can strike a target but cannot conduct PID or a CDE due to system limitations, degraded systems, or environmental factors. In this case, the SCAR aircrew must ensure PID and collateral damage requirements are met through any available means prior to tasking the striker asset. If SMACK authority has been granted, the SCAR aircrew may issue a SMACK tasking to the striker aircraft once SMACK criteria are met. If SMACK authority has been restricted by ROE, SPINS, or a C2 element, it must be requested and granted prior to the SCAR aircrew issuing a SMACK tasking.

4. Unplanned SCAR

a. There may be times when assets not originally tasked to conduct a SCAR mission are retasked to execute a SCAR mission in real time, either as the SCAR aircrew or a striker in support of a SCAR mission.

b. It is possible for the SCAR aircrew to be identified in the absence of an ATO assignment (i.e., a SCAR mission in response to immediate hostilities). In this case, aircrews will be unable to receive a mission brief prior to execution. The SCAR aircrew will be the first asset capable of leading the SCAR mission IAW the responsibilities outlined in chapter I.

c. The SCAR aircrew responsibilities may be transferred, in real time, to any capable aircrew and asset that can better provide the following:

(1) Situational awareness.

(2) Sensor capability.

(3) Communication capability.

(4) TOS.

d. The SCAR aircrew should request striker assets that are appropriate for the tactical situation and AO. Asset allocation may be the responsibility of tactical C2. Assigning an asset as a striker in support of a SCAR mission, when not tasked on the ATO, may require detailed coordination between operational C2 and tactical C2.

e. The SCAR aircrew should have a plan to integrate supporting assets. SCAR mission flow, at a minimum, should include:

(1) AO entry and check-in.

(2) Situation update/situation report (SITREP).

(3) Deconfliction.

(4) Asset tasking.

(5) BDA/inflight report (INFLTREP).

(6) Battlefield handover.

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Chapter IV
EXECUTION

1. Introduction

Prior to execution, the SCAR aircrew must have a thorough understanding of the prioritized target list, enemy threat, SCAR mission working area, FSCMs and ACMs, and the capabilities and limitations of the supporting assets.

2. Mission Brief

a. The SCAR mission brief is used to convey the SCAR aircrew's plan to all assets. Issuing timely mission information is a key factor for success and will allow assets to incorporate the information from the SCAR aircrew into mission planning. The SCAR aircrew should conduct a face-to-face mission brief, video teleconference, or phone conversation to allow for questions and discussion among all assets. At times, it will not be feasible for the SCAR aircrew to conduct a face-to-face briefing. In this instance, the SCAR aircrew should attempt to pass needed information via the most effective means available to ensure widest dissemination.

b. The mission brief should begin with an overview of the ATO mission, objectives, package composition and load out, and administrative procedures (e.g., routing, holding, and communicating within the SCAR aircrew's assigned area). The overview will aid in the smooth transition during real-time SCAR aircrew battlefield handovers. Refer to Appendix B, "SCAR Mission Briefing Guide", for mission briefing specifics.

c. The mission brief should cover an overall plan commensurate with the tactical situation, target sets, and known threats. The SCAR aircrew may have assets check-in to the AO without having received the mission brief and plan for information to be passed real time on the battlefield.

d. The SCAR aircrew is expected to have a plan for integrating assets, within the assigned AO, immediately upon first contact. SCAR mission flow should include, at a minimum:

(1) AO entry/check-in.

(2) Situation update/SITREP.

(3) Deconfliction.

(4) Asset tasking.

(5) BDA/INFLTREP.

(6) Battlefield handover.

e. Appendix B outlines a briefing template to facilitate SCAR execution. Refer to current SPINS for area of responsibility-specific briefing considerations and include SCAR information in aircrew smart packs in case missions are retasked while airborne. In addition, the SCAR aircrew should carry a coordination card to effectively manage assets, allocate targets, and record BDA. (See appendix C for SCAR product examples.)

3. The SCAR Aircrew Communication Requirements

a. En Route. The SCAR aircrew should attempt to gain as much information as possible from C2 assets and the current or outgoing SCAR aircrew. The SCAR aircrew should receive a situation update/battlefield handover as listed in table 2.

b. SCAR Aircrew Check-in. The initial situation update brief from the C2 element to the SCAR aircrew should be as thorough as possible to increase the SCAR aircrew's situational awareness. Close coordination and pre-mission planning between the SCAR aircrew and C2 element will reduce the SCAR aircrew's workload. See table 2 for an example of a situation update brief.

4. Striker Asset Communication Requirements

a. En route, striker assets will check-in with C2 elements IAW the SPINS. Before entering the SCAR mission working area, flights must check-in with the SCAR aircrew to receive routing and safety of flight. Entry into the SCAR mission working area without clearance from C2 is prohibited. In the event the SCAR aircrew is not established, the first asset on station should deconflict airspace within their capabilities if unable to assume the SCAR aircrew role.

b. When requested, the striker asset will check-in with the SCAR aircrew using the format in table 3. It is important that the SCAR aircrew evaluate information, such as TOS and ordnance, to maximize the efficiency of assigning aircraft capabilities for desired target effects. The intent is to generate and sustain the highest tempo possible against the enemy, and to degrade the enemy's ability to displace or maneuver against friendly forces.

c. Following check-in, the SCAR aircrew should pass a situation update to the striker assets, as indicated in table 2. If communication time is limited, the SCAR aircrew should only pass mission-critical information.

d. When working with UA, the SCAR aircrew should expect the UAS operators to pass a lost link plan in the remarks section at check-in. For example, "If lost link, Grim 21 will loiter for 15 minutes in 069SH quadrant 1 at flight level (FL) 230 then return to base (RTB) via minimum risk route (MRR) 3." The lost link plan may change throughout the mission. A more complex airspace construct and deconfliction plan may require a more complicated lost link plan. Lost link plan information may be passed over voice or data link. In the event of a lost link situation, the UAS operator may be unable to pass the status of the UA to the SCAR aircrew via voice. In these situations, this information must be routed through a C2 element.

Table 2. Situation Update and Battlefield Handover		
Situation Update Line	**Situation Update**	**Battlefield Handover** (In addition to information passed in the Situation Update)
Threat Activity	Threats to aviation observed: who, what, where, when.	
Enemy Situation	Give the current enemy disposition. Provide the size, activity, and location. Give specific targets and locations if available.	Include target priorities. Include target location grids. This may require breaking up the transmission.
Friendly Situation/ Commander's Intent	SCAR, striker assets on station (time on station and ordnance). Ongoing electronic warfare/suppression of enemy air defenses/nonlethal effect activity.	
Artillery (Indirect Fire) Activity	Provide friendly indirect fire assets and considerations (e.g., azimuth of fire, maximum ordinate, priority of fire, and any active mission), if applicable.	Firing unit location, call sign, frequency, status.
Coordination	Advise all players on the net/tactical air direction. If not prebriefed or planned, determine who has engagement authority.	
Hazards	Weather, terrain, or obstructions.	
Remarks/Restrictions	Active/updates to FSCMs and ACMs, current aircraft location (i.e., quadrant, keypad, altitude) and tasking.	Sensor management plan.
Note: The SCAR aircrew shall immediately pass routing/deconfliction for inbound striker assets prior to situation update. Use either the threat, enemy situation, friendly update, artillery, clearance authority, hazards, remarks/restrictions; (TEFACHR), or targets, threats, friendlies, artillery, clearance, ordnance, restrictions (TTFACOR) format.		
Legend: ACM—airspace coordinating measure FSCM—fire support coordination measure SCAR—strike coordination and reconnaissance		

Table 3. Check-in Briefing Format (Aircraft Transmits to the SCAR Aircrew)
Aircraft: "____________________, this is _______________________" (SCAR aircrew call sign) (striker call sign) Mission number: "______________________________"
Note: Authentication (initiated by the net control agency) and appropriate response suggested here. The brief may be abbreviated for brevity/security ("as fragged" or "with exception").
Number and type of aircraft: "_________________________________" Position and altitude: "_______________________________________" Ordnance: "____" (include fuzing or laser code) Playtime or time on station: "_________________________________" Capabilities: "____" (e.g., targeting pod, laser, laser spot tracker, infrared marker, video downlink, or digital capabilities) Abort code: "____" Remarks: "____" (e.g., Unmanned aircraft system lost-link routing)
Legend: SCAR—strike coordination and reconnaissance

5. Asset Deconfliction

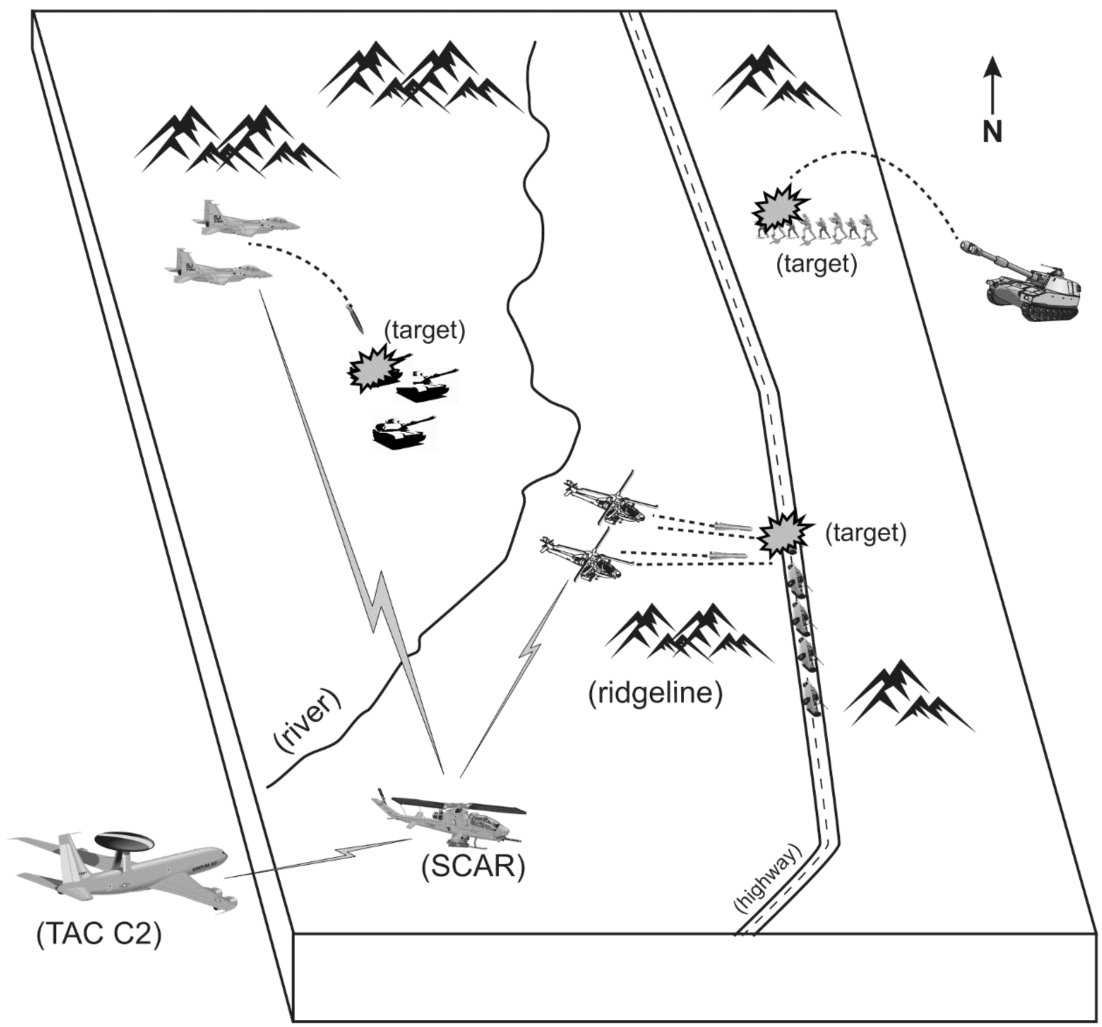
a. Deconfliction Methods. The SCAR aircrew has the option of deconflicting assets by using one of three different methods or a combination thereof. The three deconfliction methods are lateral, altitude, and timing.

(1) Lateral Deconfliction. Lateral deconfliction separates assets in geographic space. References that may be used to separate assets include terrain features, GARS/kill box, MGRS, common grid reference system (CGRS), or other commonly understood FSCMs and ACMs. Refer to figures 12–14 for examples of lateral deconfliction.

(2) Altitude Deconfliction. Altitude deconfliction separates assets in vertical space. If operating in a constricted area, it may be necessary to stack multiple elements in the same airspace and deconflict them by altitude. The SCAR aircrew should stack all aircraft to optimize sensor performance, expedite deconfliction of fires, and mitigate the threat. Refer to figure 15 for an example of altitude deconfliction.

(a) SCAR Stacked Low. This technique usually increases the probability of achieving target PID and enables immediate attack. The SCAR aircrew is required to deconflict from fires transiting through the SCAR aircrew's altitude.

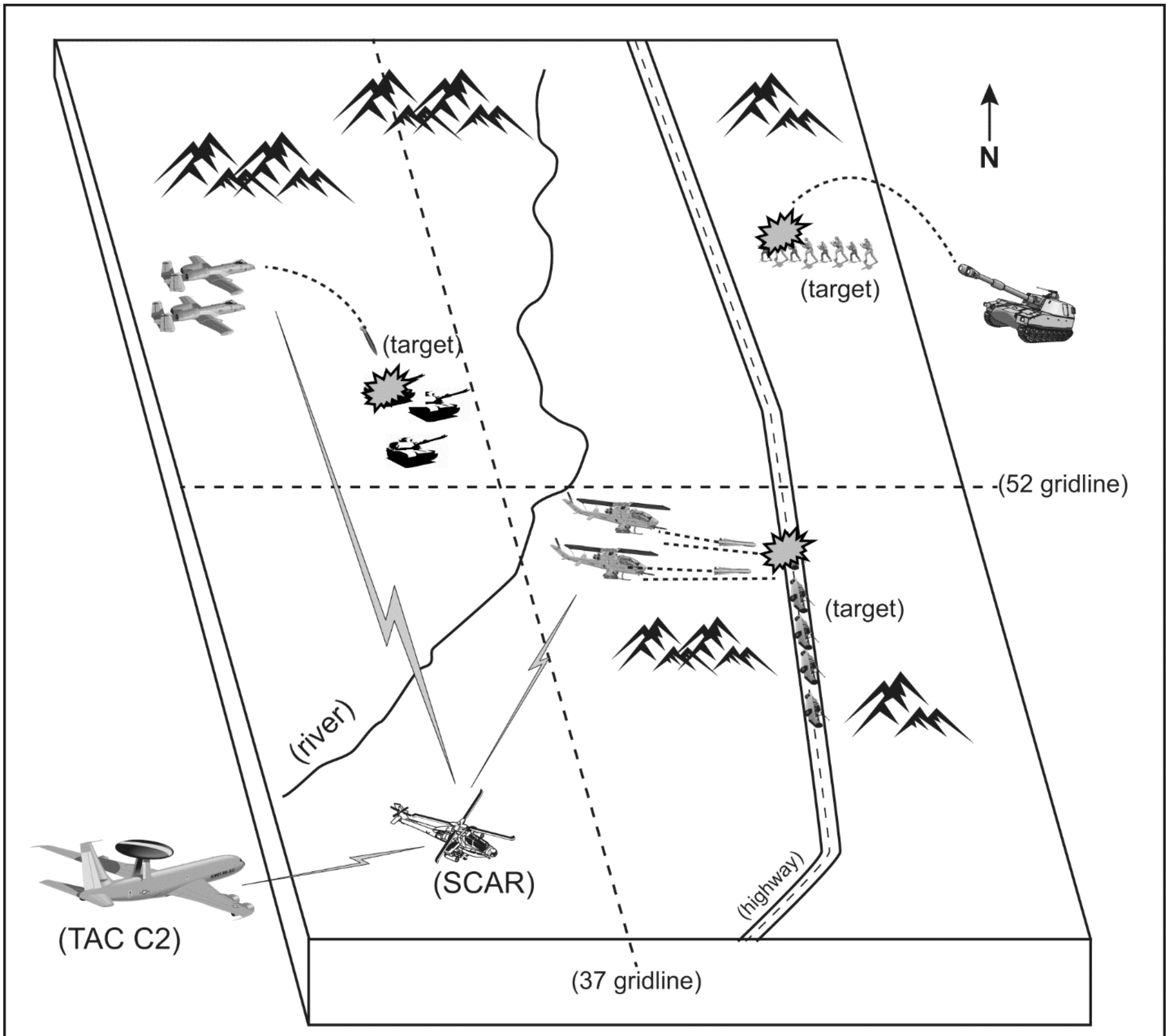
(b) SCAR Stacked High. Stacking high provides the SCAR aircrew the autonomy to move within the SCAR mission working area with minimal deconfliction. However, weapons deconfliction must be ensured should the SCAR aircrew conduct strikes through assets assigned to lower altitudes.



Lateral Deconfliction Plan:
1. The SCAR aircrew remains east of the river, west of the highway, and south of the ridgeline.
2. F-15Es remain west of the river.
3. AH-64s remain east of the river, west of the highway, and north of the ridgeline.
4. Surface-to-surface fires are requested to engage target locations east of the highway.

Legend:
C2—command and control
TAC—tactical
SCAR—strike coordination and reconnaissance

Figure 12. Lateral Deconfliction Using Terrain Features



Lateral Deconfliction Plan:

1. The SCAR remains west of the 37 gridline and south of the 52 gridline.
2. A-10s remain west of the 37 gridline and north of the 52 gridline.
3. AH-1s remain east of the 37 gridline and south of the 52 gridline.
4. Surface-to-surface fires are requested to engage target locations that are north of the 52 gridline and east of the 37 gridline.

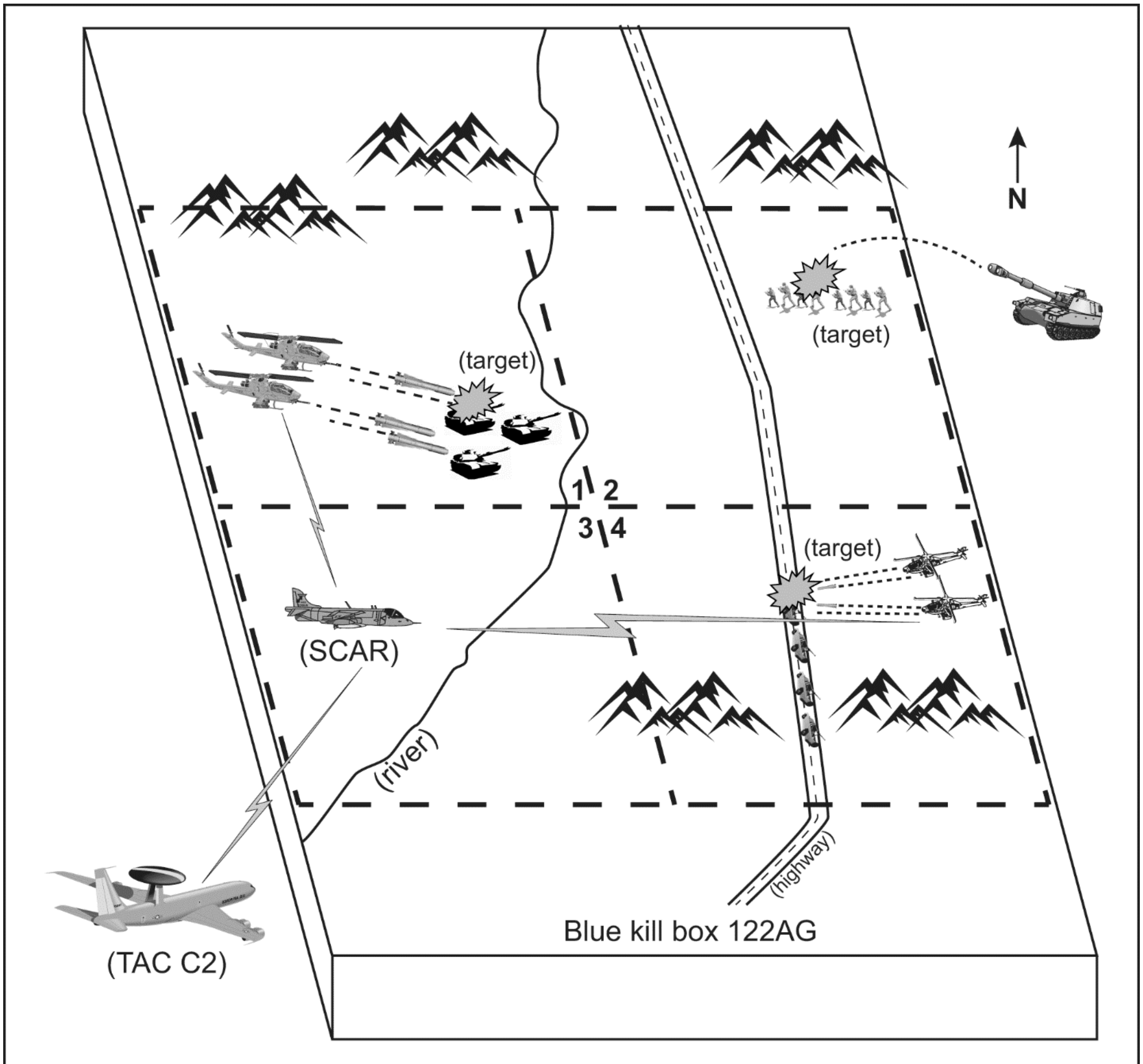
Legend:

C2—command and control

TAC—tactical

SCAR—strike coordination and reconnaissance

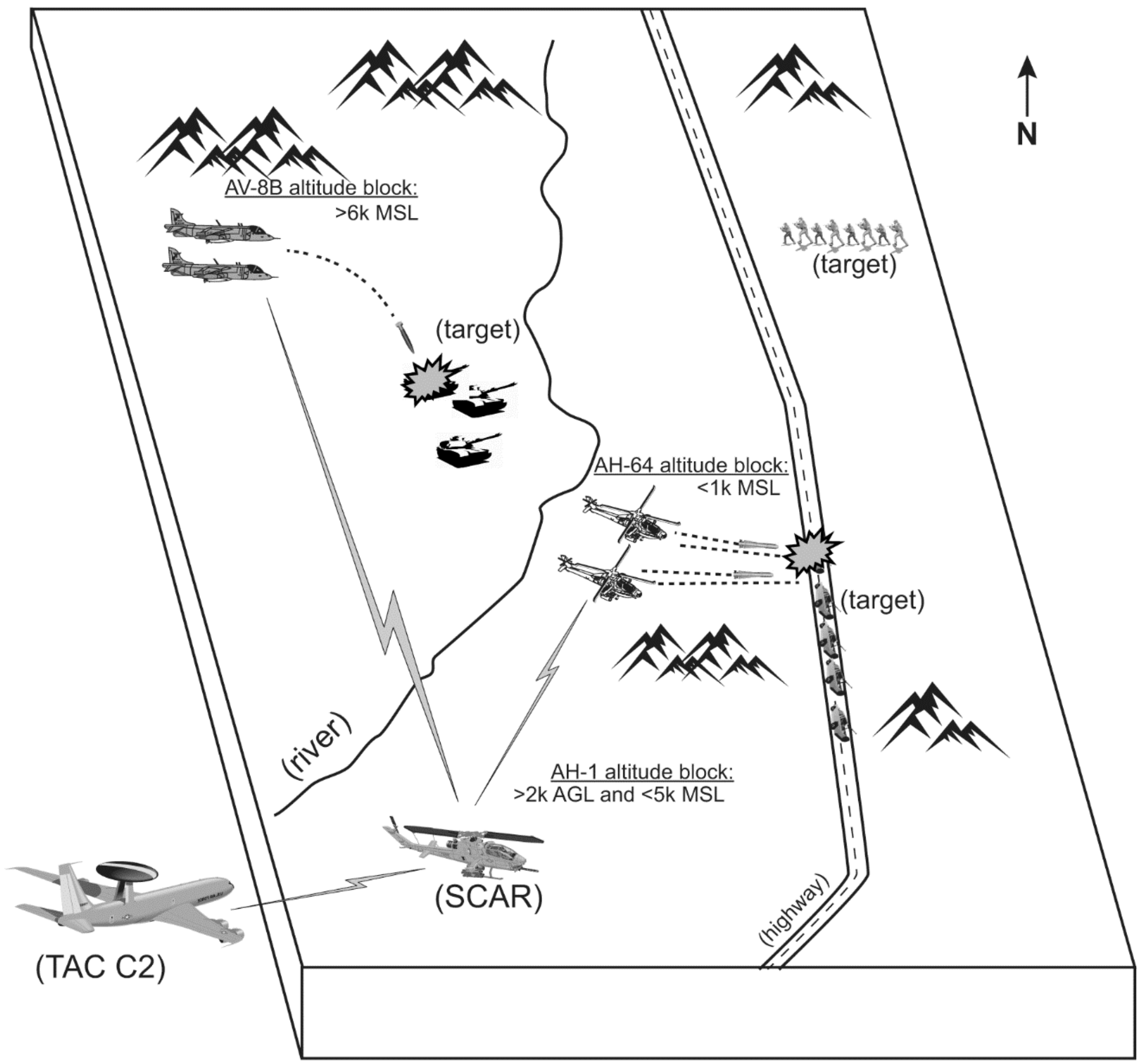
Figure 13. Lateral Deconfliction Using MGRS



Lateral Deconfliction Plan:
1. The SCAR aircrew remains in kill box 122AG, quadrant 3.
2. AH-1s remain in kill box 122AG, quadrant 1.
3. AH-64s remain in kill box 122AG, quadrant 4.
4. Surface-to-surface fires are requested to engage target locations in kill box 122AG, quadrant 2. Since the SCAR aircrew is assigned a blue kill box, the SCAR aircrew updates 122AG, quadrant 2, to a "CLOSED, HOT" kill box status to facilitate surface-to-surface fires.

Legend:
C2—command and control
TAC—tactical
SCAR—strike coordination and reconnaissance

Figure 14. Lateral Deconfliction Using a Kill Box



Note: all altitudes are expressed in feet

CAUTION: Deconfliction from aircraft munition trajectory must be ensured.

Altitude Deconfliction Plan:

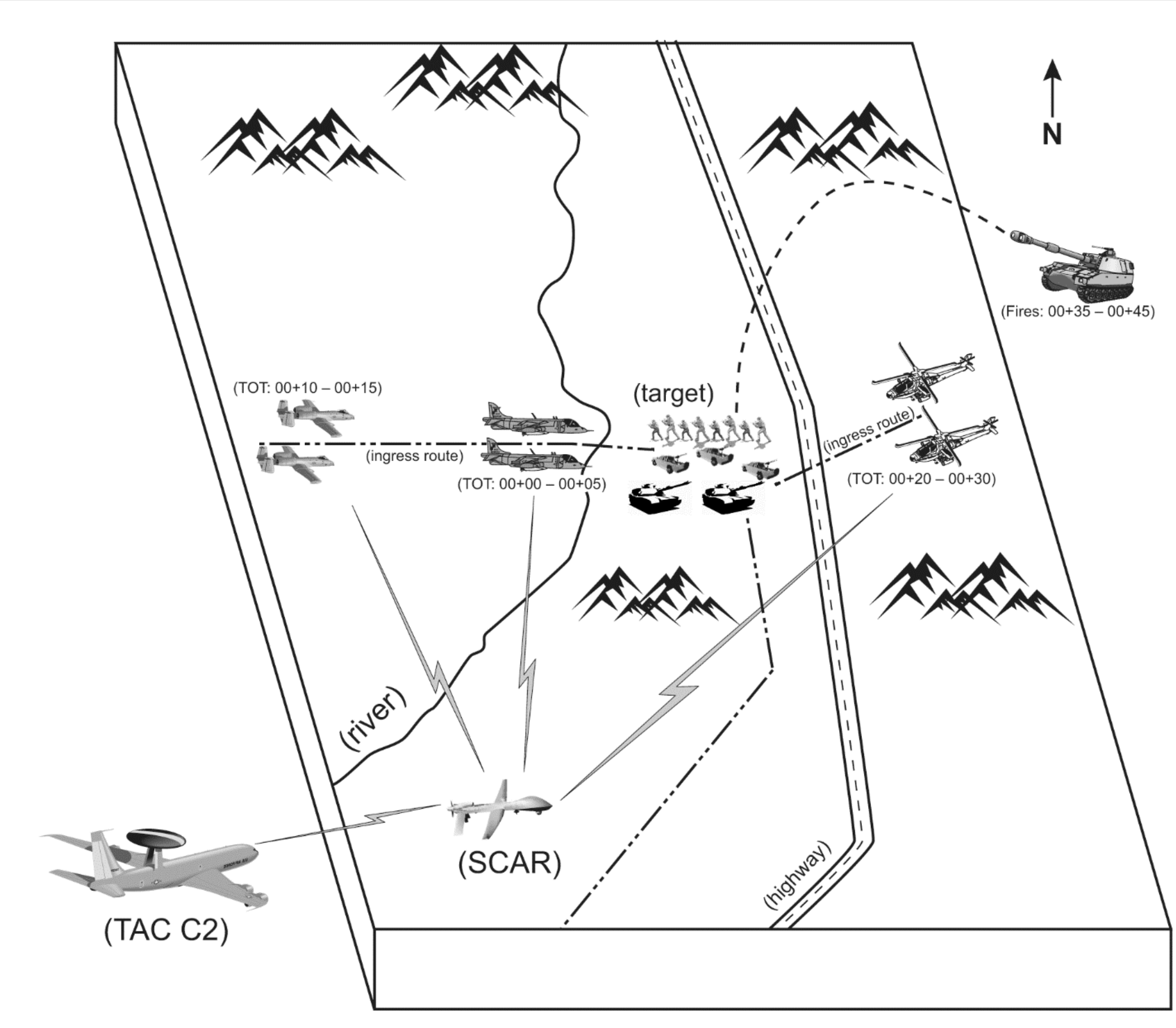
1. The SCAR aircrew remains between 2k AGL and 5k MSL.
2. AV-8Bs remain above 6k MSL.
3. AH-64s remain below 1k AGL.
4. Since surface-to-surface fires are not available, the SCAR may deconflict aircraft only via altitude separation. If surface-to-surface fires were available, the SCAR would use an alternate or additional deconfliction method to ensure deconfliction.

Legend:

AGL—above ground level
C2—command and control
k—thousand
MSL — mean sea level
SCAR—strike coordination and reconnaissance
TAC—tactical

Figure 15. Altitude Deconfliction

(3) Timing Deconfliction. Timing deconfliction separates assets by time. This method may be necessary if lateral and altitude deconfliction options are limited. Refer to figure 16 for an example of timing deconfliction.



CAUTION: Utilizing a timing deconfliction method may also require additional deconfliction methods to ensure the deconfliction plan is adequate.

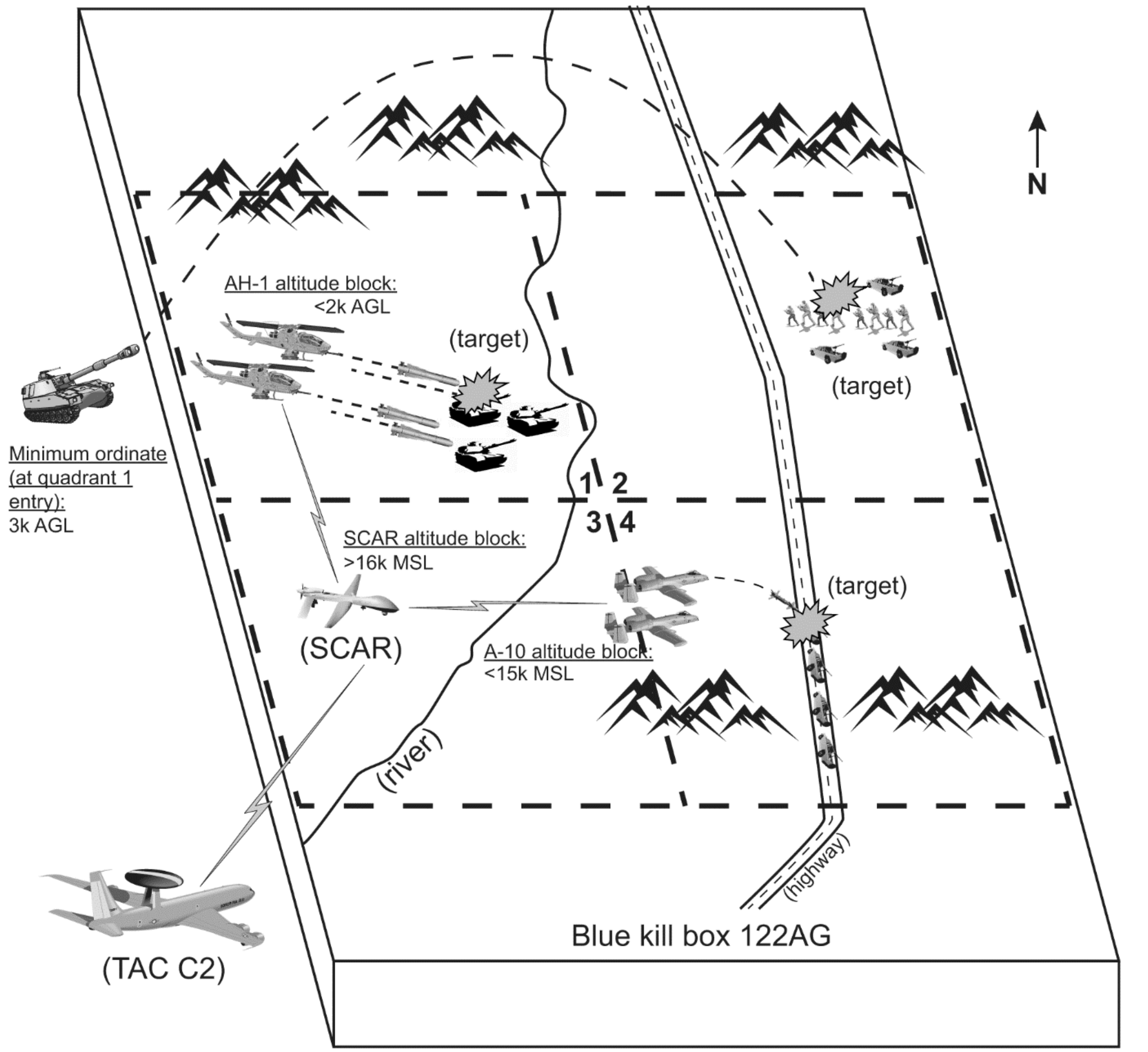
Timing Deconfliction Plan:

1. The SCAR will deconflict from the striker assets as required.
2. AV-8Bs have a TOT window of 00+00 to 00+05.
3. A-10s have a TOT window of 00+10 to 00+15.
4. AH-64s have a TOT window of 00+20 to 00+30.
5. Surface-to-surface fires have a TOT window of 00+35 to 00+45.

Legend:
C2—command and control
SCAR—strike coordination and reconnaissance
TAC—tactical
TOT—time on target

Figure 16. Timing Deconfliction

(4) Combined Lateral and Altitude Deconfliction. Deconfliction methods may be combined to increase the efficiency and effectiveness of the mission. Refer to figure 17 for an example of combining lateral and altitude deconfliction methods.



Note: all altitudes are expressed in feet

Combined Lateral and Altitude Deconfliction Plan:
1. The SCAR aircrew remains in kill box quadrants 3 and 4 >16k MSL.
2. A-10s remain in kill box quadrants 3 and 4 <15k MSL.
3. AH-1s remain in kill box quadrant 1 <2k AGL.
4. Surface-to-surface fires are requested to engage target locations in kill box quadrant 2. Since the SCAR aircrew is assigned a blue kill box, the SCAR aircrew updates 122AG, quadrant 2, to a “CLOSED, HOT” kill box status to facilitate surface-to-surface fires. Minimum ordinate altitude in quadrant 1 is 3k AGL, which drives the AH-1s to maintain <2k AGL.

Legend:
AGL—above ground level
C2—command and control
k—thousand
MSL—mean sea level
SCAR—strike coordination and reconnaissance
TAC—tactical

Figure 17. Combined Lateral and Altitude Deconfliction

b. Aviation Asset Deconfliction Considerations.

(1) The SCAR aircrew's holding and deconfliction plan should consider enemy capabilities, target orientation, weather, threats, and striker asset capabilities. The SCAR aircrew should avoid leaving striker assets idle and use the available striker assets' sensor/weapon capabilities during TOS, whether in holding or during task execution, and in transit.

(2) In addition to the main deconfliction methods, initial points (IPs), keyhole template, and geographical references (GEOREFs) are available to assist in deconfliction. Refer to ATP 3-09.32, MCRP 3-31.6, NTTP 3-09.2, AFTTP 3-2.6, *MTTP for Joint Application of Firepower*, for an example of the keyhole template.

(3) Figure 18 is an example of an aircrew card that deconflicts supporting assets using altitude and lateral deconfliction techniques. The SCAR aircrew should be prepared to provide immediate asset and fires deconfliction from any neutral or friendly air or surface contacts that enter the SCAR aircrew's working area.

c. A/S and S/S Fires Deconfliction.

(1) Separate target arrays. Timing deconfliction is useful when both assets engage the same target array. Lateral deconfliction measures include establishing a GTL that the SCAR aircrew and supporting assets will avoid. Depending on weapons delivery parameters, altitude deconfliction may be useful in either situation.

(2) When conducting a SCAR mission within a BKB, S/S fires may be deconflicted laterally by the SCAR aircrew changing the airspace status of specific quadrants or keypads within the assigned BKB to "CLOSED", while keeping the fires status "HOT". For example, if S/S fires employment is desired in quadrant 2, the SCAR aircrew would change quadrant 2's status to "CLOSED, HOT" until weapons impact. This would prohibit aircraft from entering quadrant 2, yet, permit fires to be employed into quadrant 2.

(3) Lateral deconfliction works best when A/S and S/S fires engage targets at approximately the same time.

6. SCAR Mission Working Area Execution

F2T2EA is the preferred method for conducting SCAR missions. Refer to figure 19 for a depiction of the F2T2EA process.

a. Target Detection (Find, Fix).

(1) Use off-board sensors to acquire targets (i.e., JSTARS, Aegis, SOF reconnaissance teams, or UAS). Additionally, use on-board sensors like targeting pods, A/S radars, binoculars, night vision devices, or unaided visual searches to acquire and refine targets. Use all platforms effectively (i.e., striker and RSTA assets) to maximize search potential and find targets. Use off-board sensors and target information to increase mission tempo and reduce the requirement for strike capable assets to search for targets. The following example describes target detection: "JSTARS informs the MQ-1 SCAR aircrew of a possible tank

column moving north along a main supply route in the vicinity of 391 LE quadrant 3, in the vicinity of 33S WR 0726 6669, and south of a town (find). The MQ-1 SCAR aircrew, cues its sensors to that location and establishes PID on six T-72 main battle tanks moving north (fix)."

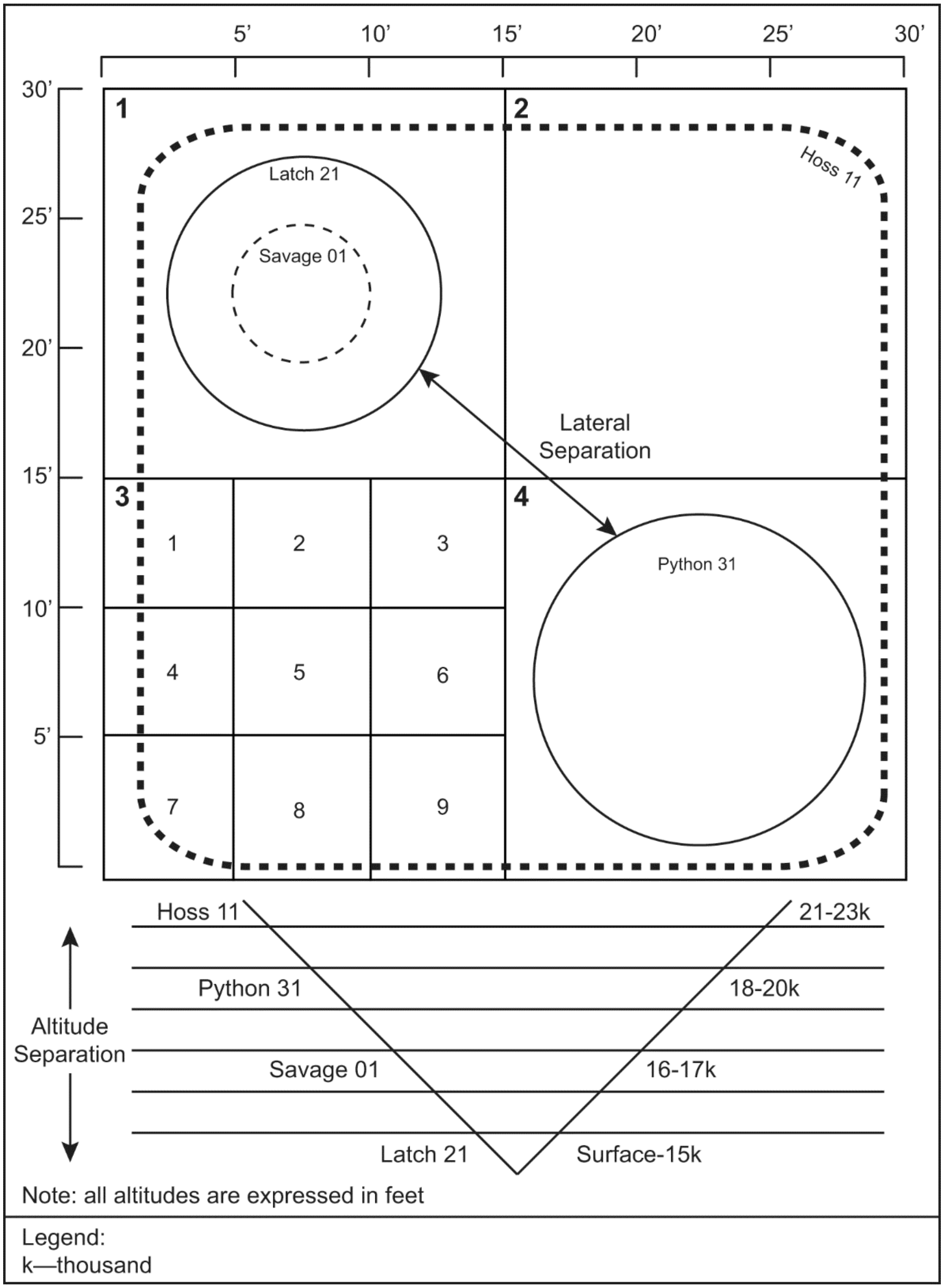


Figure 18. Deconfliction Example

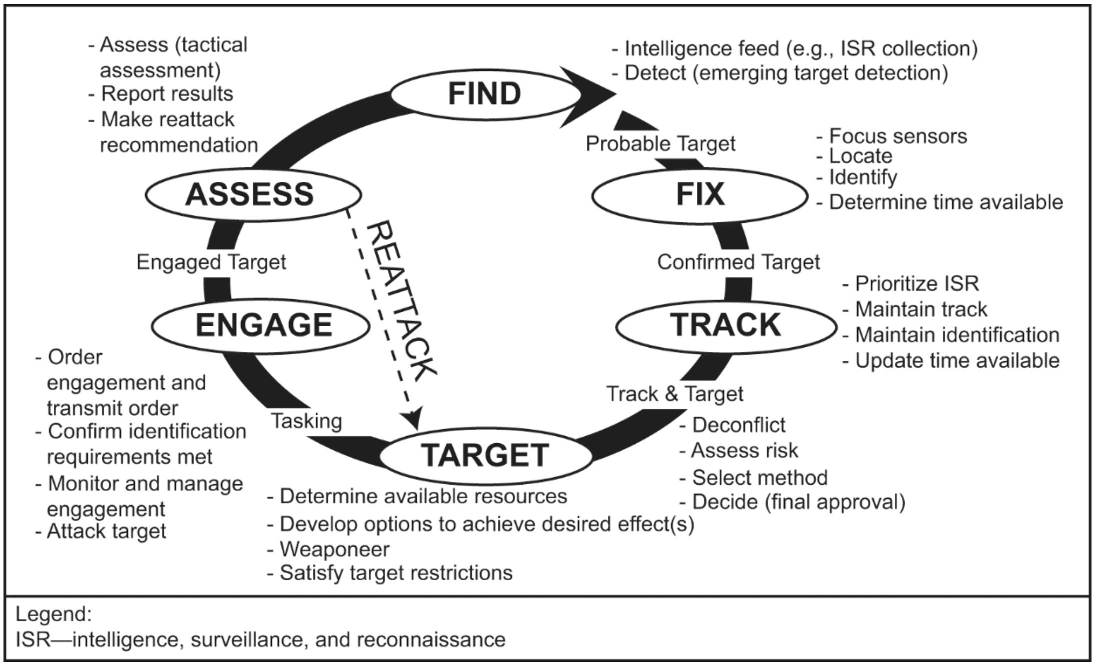


Figure 19. The F2T2EA Process

(2) The SCAR aircrew should focus on areas where the enemy may choose to conceal or fortify their position due to doctrinal or physical constraints. Identification of these focal points can be found during mission planning and during updates to the intelligence estimate. In the absence of known target locations, the SCAR aircrew should use all available assets to maximize sensor coverage. The SCAR aircrew should assign search responsibilities to assets through sensor POI, quadrants, keypads, geographic features, or LOCs.

(3) Once a target is found, the focus shifts to fixing. Fixing consists of identifying and prioritizing targets and requires organizing them. As targets are found, they need to be identified, plotted, annotated, and compared to where they fit on the prioritized target list. The SCAR aircrew should balance target engagement with target priorities, asset and ordnance availability, and remaining TOS. For example, a prioritized target list may contain artillery, tanks, and infantry. If these targets are all located in the assigned geographic area, the SCAR aircrew should focus the search to locate the highest priority targets. If the searches fail to acquire artillery targets, the SCAR aircrew may begin engaging lower priority targets. If there is a question regarding target priority, the SCAR aircrew must contact C2 for clarification, which may lengthen the kill chain.

(4) PID procedures and requirements are described in the theater ROE and SPINS. Achieving PID of a target allows the SCAR aircrew to match it against the prioritized target list.

(5) The next consideration is to refine target location to the accuracy required for strike coordination. Use of sensors, avionics or map plots can facilitate target location refinement. For visual or sensor talk-on, rough coordinates may suffice

and, if necessary, the SCAR aircrew can communicate the location via a C2 asset.

b. Decision to Engage (Track, Target).

(1) Track and Target Overview. The SCAR aircrew should use sufficient sensors to continuously track the target and maintain PID. Reasons to delay engagement and continue tracking include the need to: gather more or exploit current information, observe enemy activity, or to obtain guidance. Collateral damage considerations, inappropriate weapon-to-target match, or prosecuting higher priority targets also can cause engagement delays. The SCAR aircrew maintains situational awareness by recording the target location to revisit it later, assigning a strike asset to monitor it, or handing the track off to a C2 element. The SCAR aircrew is required to evaluate the target's window of vulnerability throughout the tracking process. Mobile targets may require immediate engagement. Destruction of all targets may not be required. Timely and accurate reporting of all targets will help build situational awareness and provide intelligence for further exploitation. The SCAR aircrew must be able to report suitable detail for every confirmed target. Collecting and sharing information while tracking targets may provide important intelligence on enemy activity or intentions. This intelligence can assist the supported commander in making decisions about enemy courses of action.

(2) Asset-to-Target Pairing. The SCAR aircrew should pair striker assets and weapons with targets so they will achieve the highest probability of success within the desired timeframe. This should be based on an understanding of specific striker asset and weapon capabilities and limitations in addition to the desired weapons effects for the target. Refer to the ATP 3-09.32, MCRP 3-31.6, NTTP 3-09.2, AFTTP 3-2.6, *MTTP for Joint Application of Firepower* for striker asset capabilities. Other considerations include:

(a) Striker asset situational awareness and location.

(b) FSCMs and ACMs.

(c) TOS.

(d) The target's window of vulnerability.

(e) Collateral damage.

(f) Assets and fires deconfliction in the target area.

(g) Required coordinate accuracy.

(3) Threat Considerations. The SCAR aircrew's situational awareness of threats will affect how strikers are tasked or packaged to create desired effects.

(a) Threat Relay. Threat information will be passed to the SCAR aircrew via C2. If a threat pops up within the assigned geographic area, the SCAR aircrew should retrograde all strike assets away from the threat. Once assets are clear of the threat, it is the SCAR aircrew's responsibility to determine if operations can continue. Perform the following when considering threats.

- Determine the extent of the threat's effect on the SCAR mission working area.

- Determine whether the threat can be isolated by altitude or lateral separation.

- Identify SEAD options.

- Notify the appropriate C2 element of the threat, or prohibitive interference, to inform incoming assets and, potentially, receive additional support assets.

(b) SEAD Execution. The SCAR aircrew should coordinate with C2 to request assets to conduct SEAD. Ultimately, the SCAR aircrew will determine if the SCAR mission should continue or be aborted. If the threat is affecting the mission the SCAR aircrew should consider neutralizing the threat prior to continuing. The SCAR aircrew should retain the flexibility to prioritize and target emerging threats; however, the prioritized target list should be followed to the maximum extent possible to achieve the supported commander's intent.

c. Attack (Engage).

(1) Target Engagement. There are three types of tasking the SCAR aircrew should use to facilitate rapid target engagement: INVESTIGATE, TARGET, and SMACK. (Refer to chapter I for specific definitions of INVESTIGATE, TARGET, and SMACK brevity terms.)

(a) INVESTIGATE. This is used to optimize sensors to gather target information such as location, PID, composition, and disposition. It is also used to gain situational awareness of the mission working area.

(b) TARGET. This is used to create effects on known or suspected targets, IAW the prioritized target list, when the striker asset can satisfy all requirements prior to target engagement.

(c) SMACK. This is used to create effects on known targets IAW the prioritized target list when the strike asset cannot satisfy all requirements prior to target engagement. SMACK authority must be delegated to the SCAR aircrew.

(2) Attack Briefs. Ways to pass a target location to a striker include data link, NAI, TAI, GEOREF, or coordinates. In all cases, the SCAR aircrew will confirm target information with the striking asset via an attack brief. The attack brief may be communicated via a DLER or 9-line CAS brief format. DLER is the preferred attack brief format during SCAR mission operations. Table 4 describes the DLER format in detail. Refer to JP 3-09.3, *Close Air Support*, for a description of the 9-line CAS brief format. The following is an example of an attack brief:

(a) Hawg 11 (the SCAR aircrew):"Savage 21, Hawg 11 advise when ready for TARGET tasking."

(b) Savage 21 (striker): "Savage 21 ready."

(c) Hawg 11: “Savage 21 TARGET, 1 main battle tank (MBT), 11S PA 5546 4682, elevation 785 ft, make attack south to north, remain west of alternate supply route (ASR) Atlantic.”

(d) Savage 21: “Savage 21.”

Note: During SCAR MTTP publication revision, a North Atlantic Treaty Organization (NATO) standardization agreement (STANAG) was in place, which prescribes the 9-line CAS brief format as the agreed upon attack brief format for passing target information during a SCAR mission supported by NATO assets. While use of the DLER attack brief format is preferred during SCAR mission operations, the 9-line CAS brief format may be required when operating with NATO supporting assets.

Table 4. DLER Attack Briefing Format

Information	Description
Description	May include target type, size, disposition, and other amplifying data.
Location	Format must be compatible with the receiving asset system. Options include Global Area Reference System, latitude and longitude coordinates, talk-ons, and BULLSEYE.
Elevation	Given in feet mean sea level.
Remarks/Restrictions	Include deconfliction measures, ordnance, final attack heading, and time on target (TOT) windows, as required. List restrictions.

(3) Coordinated Attacks. The SCAR aircrew may use coordinated attacks to integrate and synchronize fires and maximize lethal effects on a target set. These attacks include multiple flights of aircraft engaging a target using either combined or sectored attack methods with a firepower timing option. This method of employment is sometimes referred to as “massing fires”. The SCAR aircrew must approve using coordinated attacks. A striker may be assigned as the tactical lead of the attacks with all participating flight/sections. An aircrew given the tactical lead should ensure there is deconfliction between aircraft and recommend attack geometry and timing between flights/sections. A separate, common frequency may be used to conduct the coordinated attack if deconfliction with nonparticipating SCAR mission assets can be ensured. Table 5 provides additional information on coordinated attack types.

(a) Attack Method. The attack method describes how assets will be packaged to attack the intended target(s). Each attack method invokes a requirement for deconfliction measures, which the SCAR aircrew, or designated tac lead, will need to prescribe. The attack methods are as follows.

- Combined. In a combined attack, participants share the same avenue of attack. If participants cannot remain in VISUAL contact, a deconfliction measure is required since the participants share the same volume of airspace. An advantage of a combined attack is that it may be easier to maintain VISUAL with other participants, which may help synchronize fires. A disadvantage is that it is difficult to maintain deconfliction, particularly if using dissimilar striker aircraft.

• Sectored. In a sectored attack, participants are laterally deconflicted. Since participants do not share the same volume of airspace during their attack, they automatically achieve deconfliction. Sectors can be defined through any means available, such as GARS, visual ground references, or MGRS. An example of the SCAR aircrew establishing a sectored attack is as follows: “Hog 11 (striker) remain in 069SH quadrant 3, Apache 21 (striker) remain in 069SH quadrant 4.” An advantage of a sectored attack is that it is easier to maintain deconfliction with other participants. A disadvantage is that it is more difficult to gain and maintain VISUAL with participating aircraft, which may make it more difficult to synchronize fires.

Table 5. Coordinated Attacks

Type of Attack		Simultaneous	Sequential	Random
Combined	Same avenue of attack	Visual or timing Simultaneous time on target or time to target	Visual or timing (visual spacing or time separation)	Not normally used
Sectored	Acknowledged sector	Visual or timing Simultaneous time on target or time to target	Visual or timing (Visual spacing or time separation)	Free flow (must ensure strafe fan or bomb and missile fragment deconfliction)
Note: The joint standard for coordinated attack deconfliction refers to the avenue of approach.				

(b) Timing. Timing sets the pace of weapons effects on a target set. These effects can be employed simultaneously, sequentially, or at random. Choosing an option depends on which one will best achieve mission objectives based on threat composition, disposition, survivability, and mobility. Each timing option can use a visual- or time-based method for employing weapons. During visual-based weapons employment, participants conduct the attack based on a visual assessment of aircraft conducting the attack. During time-based weapons employment, participants conduct the attack based on a determined timeline of either time on target (TOT) or time to target (TTT).

• Simultaneous. Simultaneous attacks occur when all elements attack at the same time. Advantages include: more opportunities to mass fires; maximizing shock effect; and complicating the enemy’s air defense artillery targeting scheme. Disadvantages include: complicating target array sorting; and impact effects potentially creating obscuration; which can interfere with BDAs and follow-on attacks.

- Sequential. Sequential attacks require all elements to attack in a predetermined sequence. Advantages include: the target area is marked for subsequent attackers; there is continuous pressure on the target over time; attackers can reposition while other attackers shoot; there may be less weapons interference for subsequent shooters; redundant target engagement is minimized. Disadvantages include: air defenses have additional time to target airborne participants; the total time to accomplish the attack is increased; there is less shock effect; and it could provide opportunities to the enemy.

- Random. Random attack options allow elements to attack at will. Advantages include: it is easiest on an aircrew since no timing is required; it reduces C2 requirements; and is unpredictable. Disadvantages include: increased deconfliction complication with no guarantee of effects, it may deliver less pressure on the enemy, and the fire support plan may increase in complexity.

- Multiple Striker Asset Attacks. If the SCAR aircrew or tactical lead wants multiple striker assets to attack the same target array/area, the SCAR aircrew or tactical lead should assign a TOT to each asset. Assigning a TOT will ensure a time separation for each aircraft conducting attacks. The SCAR aircrew or tactical lead may assign an attack window to each striking asset that supports the type and number of attacks required. The size of the flight and ordnance expended will dictate the timing deconfliction. If deconflicting strike assets with significantly different ground speeds, more separation may be required. This prevents timing mistakes and target obscuration from reducing the striker asset's effectiveness. If slower tempos are acceptable, the SCAR aircrew or tactical lead may decide to employ striker assets one at a time with each element working until no ordnance remains.

(c) Mobile Target Considerations. If a target set is (or becomes) mobile, the SCAR aircrew must be able to adapt to the changing conditions. Mobile targets may require the SCAR aircrew to adjust the airspace allotted for the coordinated attack. The SCAR aircrew should anticipate these changes and coordinate additional airspace or alter the deconfliction plan. Should the SCAR aircrew need to alter the attack method, timing, or assets conducting a coordinated attack during an engagement, the SCAR aircrew may need to cease the coordinated attack prior to making the changes, to ensure safety of all participants.

(d) Egress. Upon attack completion, coordinated attack participants will egress along a route that minimizes threat exposure. During egress, the SCAR aircrew can assist aircraft egressing by pointing out factor threats and provide updated egress routing instructions, if necessary.

(4) SCAR Tasking Examples. The following examples illustrate the different taskings using the DLER format given by the SCAR aircrew. Figure 20 is an area map that supports the examples.

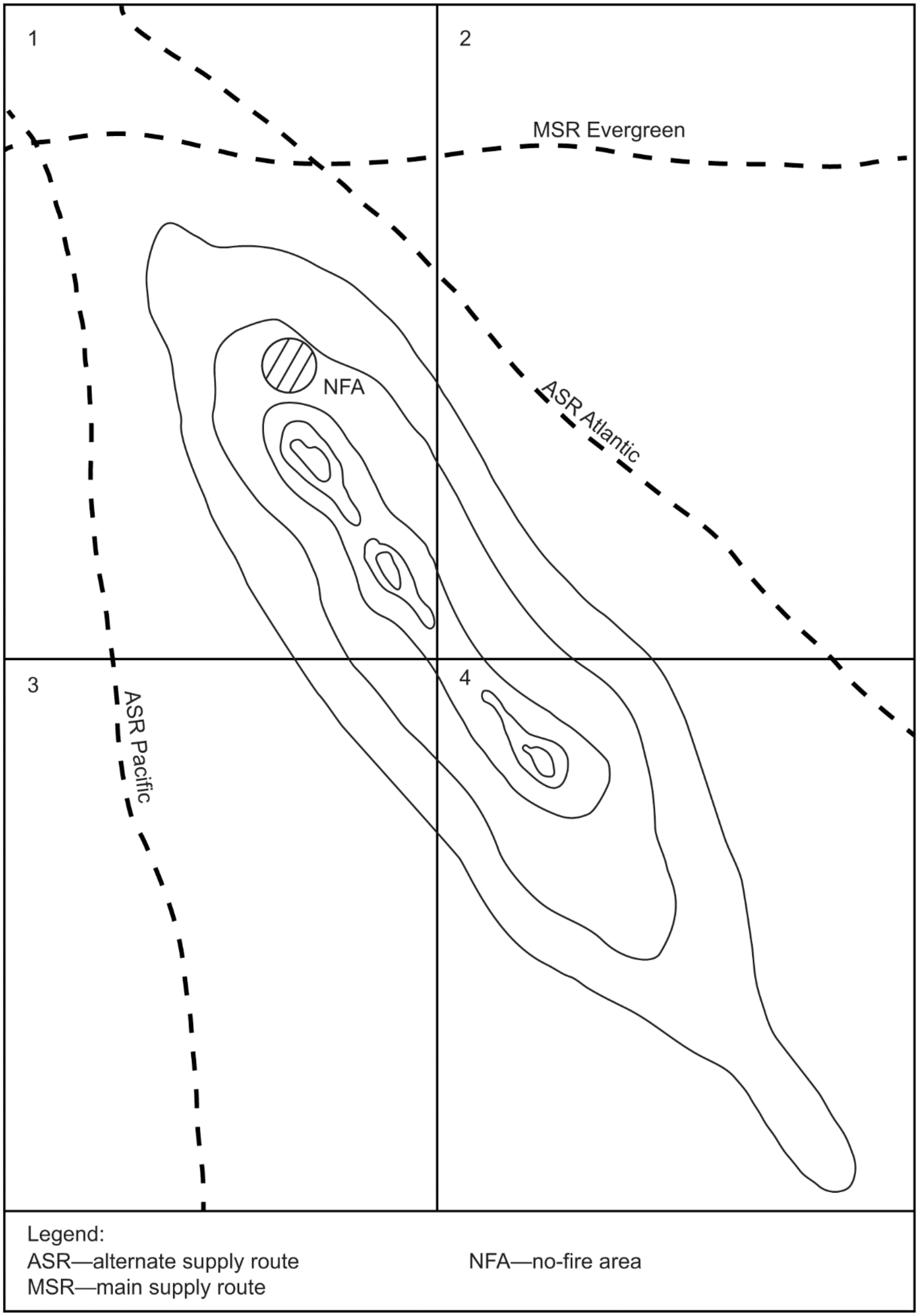


Figure 20. SCAR Execution Example

(a) INVESTIGATE.

• Example 1. "Reaper 21 (MQ-9), INVESTIGATE tasking to follow. INVESTIGATE TN 75001 in the vicinity of ASR Atlantic, suspected enemy armored column moving north to south." In this example, the engagement authority is not inherent in the INVESTIGATE tasking. This implies the strike asset will conduct a detailed route search and then report back.

Note: When a dissimilar formation of AH-64D/E and UA executing manned-unmanned teaming (MUM-T) check-in with the SCAR aircrew, maintain cooperative, two-way communication between the SCAR aircrew and AH-64D/E to ensure aircraft are deconflicted from the UA.

• Example 2. "Venom 11 (AH-1), INVESTIGATE tasking to follow. INVESTIGATE ASR Pacific for armored personnel carriers (APCs) and MBTs. If MBTs are found, you are approved to TARGET, make all attacks east to west." In this example, the SCAR aircrew has satisfied ROE on the MBTs with target correlation, PID, and CDE requirements remaining to decrease time to kill. The SCAR aircrew has not directed supporting assets to TARGET the APCs because they are ranked lower on the prioritized target list.

(b) TARGET.

• Example 1 (Single Strike Asset). "Razor 51 (AV-8B) TARGET tasking to follow. Target set is two MBTs and single BRDM-1, quadrant 1 in the vicinity of (IVO) main supply route (MSR) Evergreen/ASR Atlantic intersection, 11S QR 123 456, elevation 150 feet MSL. Remain within quadrant 1." No further coordination is required once target correlation and CDE have been satisfied. The SCAR aircrew has issued only those restrictions necessary to make the attacks successful.

• Example 2 (Single Strike Asset). "Hawg 22 (A-10), Knight 11 (MH-60S) TARGET tasking to follow. Target set is two Houdong patrol craft BULLSEYE 180, 20 miles, tracking north, fast." This example is in a maritime environment, where target location is provided by a fixed point Link 16 track or BULLSEYE. The target location also may be a bearing and range referenced off a maritime asset.

• Example 3 (Multiple Strike Assets, Sectored Sequential Coordinated Attack). "Salem 61 (F/A-18) and Venom 11 (AH-1), TARGET tasking to follow. Target set is four BMP-2s, two T-72s, one ZSU 23-4 traveling east to west on MSR Evergreen, 3 miles east of the Evergreen-Pacific intersection. Coordinated attack, sectored-sequential-timing, Salem followed by Venom. Salem 61, make attack east to west on ZSU 23-4, egress Charlie 4. Venom 11, make all attacks south to north on remaining targets, remain Charlie 4 and south until Salem 61's attacks are complete, TOT 50." In this example, target correlation and CDE must be satisfied prior to engagement. Venom 11 is the closest friendly asset. Due to the ZSU 23-4 threat, Hawg 22, the SCAR aircrew, issued complete attack briefs to integrate fires in the most expeditious manner. Consider switching attack assets to a discrete frequency for coordination during the coordinated attack as in the following: "Salem 61 (F/A-18) and Venom 11 (AH-1), TARGET tasking to follow. Target set is four BMP-2s, two T-72s, one ZSU 23-4 traveling west to east on MSR Evergreen, three miles east of the Evergreen-Pacific intersection, maintain quadrant 1, push Salem interflight for coordination, report when attack is complete."

• Example 4 (Multiple Strike Assets, Sectored Sequential Coordinated Attack). "Showtime 11 (F/A-18), Tigershark 21 (AH-64D/E, the SCAR aircrew) TARGET tasking to follow. Target array is four BMP-2s, two T-90s, one ZSU 23-4 stationary on MSR Evergreen, IVO Evergreen-Pacific intersection. Coordinated attack, sectored-sequential-timing, Redleg (S/S fires), Showtime, Tigershark. Redleg target is ZSU 23-4, GTL 020, Max Ord 10k', TOT 45. Showtime target is BMP-2s on rounds complete call, followed by Tigershark. Showtime stay above 10k', make attacks west to east. Tigershark 21 will attack south to north on the four BMP-2s." In this example, Tigershark 21, the CFF-capable SCAR aircrew, previously coordinated a CFF over a different net.

(c) SMACK.

• Example 1. "Latch 62 (F/A-18), Savage 81 (MQ-9) SMACK tasking to follow. Single piece of artillery, 11S QR 12345 67890, elevation 1,310 ft MSL." In this example, the SCAR aircrew satisfied PID, CDE, and ROE, and no further target correlation is required. Engagement authority has been granted and no further communication is required between the strikers and the SCAR aircrew. If the strike asset has more than one type of munition, a SMACK tasking could also specify a weapon, in the remarks section, to comply with CDE.

• Example 2. The Striker requires support. "Showtime 31 (F/A-18), Venom 21 (AH-1) SMACK tasking to follow. Single BMP-2 static on ASR Atlantic, 11S QR 123 456; elevation 1,350 ft MSL, request 1xGBU-54, my laser 1688, final attack heading 170 to 190, laser-to-target line (LTL) 150." In this example, the SCAR aircrew has decided to accomplish target correlation and guide or buddy-lase the weapon. The SCAR aircrew, as a RW aircraft, has completed all requirements to employ ordnance/fires for the target set, including passing a LTL and final attack headings to ensure deconfliction.

(d) S/S Fires Request. The SCAR aircrew requests S/S fire support through C2 using the DLER format: "Chalice (AWACS, tactical C2), Hawg 22 (A-10, the SCAR aircrew) fires request to follow. Request indirect fires (DLER format)." In this example, authorization for the S/S fires request rests with the appropriate C2 element. If authorized, tactical C2 will pass the GTL, and maximum or minimum ordinate altitude. The SCAR aircrew will deconflict assets from the GTL laterally or with altitude. C2 will advise when the firing agency is ready and will need verification that the SCAR aircrew's airspace is deconflicted. After the SCAR aircrew confirms deconfliction, the SCAR aircrew will command execution of the S/S fire mission via tactical C2. The SCAR aircrew will pass when desired effects are met or may request additional rounds.

(5) Target/Track Correlation. There are varying methods for target/track correlation, depending on the tactical situation. This can be as simple as identifying a valid target, per the prioritized target list, within a specified geographic area or as complex as a visual or sensor talk-on with the SCAR aircrew.

(a) The most expeditious way to begin the visual or sensor talk-on is to provide a mark or common reference point. Some examples include laser handoff, lethal strike, GEOREF, and IR marker. Some assets can correlate via data link, negating the need for additional target/track correlation.

(b) Marks can be provided by target designators/markers, A/S fires, or S/S fires. There is no need to mark a target if target correlation is complete or not required. Marks should be planned to include sufficient time before weapons employment to ensure attacking aircraft acquire and correlate the targets.

(6) IAMs. Based on the platform and ordnance, the SCAR aircrew may have the most suitable platform to generate coordinates. The process should be completed for as many targets as time allows. Based on the desired effect, a varying level of coordinate accuracy may be required. If the strike assets can self-target with IAMs, the SCAR aircrew may be able to bound a large target set with one or two target coordinates and a description of the target set. If the strike assets cannot self-target, the SCAR aircrew may be required to provide coordinates for each desired point of impact in the target set. The SCAR aircrew should be aware that some platforms can pattern manage, which could be useful for a closely arrayed target set.

(7) LGWs. Conditions may require the SCAR aircrew to provide a laser target marker for the striker. If this occurs, the SCAR aircrew should be aware of factors involved in such attacks (e.g., proper geometry required, timing, communication, laser codes, and deconfliction). Refer to the ATP 3-09.32, MCRP 3-31.6, NTTP 3-09.2, AFTTP 3-2.6, *MTTP for Joint Application of Firepower,* for laser geometry and communications considerations.

d. Assess. Assessment begins when the weapon impacts the target. The assessment is situation dependent and may be conducted by the SCAR aircrew, a strike asset, or a reconnaissance asset. SCAR mission feedback is essential to optimize subsequent tasking and update intelligence. Unlike other types of mission flow, SCAR mission feedback needs to be dynamic and continuous to ensure continuous pressure on the enemy.

(1) Striker Responsibilities. After a striker engages a target, the striker will conduct a BDA to determine if additional attacks are required to create desired effects, per the attack brief. The striker will report the results to the SCAR aircrew at completion. After checking out with the SCAR aircrew, the striker should

provide an INFLTREP to C2 elements IAW the SPINS. (See table 6 for an INFLTREP format.) A BDA consists of phase 1, phase 2, and phase 3. Post mission, assets will report phase 1 BDA to the C2 element; they should not assume the target has been destroyed. The aircrew must use judgment and precision when reporting a BDA.

(a) Phase 1: Physical Damage Assessment. A physical damage assessment is an estimate of the quantitative extent of physical damage to a target element based on observed or interpreted damage. (Damage is done through munitions blast, fragmentation, and fire damage effects.) This is the aircrew's initial physical damage assessment.

(b) Phase 2: Functional Damage Assessment. The functional damage assessment is an estimate of the effect of military force to degrade or destroy the functional or operational capability of a target to perform its intended mission. At higher echelons, BDA analysts need to compare the original objective with the target's current status to determine if the attack met the objective.

(c) Phase 3: Target System Assessment. The target system assessment is an estimate of the overall impact of the military force's effectiveness applied against an adversary's target system relative to the established operational objectives.

Table 6. INFLTREP Format
Aircrew transmits: "________________, this is___________________, INFLTREP, over." (Addressee) (Aircrew call sign) ***(Authentication requested here, as required)*** Line One/Call Sign ____ Line Two/Mission Number___________________________________ Line Three/Location____ (Latitude/longitude, universal transverse Mercator grid, military grid reference system, place name) Line Four/Time-on-Target___________________________________ Line Five/Results____ Remarks____ (Target area weather, significant sightings, essential elements of information)

(2) SCAR Assessment Responsibilities. The SCAR aircrew will determine if additional attacks are necessary based on striker BDA. If additional attacks are

required, the SCAR aircrew should task on-station strikers or request additional strike assets from C2. The SCAR aircrew is responsible for informing C2 of remaining targets in the SCAR mission working area.

(3) S/S Fires Assessment. The SCAR aircrew will assess the effects of S/S fires engagement. If the fires' impact location should be adjusted, the SCAR aircrew may determine the impact location through the SCAR aircrew's available sensors and transmit the location to C2. The fires element will adjust fires as required.

(4) Periodic Situation Updates. The SCAR aircrew is responsible for keeping C2 informed of the SCAR aircrew's working area status. This can be accomplished by providing periodic situation updates. The situation update can be passed via plain language or by using the situation update format depicted in table 2. These situation updates should only include critical updates that arose during the SCAR mission (e.g., surface-to-air threat, mission prohibitive weather, or location of large elements of enemy forces). If required, the SCAR aircrew may pass situation updates to a wingman or strike asset, who will forward it to C2. It is important to report priority targets remaining, since remaining priority targets will likely influence C2's decision to re-task aircraft to engage those targets.

(5) C2 Responsibilities. Tactical C2 is the primary link between the INFLTREP provided by the SCAR aircrew or striker and the JAOC. In the event of a gap in SCAR aircrew coverage, airborne C2 can accept a battlefield handover from the outgoing SCAR aircrew and provide a situation update to the incoming SCAR aircrew. This facilitates situational awareness on the battlefield and reduces redundant target engagement.

7. Battlefield Handover

a. The SCAR aircrew will execute a battlefield handover IAW table 2 when transferring the role to another SCAR aircrew. Considerations for selecting a replacement SCAR aircrew should include situational awareness, TOS, and available sensors and communications capability. If a replacement SCAR aircrew is unavailable, the SCAR aircrew will execute a battlefield handover with C2. Whenever possible, accomplish handovers on lower-density communications networks that do not detract from ongoing operations.

b. The outgoing SCAR should tailor the battlefield handover to the situation. If passing the SCAR aircrew role to an aircrew that has been on station for a significant period of time, with sufficient situational awareness, the battlefield handover should be brief. However, if the handover is passed to an inbound SCAR aircrew that has yet to build sufficient situational awareness, the battlefield handover should be robust. The incoming SCAR aircrew should avoid slowing the tactical tempo to build situational awareness or conduct their own attacks as the strike assets hold idle.

Appendix A
STRIKE COORDINATION AND RECONNAISSANCE (SCAR) MISSION PLANNING GUIDE

1. Overview

a. Review the commander's intent and acceptable level of risk (ALR).

b. Review rules of engagement (ROE), special instructions (SPINS), the collateral damage limitations for the area of operations (AO) or target area, air operations directive (AOD), and the supported commander's prioritized target list.

c. Review the airspace control order (ACO) for kill boxes, fire support coordination measures (FSCMs), airspace coordinating measures (ACMs), and restricted operating zones (ROZs).

2. Friendly Situation

a. Scrub the air tasking order (ATO) for mission-pertinent data to include determining:

(1) The SCAR mission working area.

(2) The SCAR, strikers, and supporting assets.

(3) Reconnaissance assets.

(4) Offensive counterair (OCA) and defensive counterair (DCA) assets.

(5) Electronic warfare (EW) assets.

(6) Suppression of enemy air defense (SEAD) assets.

(7) Refueling assets.

(8) The battlefield airborne communications node.

b. Contact the ground liaison officer (GLO) for the following information.

(a) The current friendly disposition.

(b) FSCMs/ACMs.

(c) The ground scheme of maneuver.

(d) Indirect fire assets.

c. Determine command and control communication requirements.

3. Communications

a. Check in and out and inflight report procedures (i.e., agency and frequency).

b. SCAR working frequencies.

c. Data link considerations.

4. Intelligence Update

a. Review suspected enemy locations and identify points of interest (POIs).

b. Review reconnaissance guides for the enemy's order of battle.

c. Assess the enemy's air and missile orders of battle.

d. Review the enemy's most likely and most dangerous courses of action.

5. Weaponeering

a. Review the desired weapons effects to meet the commander's intent.

b. Identify potential targets and select appropriate weapons.

c. Develop a load plan or select a standard conventional load (SCL).

d. Plan valid deliveries for ordnance.

6. Reconnaissance

Develop a search profile considering the following.

a. Target priorities.

b. Expected target locations.

c. The threat.

d. Predicted sensor detection ranges.

7. Target Area Tactics

a. Study target area imagery.

b. Assess the impact of the following on target area tactics.

(1) Sensor capabilities and limitations.

(2) Threat.

(3) Weather.

(4) Terrain.

c. Develop a holding/deconfliction plan based on ATO-assigned airspace and assets.

d. Consider optimum attack axes based on expected target orientations and environmental influences on target acquisition.

e. Prepare attack brief templates for all striker assets.

f. Develop a fires integration plan.

8. Navigation Plan

a. Waypoints.

(1) Ingress and egress routing.

(2) Sensor POIs, named area of interest (NAI), and target area of interest (TAI).

(3) BULLSEYE.

(4) Initial points (IPs).

(5) Holding Points.

b. Overlays.

c. Communicate routing requirements/plan to fuels planner.

9. Go/No-Go Criteria

a. Aircraft system requirements.

b. Weather.

c. Supporting assets.

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Appendix B
STRIKE COORDINATION AND RECONNAISSANCE (SCAR) MISSION BRIEFING GUIDE

1. Mission Objectives

a. Commander's intent/desired effects in the SCAR mission working area.

b. Responsibilities.

(1) Navigation.

(2) Fires and airspace Deconfliction.

(3) Check-in briefs.

(4) Stack management.

c. Systems and sensors.

(1) Visual lookout.

(2) Targeting pod.

(3) Radar.

2. Threat Counter-Tactics

a. Surface-to-air.

(1) Surface-to-air missile (SAM).

(2) Air defense artillery.

b. Air-to-air.

3. Air Tasking Order (ATO)/Schedule Overview

a. Supporting Air Assets.

(1) Event/mission number.

(2) Call sign.

(3) Load out.

(4) Time on station.

(5) Capabilities and limitations.

b. Surface-to-surface Fires Assets.

(1) Unit.

(2) Location.

(3) Communication plan.

(4) Capabilities.

c. Command, Control, and Communications Specifics.

(1) Requirements.

(2) Controlling agencies.

(3) SCAR common.

(4) Primary and alternate frequencies.

4. Working Area/Map/Imagery Study

a. Global Area Reference System (GARS)/common grid reference system (CGRS).

b. SCAR working area/named area of interest (NAI)/target area of interest (TAI).

c. Fire support coordination measure (FSCM)/airspace coordinating measure (ACM).

d. Datum shifts.

5. Cockpit Management

a. Maps and smart packs.

b. Personal equipment, night vision devices (NVDs), binoculars.

c. Storage and accessibility plan.

6. Ingress Routing

a. Route.

b. Formation/altitude/airspeed.

c. Aircraft lighting.

d. Timing.

e. Fuel.

f. Sensor/navigation update plan.

g. Command, control, and communication requirements.

h. Threat updates.

i. Coordination.

j. Deconfliction.

k. Reactions to threats.

l. Emissions control.

7. SCAR Plan

a. Overview.

(1) Assigned SCAR working area.

(2) Geographic orientation.

(3) Target priorities.

(4) Expected target locations.

(5) Applicable commander’s critical information requirements (CCIRs).

b. Reconnaissance profile.

(1) System Preparation.

(a) Waypoints.

(b) Sensors.

(2) Sanitation pass.

(3) Search.

(a) Formation/altitude/airspeed.

(b) Aircraft lighting.

(c) Lead/wingman contracts.

(d) Visual/NVDs/binoculars.

(e) Sensor plan.

(f) Sensor predictions.

c. Targeting.

d. Rules of engagement (ROE), positive identification (PID), and collateral damage estimate (CDE).

e. Passing/entering coordinates.

f. Sorting.

g. Talk-on/marks (laser codes).

8. SCAR Coordination

a. SCAR Asset Check-in.

(1) Call sign.

(2) Mission number.

(3) Number/type.

(4) Position.

(5) Ordnance (laser code).

(6) Time on station (TOS).

(7) Type sensors.

(8) Abort.

b. Situation Update Brief.

(1) Threat.

(2) Enemy situation.

(3) Friendly situation.

(4) Artillery.

(5) Coordination.

(6) Hazards.

(7) Remarks/restrictions.

c. Reconnaissance.

(1) Geographic Area Allocation.

(a) Grid/Terrain Feature.

(b) Vertical.

(c) Timing.

(d) Deconfliction.

(2) Priority Targets.

(a) Priority.

(b) Expected target locations.

(3) Applicable CCIRs.

d. Target Engagement.

(1) Holding.

(a) Location.

(b) Orientation.

(c) Altitude.

(2) Deconfliction.

(a) Lateral/geographic.

(b) Altitude.

(c) Timing.

(3) Targeting plan.

(4) Marks/laser codes.

9. SCAR Tasking

a. Target attack brief.

b. Attack geometry.

c. Attack/Weapons Selection.

(1) Planned weapon-to-target pairing.

(2) Reactive weapon-to-target pairing.

d. Weapons check.

e. Formation/altitude/airspeed.

f. Sensor plan.

(1) Visual/NVDs.

(2) Laser spot tracker.

(3) Electro-optical/Infrared (EO/IR) sensor.

(4) Radar.

g. Delivery mode (primary/alternate).

h. Moving target considerations.

i. Expendable use.

j. Communication cadence.

k. Reaction to threats.

l. Abort criteria.

(1) ROE, PID, CDE.

(2) CAPTURE/SPOT.

(3) Expendables BINGO.

m. Off-target.

(1) Geometry and flow.

(2) Expendable use.

(3) Communication cadence.

(4) Deconfliction.

(5) Shooter-cover role-swap.

(6) Sensor plan.

(7) Visual/NVDs.

n. Reattack criteria/plan.

o. Battle damage assessment (BDA).

10. Battlefield Handover

a. Threat.

b. Enemy situation.

c. Friendly situation.

d. Artillery.

e. Coordination.

f. Hazards.

g. Remarks/restrictions.

11. Egress

a. Flow/rejoin.

b. Route/deconfliction.

c. Formation/altitude/airspeed

d. Aircraft lighting.

e. Timing.

f. Fuel.

g. Command, control, and communication requirements.

h. Inflight report.

i. Coordination.

j. Reaction to threats.

k. Emissions control.

12. Contingencies

a. DEADEYE.

b. Asset fallout.

c. Separated wingman.

d. Weather.

e. Battle damage.

f. Hung ordnance.

g. Personnel recovery.

h. Contested, degraded, operationally limited operations

Appendix C
STRIKE COORDINATION AND RECONNAISSANCE (SCAR) PRODUCT EXAMPLES

SCAR aircrews may use Figure 21, A SCAR Deconfliction Card Example, and Table 7, SCAR Attack Brief (description, location, elevation, remarks/restrictions (DLER) Format) to assist in mission execution.

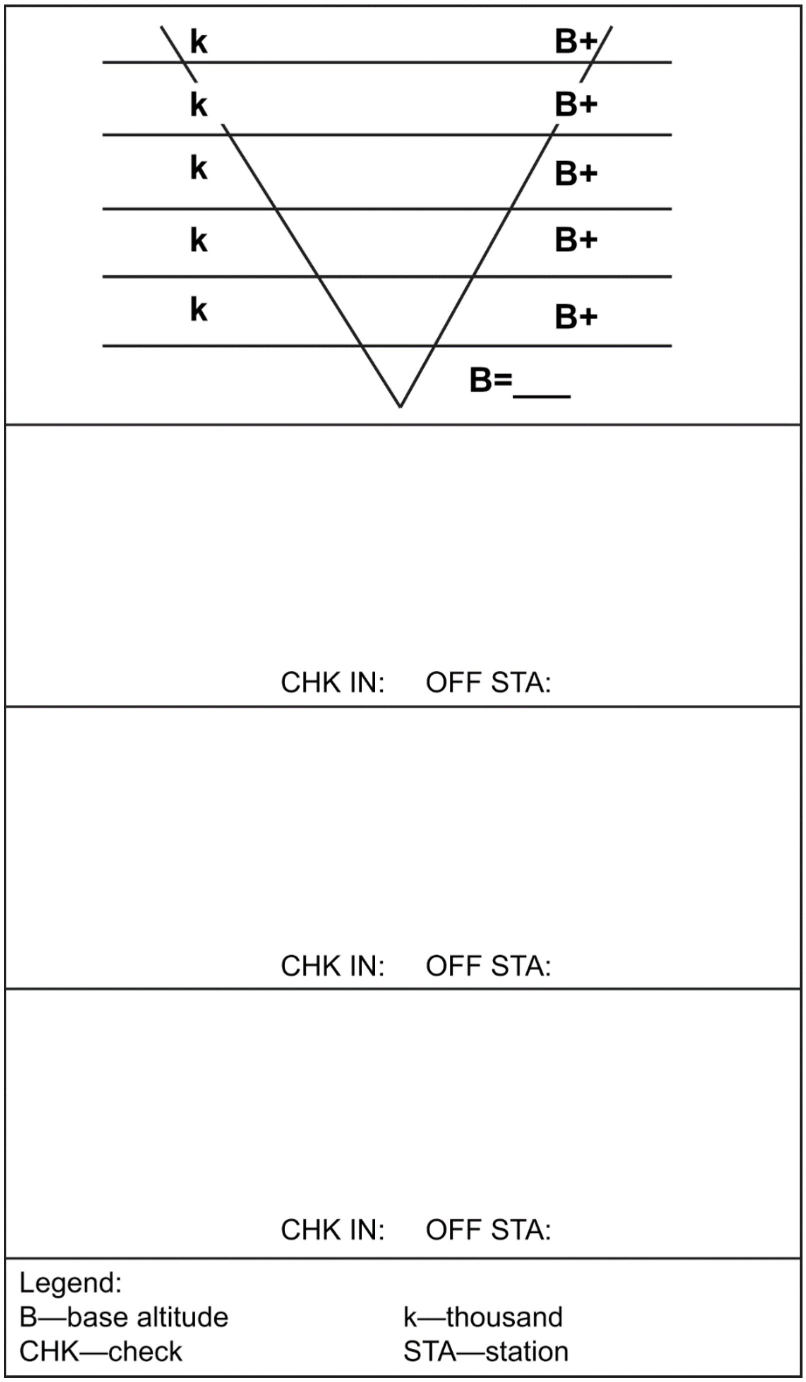

Figure 21. SCAR Deconfliction Card Example

Table 7. SCAR Attack Brief (DLER Format)						
MKPT	DESCRIPTION	LOCATION	ELEVATION	REMARKS/ RESTRICTIONS	PRIORITY	ASSIGNED AREA
Legend: DLER—description, location, elevation, remarks/restrictions MKPT—mark point						

Appendix D
COORDINATION MEASURES

Airspace coordinating measures (ACMs) are employed to effectively use airspace for mission accomplishment. Fire support coordination measures (FSCMs) facilitate engaging targets rapidly or providing safeguards for friendly forces. Maritime defense measures (MDMs) facilitate maritime offensive and defensive actions. Strike coordination and reconnaissance (SCAR) and striker assets should have a detailed understanding of which ACMs and FSCMs are in effect and the implication of those measures on the SCAR mission. Table 8 lists some ACMs, FSCMs, and MDMs commonly used during SCAR..

Table 8. ACMs, FSCMs, and MDMs

Measure	Definition/Description	Planning Considerations
Airspace Coordination Area (ACA)	A three-dimensional block of airspace in a target area, established by the appropriate commander, in which friendly aircraft are reasonably safe from friendly surface fires. (Joint Publication (JP) 3-52, *Joint Airspace Control*)	An ACA is used, primarily, in situations with high-volume fire. Friendly aircraft are reasonably safe from friendly surface fires when artillery, helicopters, and fixed winged aircraft are given specific lateral or vertical airspace within which to operate. The deconfliction burden rests with, and the ACA is established by, the ground commander.
Amphibious Objective Area (AOA)	A geographical area (delineated for command and control (C2) purposes in the initiating directive) within which is located the objective(s) to be secured by the amphibious force. This area must be of sufficient size to ensure accomplishment of the amphibious force's mission and must provide a sufficient area for conducting necessary sea, air, and land operations. (JP 3-52)	Air control procedures, within an AOA, are almost identical to high-density airspace control zone (HIDACZ) procedures. Aircraft not organic to the amphibious force must coordinate entries and exits. Deconflicting aircraft and operations within the AOA from aircraft and operations adjacent to the AOA requires continuous, active involvement by affected commanders and staffs.
Battlefield Coordination Line (BCL)	A measure established by the Marine air-ground task force (MAGTF) that facilitates the expeditious attack of surface targets of opportunity between the BCL and fire support coordination line (FSCL). When established, the primary purpose is to allow MAGTF aviation to attack surface targets without further approval from the ground combat element (GCE) commander in whose area the targets are located. (Marine Corps Tactics Publication (MCTP) 3-10F, *Fire Support Coordination in the Ground Combat Element*)	An ACA will always overlie the area between the BCL and FSCL.

Table 8. ACMs, FSCMs, and MDMs (Cont'd.)		
Measure	**Definition/Description**	**Planning Considerations**
Combat Air Patrol (CAP)	An aircraft patrol provided over an objective area, the force protected, the critical area of a combat zone, or in an air defense area, for the purpose of intercepting and destroying hostile aircraft before they reach their targets. (JP 3-01, *Countering Air And Missile Threats*)	A defensive counterair activity conducted in support of air operations.
Coordinated Fire Line (CFL)	A line beyond which conventional surface-to-surface (S/S) direct fire and indirect fire support means may fire at any time within the boundaries of the establishing headquarters without additional coordination (JP 3-09, *Joint Fire Support*).The purpose of the CFL is to expedite S/S target attacks beyond the CFL without coordination with the ground commander in whose area the targets are located.	Airspace must still be coordinated.
Coordinating Altitude (CA)	An airspace coordinating measure that uses altitude to separate users and as the transition between different airspace control elements. (JP 3-52)	The airspace coordinating entities should be included in the airspace control plan (ACP) and promulgated in the airspace control order (ACO). Army echelons incorporate ACP guidance and integrate the ACO, area air defense plan (AADP), special instructions (SPINS), and air tasking order (ATO) via operations orders. All airspace users should coordinate with the appropriate airspace coordinating entities when transitioning or firing through the CA.
Coordination Level (CL)	A procedural method to separate fixed- and rotary-wing aircraft by determining an altitude below which fixed-wing aircraft normally will not fly. (JP 3-52)	
Fleet Air Defense Identification Zone (FADIZ) (United States Message Text Format [USMTF] refers to this as a positive identification radar advisory zone)	An FADIZ is a specified area established for identification and flight following of aircraft in the vicinity of a fleet-defended area. (JP 3-52)	An FADIZ provides tracking, control, and assistance to friendly aircraft within the antiair coverage of the battle group. It may be used during SCAR operations in a maritime environment.

Table 8. ACMs, FSCMs, and MDMs (Cont'd.)		
Measure	**Definition/Description**	**Planning Considerations**
Fire Support Coordination Line (FSCL)	A fire support coordination measure established by the land or amphibious force commander to support common objectives within an area of operation; beyond which all fires must be coordinated with affected commanders prior to engagement, and short of the line, all fires must be coordinated with the establishing commander prior to engagement. (JP 3-09)	Supporting elements attacking targets beyond the FSCL must ensure the attack will not produce adverse effects short of the line. Short of the FSCL, all air-to-ground and S/S attack operations are controlled by the responsible land or amphibious force commander. The JFLCC and JFMCC cannot employ S/S fires long of the FSCL without coordination with affected commanders, and the JFACC cannot employ fires short of the FSCL without coordination with the JFLCC or JFMCC. The FSCL is a significant consideration during interdiction operations. The FSCL is, primarily, used to establish C2 procedures for planning and execution purposes. It does not define mission types. Interdiction can occur short of, and beyond, the FSCL. Attacks on surface targets short of the FSCL, while conducting joint interdiction operations, must be controlled by (or coordinated with) the responsible land or amphibious force commander.
High-Density Airspace Control Zone (HIDACZ)	Airspace designated in ACP or ACO in which there is a concentrated employment of numerous and varied weapons and airspace users. A HIDACZ has defined dimensions, which usually coincide with geographical features or navigational aids. Access to a HIDACZ is normally controlled by the maneuver commander. The maneuver commander can also direct a more restrictive weapons status within the HIDACZ. (JP 3-52)	If SCAR operations are to be conducted in an HIDACZ, detailed coordination will be required between the SCAR aircrew and the HIDACZ control authority.
Joint Special Operations Area (JSOA)	An area of land, sea, and airspace assigned by a joint force commander, to the commander of a joint special operations force, to conduct special operations activities. (JP 3-52)	The JSOA may be limited in size to accommodate a discrete direct-action mission or may be extensive enough to allow a continuing, broad range of unconventional warfare operations. If SCAR operations are to be conducted in a JSOA, detailed coordination will be required between the SCAR aircrew and the JSOA control authority.

Table 8. ACMs, FSCMs, and MDMs (Cont'd.)		
Measure	**Definition/Description**	**Planning Considerations**
Kill Box	A kill box is a three-dimensional permissive fire support coordination measure with an associated airspace coordinating measure used to facilitate the integration of fires and the coordination of the airspace within. A kill box is a measure, not a mission. (ATP 3-09.34, MCRP 3-31.4, NTTP 3-09.2.1, AFTTP 3-2.59, *Multi-Service Tactics, Techniques, and Procedures for Kill Box Planning and Employment*.)	Blue kill boxes facilitate air-to-surface (A/S) fires. During SCAR mission planning and execution, the SCAR aircrew may elect to clear S/S fires into a blue kill box, assuming proper deconfliction and coordination have been conducted. Purple kill boxes allow A/S and S/S fires to be conducted without coordination between A/S and S/S assets. To protect friendly aircraft operating in a purple kill box, the measure must have a floor. Purple kill boxes will restrict a SCAR mission's airborne assets to descend below the purple kill box floor without prior deconfliction and coordination.
Minimum-Risk Route (MRR)	A temporary corridor of defined dimensions recommended for use by high-speed, fixed-wing aircraft that presents the minimum known hazards to low-flying aircraft transiting the combat zone. (JP 3-52)	MRRs are used primarily for operations that cross the forward line of own troops. MRRs are established based on known threats.
No-Fire Area (NFA)	An area designated by the appropriate commander into which fires or their effects are prohibited. (JP 3-09.3)	There are two exceptions to NFA fires prohibition, they are: (1) When the establishing headquarters approves joint fires within the NFA on a mission-by-mission basis. (2) When an enemy force, within the NFA, engages a friendly force and the engaged commander determines there is a requirement for immediate protection and responds with the minimal force needed to defend the force.
Restrictive Fire Area (RFA)	An area where specific restrictions are imposed and into which fires that exceed those restrictions will not be delivered without coordination with the establishing headquarters. (JP 3-09)	An RFA is a restrictive measure. The purpose of an RFA is to regulate joint fires into an area according to the stated restrictions set by the establishing headquarters.

Table 8. ACMs, FSCMs, and MDMs (Cont'd.)		
Measure	**Definition/Description**	**Planning Considerations**
Restrictive Fire Line (RFL)	A line established between converging friendly surface forces that prohibits fires or their effects across that line. (JP 3-09)	An RFL is a restrictive measure. The purpose of an RFL is to prevent friendly fire incidents and duplication of engagements by converging friendly forces.
Restricted Operations Zone (ROZ)	Airspace reserved for specific activities in which the operations of one or more airspace user is restricted. (JP 3-52)	These are commonly used for: drop zones, landing zones, search and rescue areas, unmanned aircraft (UA) launch and recovery sites, UA mission areas, S/S missile launch sites, missile flight paths (if necessary), predicted missile munitions impact locations, and special electronics mission aircraft. A ROZ can be used to establish the boundaries and airspace of a SCAR mission working area. A preplanned ROZ is published in the ACP. Requests for activation of ROZs are to be made to the airspace control authority. A ROZ may be established to support personnel recovery.
Transit Corridor (TC)	(North Atlantic Treaty Organization (NATO)) A bi-directional corridor established in the rear area to route aircraft through air defenses, in the rear area, where appropriate, with minimum risk. (Allied joint publication 3.3.5, *Allied joint doctrine for airspace control*)	Established to route aircraft through air defenses, in the rear area, where appropriate, with minimum risk. Pre-planned TCs will be published in ACPs, as will their horizontal and vertical dimensions. Air traffic services not provided.
Transit Route (TR)	(NATO) In air operations, a temporary air corridor of defined dimensions established in the forward area to minimize the risks to friendly aircraft from friendly air defenses or surface forces. (Allied Administrative Publication (AAP) 06, *NATO Glossary of Terms and Definitions*)	TRs may link with TCs or appear independently. Horizontal and vertical dimensions of TRs will be published in the ACP. Requests for activation of TRs are to be submitted to the airspace control authority. Activated TRs will be published in the ACO.
Unmanned Aircraft Area	Airspace of defined dimensions is created specifically for unmanned aircraft operations. Generally, this airspace will consist of the area in which unmanned aircraft missions are conducted, not en route airspace. (JP 3-52)	

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GLOSSARY

PART I – ABBREVIATIONS AND ACRONYMS

A

A/S	air-to-surface
AAGS	Army air-ground system
ABCT	armored brigade combat team
ACA	airspace coordination area
ACE	aviation combat element (USMC)
ACM	airspace coordinating measure
ACO	airspace control order
ACP	airspace control plan
ACU	aircraft control unit
ADAM	air defense airspace management
ADC	air defense commander
AE	airspace element
AEZ	air engagement zone
AFATDS	Advanced Field Artillery Tactical Data System
AFB	Air Force base
AFTTP	Air Force tactics, techniques, and procedures
AGL	above ground level
AI	air interdiction
AIC	air intercept control
AIR-MAR	maritime air
ALR	acceptable level of risk
ALSA	Air Land Sea Application Center
AMD	air and missile defense
AO	area of operations
AOA	amphibious objective area
AOD	air operations directive
APC	armored personnel carrier
AR	armed reconnaissance
ARG	amphibious ready group

ARM	antiradiation missile
ASOC	air support operations center
ASR	alternate supply route
ASWC	antisubmarine warfare commander
ATO	air tasking order
ATP	Army tactical publication
AWACS	Airborne Warning and Control System

B

BAE	brigade aviation element
BCD	battlefield coordination detachment
BCL	battlefield coordination line
BCT	brigade combat team
BDA	battle damage assessment
BKB	blue kill box
BN	battalion

C

C2	command and control
CA	coordinating altitude
CAS	close air support
CATF	commander, amphibious task force
CCIR	commander's critical information requirement
CDE	collateral damage estimation
CFF	call for fire
CFL	coordinated fire line
CGRS	common geographic reference system
CIEA	classification, identification, and engagement area
COD	combat operations division
COI	contact of interest
COIC	current operations integration cell
COMAFFOR	commander, Air Force forces
CONOPS	concept of operations
COP	common operational picture

CP	command post
CPD	combat plans division
CRC	control and reporting center
CSG	carrier strike group
CWC	composite warfare commander

D

DASC	direct air support center
DC, CD&I	Deputy Commandant, Combat Development and Integration
DCA	defensive counterair
DEZ	dual engagement zone
DLER	description, location, elevation, remarks/restrictions

E

EO	electro-optical
EW	electronic warfare

F

F2T2EA	find, fix, track, target, engage, and assess
FA	field artillery
FADIZ	fleet air defense identification zone
FC	fires cell (Army)
FL	flight level
FLOT	forward line of own troops
FM	field manual (Army)
FSCC	fire support coordination center
FSCL	fire support coordination line
FSCM	fire support coordination measure
ft	foot
FW	fixed-wing

G

GARS	Global Area Reference System
GBU	guided bomb unit
GCE	ground combat element
GEOREF	geographic reference

GLD	ground liaison detachment
GLO	ground liaison officer
GTL	gun-target line

H

HIDACZ	high-density airspace control zone
HIMARS	High Mobility Artillery Rocket System
HQ	headquarters
HQMC	Headquarters, Marine Corps

I

IADS	integrated air defense system
IAM	inertially aided munition
IAW	in accordance with
IDF	indirect fire
INFLTREP	inflight report
IR	infrared
ISR	intelligence, surveillance, and reconnaissance
IVO	in vicinity of

J

JAAT	joint air attack team
JACE	joint air coordination element
JAGIC	Joint Air-Ground Integration Center
JAOC	joint air operations center
JFACC	joint force air component commander
JFC	joint force commander
JFE	joint fires element
JFLCC	joint force land component commander
JFMCC	joint force maritime component commander
JFSOCC	joint forces special operations component commander
JIPTL	joint integrated prioritized target list
JOA	joint operations area
JOC	joint operations center
JP	joint publication

JSOA	joint special operations area
JSOACC	joint special operations air component commander
JSOTF	joint special operations task force
JSTARS	Joint Surveillance Target Attack Radar System
JTCB	joint targeting coordination board

K

k	thousand
KBC	kill box coordinator
kt	knot (nautical miles per hour)

L

LeMay Center	Curtis E. LeMay Center for Doctrine Development and Education
LGW	laser-guided weapon
LOAC	law of armed conflict
LOC	line of communication
LST	laser spot tracker
LTL	laser-to-target line

M

MAAP	master air attack plan
MAC	maritime air control
MACCS	Marine air command and control system
MAGTF	Marine air-ground task force
MARFOR	Marine Corps forces
MARLE	Marine liaison element
MATCD	Marine air traffic control detachment
MBT	main battle tank
MCAS	Marine Corps air station
MCRP	Marine Corps reference publication
MCTP	Marine Corps tactical publication
MDM	maritime defense measure
MGRS	Military Grid Reference System
MKPT	mark point

MRR	minimum-risk route
MSL	mean sea level
MSR	main supply route
MTTP	multi-Service tactics, techniques, and procedures
MUM-T	manned unmanned teaming

N

NAI	named area of interest
NALE	naval and amphibious liaison element
NAS	naval air station
NATO	North Atlantic Treaty Organization
NFA	no-fire area
nm	nautical mile
NSFS	naval surface fire support
NTACS	Navy tactical air control system
NTRP	Navy tactical reference publication
NTTP	Navy tactics, techniques, and procedures
NVD	night vision device
NWDC	Navy Warfare Development Command

O

OCA	offensive counterair
OPCON	operational control
OPORD	operation order

P, Q

PID	positive identification
PKB	purple kill box
POI	point of interest

R

RFA	restrictive fire area
RFL	restrictive fire line
ROE	rules of engagement
ROZ	restricted operations zone
RPA	remotely piloted aircraft

RSTA	reconnaissance, surveillance, and target acquisition
RTB	return to base
RW	rotary-wing

S

S/S	surface-to-surface
SA	surveillance area
SACC	supporting arms coordination center
SAG	surface action group
SAM	surface-to-air missile
SAR	synthetic aperture radar
SCAR	strike coordination and reconnaissance
SCC	sea combat commander
SCL	standard conventional load
SEAD	suppression of enemy air defenses
SEZ	ship engagement zone
SFC	surface
SITREP	situation report
SOF	special operations forces
SOLE	special operations liaison element
SPINS	special instructions
STANAG	standardization agreement (NATO)
SUWC	surface warfare commander

T

TACC (USMC)	tactical air command center
TACC (USN)	tactical air control center
TACON	tactical control
TACP	tactical air control party
TACS	theater air control system
TADC	tactical air direction center
TAGS	theater air-ground system
TAI	target area of interest
TAOC	tactical air operations center (USMC)

TBMCS	theater battle management core system
TC	transit corridor
TEFACHR	threat, enemy situation, friendly update, artillery, clearance authority, hazards, remarks/restrictions
TOS	time on station
TOT	time on target
TR	transit route
TRADOC	United States Army Training and Doctrine Command
TST	time-sensitive target
TTFACOR	targets, threats, friendlies, artillery, clearance, ordnance, restrictions
TTP	tactics, techniques, and procedures
TTT	time to target

U

UA	unmanned aircraft
UAS	unmanned aircraft system
US	United States
USAF	United States Air Force
USMC	United States Marine Corps

V

VA	vital area

W

WOC	wing operations center

X, Y,Z

XAI	airborne alert air interdiction
XINT	alert interdiction

PART II – TERMS AND DEFINITIONS

aviation combat element (USMC)—The core element of a Marine air-ground task force (MAGTF) that is task-organized to conduct aviation operations. The aviation combat element (ACE) provides all or a portion of the six functions of Marine aviation necessary to accomplish the MAGTF's mission. These functions are antiair warfare, offensive air support, assault support, electronic warfare, air reconnaissance, and control of aircraft and missiles. The ACE is usually composed of an aviation unit headquarters and various other aviation units or their detachments. It can vary in size from a small aviation detachment of specifically required aircraft to one or more Marine aircraft wings. In a joint or multinational environment, the ACE may contain other Service or multinational forces assigned or attached to the MAGTF. The ACE itself is not a formal command. Also called ACE. See also command element; ground combat element; logistics combat element; Marine air-ground task force; Marine expeditionary brigade; Marine expeditionary force; Marine expeditionary force (Forward); Marine expeditionary unit; special purpose Marine air-ground task force. (Source: MCRP 1-10.2)

air interdiction—Air operations conducted to divert, disrupt, delay, or destroy the enemy's military surface capabilities before it can be brought to bear effectively against friendly forces, or to otherwise achieve objectives that are conducted at such distances from friendly forces that detailed integration of each air mission with the fire and movement of friendly forces is not required. Also called AI (DOD Dictionary of Military and Associated Terms. Source: JP 3-03)

airborne alert air interdiction—Airborne alert air interdiction identifies an AI airborne alert mission tasked for on-call targets that may be retasked during execution for targets of opportunity (also referred to as armed reconnaissance). Also called XAI. (Source: Air Force Doctrine, Annex 3-03)

aircraft control unit—A unit exercising tactical control of an aircraft, or a unit with facilities and controllers for conducting aircraft control, usually titled according to the mission or type of aircraft being controlled. Also known as an air control unit. (Source: NTRP 1-02)

airspace control order—An order implementing the airspace control plan that provides the details of the approved requests for airspace coordinating measures. It is published either as part of the air tasking order or as a separate document. Also called ACO. (DOD Dictionary of Military and Associated Terms. Source: JP 3-52)

airspace coordinating measures—Measures employed to facilitate the efficient use of airspace to accomplish missions and simultaneously provide safeguards for friendly forces. Also called ACMs. See also airspace control area; airspace control sector; airspace coordination area; high-density airspace control zone; weapons engagement zone. (DOD Dictionary of Military and Associated Terms. Source: JP 3-52)

air support operations center—The principal air control agency of the theater air control system responsible for the direction and control of air operations directly supporting the ground combat element. Also called ASOC. See also close air support; operation; tactical air control center. (DOD Dictionary of Military and Associated Terms. Source: JP 3-09.3)

air tasking order—A method used to task and disseminate to components, subordinate units, and command and control agencies projected sorties, capabilities and/or forces to targets and specific missions. Normally, provides specific instructions to include call signs, targets, controlling agencies, etc., as well as general instructions. Also called ATO. (DOD Dictionary of Military and Associated Terms. Source JP 3-30)

area of operations—An operational area defined by the joint force commander for land and maritime forces that should be large enough to accomplish their missions and protect their forces. Also called AO. See also, area of responsibility; joint operations area; joint special operations area. (DOD Dictionary of Military and Associated Terms. Source: JP 3-0)

armed reconnaissance—A mission with the primary purpose of locating and attacking targets of opportunity (i.e., enemy materiel, personnel, and facilities), in assigned general areas or along assigned communications routes, and not for the purpose of attacking specific briefed targets. (Source: MCTP 3-20D)

battle damage assessment—The estimate of damage composed of physical and functional damage assessments, as well as target system assessment, resulting from the application of lethal or nonlethal military force. Also called BDA. See also combat assessment. (DOD Dictionary of Military and Associated Terms. Source: JP 3-0)

battlefield coordination detachment—An Army liaison located in the air operations center that provides selected operational functions between the Army forces and the air component commander. Also called BCD. See also air and space operations center; liaison. (DOD Dictionary of Military and Associated Terms. Source: JP 3-03)

BINGO—Fuel state needed for [aircraft] recovery. (Source: ATP 1-02.1, MCRP 3-30B.1, NTTP 6-02.1, AFTTP 3-2.5)

blue kill box—A fire support and airspace coordination measure used to facilitate the attack of surface targets with air-to-surface munitions without further coordination with the establishing headquarters. (Source: ATP 3-09.34, MCRP 3-31.4, NTTP 3-09.2.1, AFTTP 3-2.59)

BULLSEYE—An established reference point from which the position of an object can be referenced by bearing (magnetic) and range (nautical mile) from this point. (Source: ATP 1-02.1, MCRP 3-30B.1, NTTP 6-02.1, AFTTP 3-2.5)

(target/object) CAPTURED—[A/S] A specified target or object has been acquired and is being tracked with an onboard sensor. (Source: ATP 1-02.1, MCRP 3-30B.1, NTTP 6-02.1, AFTTP 3-2.5)

classification, identification, and engagement area—In maritime operations, the area within the surveillance area and surrounding the vital area(s) in which all objects detected must be classified, identified, and monitored; and the capability maintained to escort, cover, or engage. (Source: NTRP 1-02)

close air support—Air action by fixed- and rotary-wing aircraft against hostile targets that are in close proximity to friendly forces and that require detailed integration of each air mission with the fire and movement of those forces. Also called CAS. See also air interdiction. (DOD Dictionary of Military and Associated Terms. Source: JP 3-0)

collateral damage—Unintentional or incidental injury or damage to persons or objects that would not be lawful military targets in the circumstances ruling at the time. (DOD Dictionary of Military and Associated Terms. Source: JP 3-60)

CONTACT—1. Sensor situational awareness at the stated position. 2. Air-to-surface [A/S]. Acknowledges sighting of a specified reference point (either visually or via sensor). 3. Air-to-air [A/A] Individual radar return within a GROUP or ARM. (Source: : ATP 1-02.1, MCRP 3-30B.1, NTTP 6-02.1, AFTTP 3-2.5)

contact of interest—In surface warfare, a naval combatant or unknown surface contact operating in a designated area whose presence may have tactical significance but should have no real impact on mission completion. (Source: NTRP 1-02)

DEADEYE—Laser designator system inoperative. (Source: : ATP 1-02.1, MCRP 3-30B.1, NTTP 6-02.1, AFTTP 3-2.5)

direct air support center—The principal air control agency of the United States Marine air command and control system responsible for the direction and control of air operations directly supporting the ground combat element. Also called DASC. (DOD Dictionary of Military and Associated Terms. Source: JP 3-09.3)

fire support coordination center—A single location in which are centralized communications facilities and personnel incident to the coordination of all forms of fire support for Marine forces. Also called FSCC. See also fire support; fire support coordination; support; supporting arms coordination center. (DOD Dictionary of Military and Associated Terms. Source: JP 3-09)

fire support coordination measure—A measure employed by commanders to facilitate the rapid engagement of targets and simultaneously provide safeguards for friendly forces. Also called FSCM. See also, fire support coordination. (DOD Dictionary of Military and Associated Terms. Source: JP 3-0)

forward line of own troops—A line that indicates the most forward positions of friendly forces in any kind of military operation at a specific time. Also called FLOT. (DOD Dictionary of Military and Associated Terms. Source: JP 3-03)

ground combat element—The core element of a Marine air-ground task force that is task-organized to conduct ground operations. It is usually constructed around an infantry organization but can vary in size from a small ground unit of any type to one or more Marine divisions that can be independently maneuvered under the direction of the MAGTF commander. It includes appropriate ground combat and combat support forces, and in a joint or multinational environment, it may also contain other Service or multinational forces assigned or attached to the MAGTF.

The ground combat element itself is not a formal command. Also called GCE. See also aviation combat element; command element; logistics combat element; Marine air-ground task force; Marine expeditionary brigade; Marine expeditionary force; Marine expeditionary force (Forward); Marine expeditionary unit; special purpose Marine air-ground task force. (Source: MCRP 1-10.2)

identification—1. The process of determining the friendly or hostile character of an unknown detected contact. 2. In arms control, the process of determining which nation is responsible for the detected violations of any arms control measure. 3. In ground combat operations, discrimination between recognizable objects as being friendly or enemy, or the name that belongs to the object as a member of a class. Also called ID. (DOD Dictionary of Military and Associated Terms. Source JP 3-01)

intelligence, surveillance, and reconnaissance—1. An integrated operations and intelligence activity that synchronizes and integrates the planning and operation of sensors, assets, and processing, exploitation, and dissemination systems in direct support of current and future operations. 2. The organizations or assets conducting such activities. Also called ISR. (DOD Dictionary of Military and Associated Terms. Source: JP 2-01)

interdiction—An action to divert, disrupt, delay, or destroy the enemy's military surface capability before it can be used effectively against friendly forces, or to achieve enemy objectives. (DOD Dictionary of Military and Associated Terms. Source: JP 3-03)

INVESTIGATE—1. [A/S] Verify specified element(s) of rules of engagement (ROE), positive identification (PID), collateral damage estimation (CDE), and/or coordination of forces on the referenced target or track. 2. [AIR-MAR] Verify specified element(s) of ROE, PID, clear field of fire, and/or coordination of forces on the referenced target or track. (Source: : ATP 1-02.1, MCRP 3-30B.1, NTTP 6-02.1, AFTTP 3-2.5)

joint air-ground integration center—A staff organization designed to enhance joint collaborative efforts to deconflict joint air-ground assets in the division's airspace. Also called JAGIC. (DOD Dictionary of Military and Associated Terms. Source: JP 3-09.3)

joint air operations center—A jointly staffed facility established for planning, directing, and executing joint air operations in support of the joint force commander's operation or campaign objectives. Also called JAOC. See also joint air operations. (DOD Dictionary of Military and Associated Terms. Source: JP 3-30)

joint fires—Fires delivered during the employment of forces from two or more components in coordinated action to produce desired effects in support of a common objective. See also fires. (DOD Dictionary of Military and Associated Terms. Source: JP 3-0)

joint fires element—An optional staff element that provides recommendations to the operations directorate to accomplish fires planning and synchronization. Also called JFE. See also fire support; joint fires. (DOD Dictionary of Military and Associated Terms. Source: JP 3-60)

joint force air component commander—The commander within a unified command, subordinate unified command, or joint task force responsible to the establishing commander for recommending the proper employment of assigned, attached, and/or made available for tasking air forces; planning and coordinating air operations; or accomplishing such operational missions as may be assigned. Also called JFACC. See also joint force commander. (DOD Dictionary of Military and Associated Terms. Source: JP 3-0)

joint force commander—A general term applied to a combatant commander, subunified commander, or joint task force commander authorized to exercise combatant command (command authority) or operational control over a joint force. Also called JFC. See also joint force. (DOD Dictionary of Military and Associated Terms. Source: JP 1)

joint force land component commander—The commander within a unified command, subordinate unified command, or joint task force responsible to the establishing commander for recommending the proper employment of assigned, attached, and made available for tasking land forces; planning and coordinating land operations; or accomplishing such operational missions as may be assigned. Also called JFLCC. See also joint force commander. (DOD Dictionary of Military and Associated Terms. Source: JP 3-0)

joint force maritime component commander—The commander within a unified command, subordinate unified command, or joint task force responsible to the establishing commander for recommending the proper employment of assigned, attached, and/or made available for tasking maritime forces and assets; planning and As coordinating maritime operations; or accomplishing such operational missions as may be assigned. Also called JFMCC. See also joint force commander. (DOD Dictionary of Military and Associated Terms. Source: JP 3-0)

joint force special operations component commander—The commander within a unified command, subordinate unified command, or joint task force responsible to the establishing commander for recommending the proper employment of assigned, attached, and made available for tasking special operations forces and assets; planning and coordinating special operations; or accomplishing such operational missions as may be assigned. Also called JFSOCC. See also joint force commander. (DOD Dictionary of Military and Associated Terms. Source: JP 3-0)

joint integrated prioritized target list—A prioritized list of targets approved and maintained by the joint force commander. Also called JIPTL. See also target. (DOD Dictionary of Military and Associated Terms. Source: JP 3-60)

joint operations area—An area of land, sea, and airspace, defined by a geographic combatant commander or subordinate unified commander, in which a joint force commander (normally a joint task force commander) conducts military operations to accomplish a specific mission. Also called JOA. See also area of responsibility; joint special operations area. (DOD Dictionary of Military and Associated Terms. Source: JP 3-0)

joint special operations task force—A joint task force composed of special operations units from more than one Service, formed to carry out a specific special operation or prosecute special operations in support of a theater campaign or other operations. Also called JSOTF. (DOD Dictionary of Military and Associated Terms. Source: JP 3-05)

line of communications—A route, either land, water, and/or air, that connects an operating military force with a base of operations and along which supplies and military forces move. Also called LOC. (DOD Dictionary of Military and Associated Terms. Source: JP 2-01.3)

Marine air command and control system—A system that provides the aviation combat element commander with the means to command, coordinate, and control all air operations within an assigned sector and to coordinate air operations with other Services. Also called MACCS. See also direct air support center; tactical air operations center. (DOD Dictionary of Military and Associated Terms. Source: JP 3-09.3)

Marine air-ground task force—The Marine Corps' principal organization for all missions across a range of military operations, composed of forces task-organized under a single commander capable of responding rapidly to a contingency anywhere in the world. The types of forces in the Marine air-ground task force (MAGTF) are functionally grouped into four core elements: a command element, an aviation combat element, a ground combat element, and a logistics combat element. The four core elements are categories of forces, not formal commands. The basic structure of the MAGTF never varies, through the number, size, and type of Marine corps units comprising each of its four elements will always be mission dependent. The flexibility of the organizational structure allows for one or more subordinate MAGTFs to be assigned. In a joint or multinational environment, other Service or multinational forces may be assigned or attached. Also called MAGTF. See also aviation combat element; command element; ground combat element; logistics combat element; Marine expeditionary brigade; Marine expeditionary force; Marine expeditionary force (Forward); Marine expeditionary unit; special purpose Marine air-ground task force. (Source: MCRP 1-10.2)

named area of interest—The geospatial area or systems node or link against which information that will satisfy a specific information requirement can be collected, usually to capture indications of adversary courses of action. Also called NAI. See also area of interest. (DOD Dictionary of Military and Associated Terms. Source: JP 2-01.3)

NO JOY—[A/A][A/S][S/A] Aircrew does not have visual contact with the TARGET/BANDIT. Opposite of TALLY. (Source: : ATP 1-02.1, MCRP 3-30B.1, NTTP 6-02.1, AFTTP 3-2.5)

operational area—An overarching term encompassing more descriptive terms (such as area of responsibility and joint operations area) for geographic areas in which military operations are conducted. Also called OA. See also amphibious objective area; area of operations; area of responsibility; joint operations area; joint special operations area; theater of operations; theater of war. (DOD Dictionary of Military and Associated Terms. Source: JP 3-0)

PLAYTIME—Amount of time aircraft can remain on station, given in hours plus minutes. (e.g., ONE PLUS THIRTY equals 1 hour and 30 minutes) (Source: : ATP 1-02.1, MCRP 3-30B.1, NTTP 6-02.1, AFTTP 3-2.5)

positive identification—An identification derived from observation and analysis of target characteristics including visual recognition, electronic support systems, non-cooperative target recognition techniques, identification friend or foe systems, or other physics-based identification techniques (DOD Dictionary of Military and Associated Terms. Source: JP 3-01)

purple kill box—A fire support and airspace coordination measure used to facilitate the attack of surface targets with subsurface-, surface-to-surface, and air-to-surface munitions without further coordination with the establishing headquarters. (Source: ATP 3-09.34, MCRP 3-31.4, NTTP 3-09.2.1, AFTTP 3-2.59)

RETROGRADE (ING)—Withdraw/withdrawing, while executing defensive procedures in response to a threat. (Source: : ATP 1-02.1, MCRP 3-30B.1, NTTP 6-02.1, AFTTP 3-2.5)

rules of engagement—Directives issued by competent military authority that delineate the circumstances and limitations under which United States forces will initiate and/or continue combat engagement with other forces encountered. Also called ROE. See also law of war. (DOD Dictionary of Military and Associated Terms: Source: JP 1-04)

SCRAM (direction)—1. FRIENDLY asset is in immediate danger. Withdraw clear in the direction indicated for survival. No further mission support from the FRIENDLY asset is expected. 2. [A/A] [EW] [S/A] Cease the intercept and take immediate evasive action. Implies that the target aircraft is being engaged by SAMs or other air defense fighters. 3. [A/A] Directive term to instruct air traffic control (ATC) to clear all civilian/nonengagement aircraft outside a specific area. TERMINATE SCRAM will resume normal operations. (Source: : ATP 1-02.1, MCRP 3-30B.1, NTTP 6-02.1, AFTTP 3-2.5)

SLIDE—1. [A/A] [EW] Continue/continuing present mission while flowing from station in response to a perceived threat, implies intent to RESET. 2. [A/S] Directive or descriptive term to laterally reposition helicopter. (Source: : ATP 1-02.1, MCRP 3-30B.1, NTTP 6-02.1, AFTTP 3-2.5)

SMACK—1. [A/S] [S/S] Clearance to employ ordnance/fires on surface target coordinates. ROE, PID, CDE, coordination of forces, and commander's guidance requirements on the referenced target/track have been satisfied. Not to be utilized in a close air support engagement. 2. [AIR-MAR] Clearance to employ ordnance/fires on surface target coordinates/Link-16 track. ROE, PID, clear field of fire, coordination of forces, and commander's guidance requirements on the referenced target/track have been satisfied. (Source: : ATP 1-02.1, MCRP 3-30B.1, NTTP 6-02.1, AFTTP 3-2.5)

special operations forces—Those Active and Reserve Component forces of the Services designated by the Secretary of Defense and specifically organized, trained, and equipped to conduct and support special operations. Also called SOF. See also Air Force special operations forces; Army special operations forces; Navy special operations forces. (DOD Dictionary of Military and Associated Terms. Source: JP 3-05)

strike coordination and reconnaissance—A mission flown for the purpose of detecting targets and coordinating or performing attack or reconnaissance on those targets. [SCAR missions are flown in a specific geographic area and are an element of the C2 interface to allocate flights and assets, properly match weapons effects with targets to effectively and efficiently destroy targets per the supported commander's prioritized target list, and provide battle damage assessment (BDA).] (DOD Dictionary of Military and Associated Terms. Source JP 3-03)

supported commander—1. The commander having primary responsibility for all aspects of a task assigned by the Joint Strategic Capabilities Plan or other joint operation planning authority. 2. In the context of joint operation planning, the commander who prepares operation plans or operation orders in response to requirements of the Chairman of the Joint Chiefs of Staff. 3. In the context of a support command relationship, the commander who receives assistance from another commander's force or capabilities, and who is responsible for ensuring that the supporting commander understands the assistance required. See also support; supporting commander. (DOD Dictionary of Military and Associated Terms. Source: JP 3-0)

supporting arms coordination center—A single location on board an amphibious command ship in which all communication facilities incident to the coordination of fire support of the artillery, air, and naval gunfire are centralized. (DOD Dictionary of Military and Associated Terms. Source: 3-09.3)

surveillance area—In surface warfare, the operational environment that extends out to a range that equals the force's ability to conduct a systematic observation of a surface area using all available and practical means to detect any vessel of possible military concern. (Source: NTRP 1-02)

tactical air command center (USMC)—The principal US Marine Corps air command and control agency from which air operations and air defense warning functions are directed. Also called TACC (USMC). (DOD Dictionary of Military and Associated Terms. Source: JP 3-09.3)

tactical air control center (USN)—The principal air operations installation (ship-based) from which all aircraft and air warning functions of tactical air operations are controlled. Also called TACC (USN). (DOD Dictionary of Military and Associated Terms. Source: JP 3-09.3)

TARGET—2. [A/S] [S/S] ROE, PID, coordination of forces, and commander's guidance requirements on the referenced target/track have been satisfied. Target/track correlation and CDE must be accomplished prior to employing ordnance/fires. 3. [AIR-MAR] ROE, PID, coordination of forces, and commander's guidance requirements on the referenced target/track have been satisfied. Target/track correlation and clear field of fire must be accomplished prior to employing ordnance/fires. (Source: : ATP 1-02.1, MCRP 3-30B.1, NTTP 6-02.1, AFTTP 3-2.5)

target area of interest—The geographical area where high-value targets can be acquired and engaged by friendly forces. Also called TAI. See also area of interest; high-value target; target. (DOD Dictionary of Military and Associated Terms. Source: JP 2-01.3)

time on target—The actual time at which munitions impact the target. Also called TOT. (DOD Dictionary of Military and Associated Terms. Source: JP 3-09.3)

time-sensitive target— A joint force commander validated target or set of targets requiring immediate response because it is a highly lucrative, fleeting target of opportunity or it poses (or will soon pose) a danger to friendly forces. Also called TST. (DOD Dictionary of Military and Associated Terms. Source: JP 3-60)

unmanned aircraft—An aircraft that does not carry a human operator and is capable of flight with or without human remote control. Also called UA. (DOD Dictionary of Military and Associated Terms. Source: JP 3-30)

vital area—A designated area or installation to be defended by air defense units, (Source: NTRP 1-02)

VISUAL—[A/A][A/S] Sighting of a friendly aircraft or ground position. Opposite of BLIND. (Source: : ATP 1-02.1, MCRP 3-30B.1, NTTP 6-02.1, AFTTP 3-2.5)

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***ATP 3-60.2**
MCRP 3-20D.1
NTTP 3-03.4.3
AFTTP 3-2.72
31 Jan 2018

By Order of the Secretary of the Army

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* Supersedes FM 3-60.2/MCRP 3-20D.1/NTTP 3-03.4.3/AFTTP 3-2.72 dated 31 March 2014.

MARINE CORPS PCN: 144 000170 00 PIN: 103801-000